Dark Side of the Day: Overnight Price Jumps and Short-Term Return Predictability

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June 2026

Abstract

Using 9,283 stocks listed on the NYSE, AMEX, and NASDAQ, we document return reversals and show that overnight jump returns negatively predict short-term returns in both positive and negative episodes. This relationship remains robust across subperiods, including times of market distress, and across quintile portfolios formed on firm-specific factors. Although a frictionless long-short strategy following jump days generates 1.05% positive abnormal returns with the magnitude declining as the holding period expands, the same strategy based on extreme deciles of lagged overnight jump returns yields a 0.6% risk-adjusted loss over a one-month horizon, implying that the overreaction is transient. Overall, the findings suggest that price discovery is contingent on an active trading environment as well, and reversal patterns following overnight jumps are consistent with temporary mispricing that tends to diminish once continuous trading resumes.

JEL classification: G10; G11; G12; G14; G40

Keywords: Price Jumps; Investor Overreaction; Return Predictability; Overnight and Intraday Returns

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1 Introduction

Instant and precise reflection of new information to prices in a friction-free market has been one of the asset pricing mantras for decades. Also linked to this *frictionless markets* parlance, there should be no predictability in returns following shocks, be they news-induced or not (Frank and Sanati (2018)). Nevertheless, this assertive textbook approach is not a proper description of market practice, as the prices do not completely embody the information available to participants at a given time. The literature extensively documents drifting as well as reversing return patterns in the wake of information arrival. In one strand of the literature, these predictable patterns are linked to flaws in investors' cognitive judgments and to market inefficiency, while other lines of research tie this return behavior to varying levels of expected returns as a rational reaction to fluctuating risk levels (see Lehmann (1990), Fama (1991), Chopra et al. (1992) and McLean and Pontiff (2016) among many others). However, beyond these discussions, a panoramic picture of overreaction and underreaction studies exhibits that the literature is still indecisive about the dominant return patterns in the post-shock period, as highlighted in Frank and Sanati (2018) and Tetlock (2014).

We revisit short-term return predictability discussions by concentrating on price discontinuities observed in the overnight section. Our contribution to the discourse centers on short-term return predictability through the lens of a specific context -positive and negative overnight price jumps- in which the predictive power of the factor is not marred with the confounding effects of noise trading[1], liquidity imbalances, microstructure design, intraday learning from others and other daytime forces that might be relevant to the price discovery. Since trading is largely suspended during the night, overnight jumps provide an instrumental setting to isolate immediate investor reactions to information shocks before daytime frictions revive with the commencement of trading. Although prior literature documents return predictability linked to scheduled and unscheduled information events and their forms (whether discrete or continuous)[2], the relevance of information forms to return predictability is relatively untouched for the overnight segment. Building on the psychological notion that dramatic information events stimulate stronger responses, our research addresses this void by studying overnight price jumps and subsequent returns to examine whether the systematic short-term reversals accord with heuristic biases and

[1] "Noise trading is trading on noise as if it were information" as shortly defined by Black (1986)

[2] See for instance Da et al. (2014) and the references therein.

investor overreaction.

We find return reversals following overnight information shocks, and document that negative and positive overnight jump returns are negatively associated with the short-term cumulative future returns.[3] This empirical regularity is reconcilable with temporary mispricing mechanisms of investor overreaction. We report that negative and positive overnight jumps are followed by statistically and economically significant return reversals, concentrated to a large extent in the subsequent daytime trading session. The strength of the association and the factor loadings vary among quintile portfolios formed on firm-specific factors with statistical significance maintained across isolated and cumulative approach specifications.

The finance literature has extensively considered extreme price changes over the past decades and the implications for price discontinuities have been widely studied for single assets, portfolios, and derivative instruments in a myriad of ways: amendments in asset pricing (Merton (1976); Beckers (1981); Ball and Torous (1983); Ball and Torous (1985); Câmara (2009)), return predictability (Bali and Hovakimian (2009); Jiang and Yao (2013); Jiang and Zhu (2017)), information flow (Barclay and Litzenberger (1988); Andersen et al. (2007); Bollerslev et al. (2008); Baker et al. (2021); Jeon et al. (2022b)), liquidity shocks (Jiang et al. (2011); Christensen et al. (2014)) and overreaction/underreaction and herd behavior (Kaul and Nimalendran (1990); Jiang and Zhu (2017) Wanidwaranan and Padungsaksawasdi (2020)) are a few of the concepts analyzed in connection with stock price jumps. Using the methodology of Lee and Mykland (2008), we first detect overnight price jumps in stocks listed on the New York Stock Exchange (NYSE), American Stock Exchange (AMEX), and National Association of Securities Dealers Automated Quotations (NASDAQ) Stock Market over the entire July 1993 - December 2021 period or in-between. Like Jiang and Zhu (2017), we use jumps as a proxy for information shocks that lead to breaks in the price path. After identifying the dates with price discontinuities, we scrutinize subsequent returns to assess the consequences of investor decisions over the full sample period and across specific subperiods. We also form quintile portfolios based on different levels of firm-specific factors and document varying overreaction dynamics and return predictability. Contrarian trading strategies based on short-term holding periods following jumps, as well as on monthly jump figures,

[3] A stream of research studies investor overreaction and its reciprocal interaction with expected returns. Among others, see for instance De Bondt and Thaler (1985) and Chopra et al. (1992) for long-term return predictability discussions and Avramov et al. (2006), Lo and MacKinlay (1990), Lehmann (1990), Poterba and Summers (1988) and Barr Rosenberg and Lanstein (1998) for return predictability in the short-term.

further evince that these overreactions and return reversals are short-term market episodes.

Our findings suggest that price discovery requires not only information but also an active trading environment that facilitates a corrective impact through the incorporation of liquidity, microstructure effects, and learning from others. With trading largely inactive after market closure, overnight return components may dominantly channel information effects. Hence, overnight jump returns predicting subsequent daytime returns with negative loadings suggest that continuous trading plays an essential role in accomplishing price discovery and reveals market frictions in information processing. On the whole, our contribution to extant literature will be twofold.

First, our study concentrates on an unexplored question and connects overnight price jumps and their reversals with short-term overreaction discussions in stock markets. As opposed to Jiang and Zhu (2017) which identify daily jumps and decompose these close-to-close returns into their overnight and intraday components, we detect overnight jumps in its own time series and mark the days with overnight return surprises. Jiang and Zhu (2017) provide evidence of underreaction to information shocks, which end up as daily jumps whereas we analyze extreme price movements that become ephemeral to a certain extent after the market correction.[4] In contrast to other factors that are priced only over certain horizons, jump returns exhibit consistently strong and statistically significant predictive power across all model specifications and subperiods, maintaining robustness and economic relevance for overreaction in quintile portfolio tests.

Second, we implement daily event-based contrarian trading strategies around jump incidents by taking long positions in stocks with negative jumps and short positions in those with positive jumps. This strategy yields a highly significant risk-adjusted return of 1.05% for a one-day holding period, with the magnitude decreasing as the holding period increases. On the other hand, we also document the results of contrarian and relative strength trading strategies to see if the winners (stocks with cumulative positive jump returns in the previous month) will be the losers within the one-month investment horizon or vice versa. However, shorting the stocks in

[4]If we focus exclusively on daily jumps, the dynamics embedded in the overnight section would remain unexamined on the days when no daily jump is identified. Put differently, a substantial price discontinuity may occur in the overnight interval, yet that day may fail to be classified as a daily jump due to subsequent intraday price movements alleviating the overnight change. As a result, economically meaningful overnight dynamics may remain untouched under a daily-jump framework. Hence, our filtering methodology provides us with special information when there is no daily jump. We additionally run our detection test for close-to-close returns to see jumps in daily price movements and their alignment with overnight jumps; only 11.6% percent of overnight jump days have also jumps in daily returns.

the highest decile and buying the stocks with the most negative jump figures ended up in a statistically significant 0.6% loss. In the same fashion, short and long trading strategies, respectively in the highest and lowest decile portfolios, do not result in any risk-adjusted gain. These pricing behaviors imply that overreaction and return reversal after overnight jumps are short-term market phenomena with the predictive signal dissipating as the holding horizon extends.

While prior work establishes information as the dominant driver of overnight returns, our study does not directly classify the news events underlying these jumps. Nevertheless, we conduct specific assessments for crisis periods during which market-wide shocks are relatively more probable and report that the factor loadings are higher following both positive and negative overnight jumps in crisis periods than in non-crisis intervals. Notably, the coefficient of JR after negative overnight jumps exceeds double the corresponding non-crisis factor loading. We therefore conjecture that market-wide information shocks are likely to produce more powerful reversal patterns, though this reasoning is not a substitute for a dedicated news-based analysis, and we acknowledge the value of a dedicated study on news types. In particular, distinguishing between firm-specific and market-wide information releases could provide a more granular mapping between overnight price discontinuities and information flows and can help us identify the informational contexts in which the documented reversal patterns are most pronounced. Although we construct a proxy-based approach to differentiate between market-wide and firm-specific news days, the lack of a direct classification of news events remains a limitation of the analysis.

The rest of the paper is organized as follows. Section 2 presents the related literature and contrasts our study with the previous research. In Section 3, we provide the details of data and filtering mechanisms together with the applied methodology for jump identification and data construction. Section 4 is reserved for empirical findings. Implications for market participants and theoretical foundations are detailed in Section 5. Section 6 concludes.

2 Relevant Literature

The foundations of human psychology constrain the uniform allocation of attention across information arrivals; large and salient changes command more attention than smaller, incremental ones. Gino and Bazerman (2009) explain this dual behavioral phenomenon by studying that unethical behaviors become more acceptable when they are introduced incrementally rather than

enacted in a single one-time action. In the same vein, finance literature extensively documents that behavioral factors shape return dynamics. Indicating the relevance of psychological biases to financial markets, a recent study by Nguyen and Pham (2021) highlights that market anomalies become more pronounced when the level of attention is high. Although gradually arriving small amounts of information lead to investor inattention and underlie the formation of momentum returns, surprising news will induce a stronger reaction, resulting in return reversal. Investor attention is also weakened when there is too much information to process, leading to underreaction (among others, see Hirshleifer et al. (2009) and Michaely et al. (2014)), but their sensitivity is reinforced if the information is discrete (Da et al. (2014)). Avery and Zemsky (1998) also point out that herd behavior can lead to bubbles and market crashes when there is uncertainty about the traders' information.

The impact of overnight periods on stock price behaviors has attracted interest over decades. Among early works, Amihud and Mendelson (1987) show that open-to-open returns in Dow Jones Industrial stocks have higher variance, higher serial dependence, greater peakedness, and thicker tails compared to close-to-close returns. This was first attributed to different trading mechanisms in the opening (periodic call market) and closing hours (continuous auction). In their follow-up study, however, Amihud and Mendelson (1991) analyze 50 stocks in the Tokyo Stock Exchange in which the Exchange employs two separate periodic call markets, both in the morning and afternoon sessions. Authors this time show that higher volatility and negative autocorrelation in open-to-open returns are due to the prior nontrading section over the night and document that price reversals are stronger in the morning session compared to the afternoon period. With a similar interest in the dynamics of intraday and overnight return components, Gerety and Mulherin (1992) show that enhanced trading activity at the opening and closing hours of a typical trading day is positively related to the expected overnight return volatility. Investors abstain from carrying their positions over the night to decrease their risk exposure and that leads to a surge in trading volumes at both ends of the day. Opening volume is also positively correlated with the unexpected volatility component, which accounts for the surprise in overnight information flow. Moreover, a dedicated strand of literature rests on overnight returns to derive the firm-specific investor sentiments. Berkman et al. (2012) report that individual investors, after the markets open, descend on stocks that attracted their attention during the last closing period. In a similar vein, Barber and Odean (2008) document that periods of higher returns trigger retail

investor attention, leading them to submit buy orders predominantly at the market opening on the following day. As also discussed by Aboody et al. (2018), retail investor decisions are more prone to sentimental factors compared to institutional investors. Taking a closer look at investors' timing preferences, Lou et al. (2019) report that individual investors generally tend to initiate their trades around opening hours, whereas institutional activity is more prevalent in the second half of the day, particularly toward the close. To the extent that such features characterize the market participation dynamics, the opening session is more susceptible to behavioral forces. Hence, the literature provides a foundation for understanding how non-trading periods after market closure shape trading patterns, volatility, and volume, setting the stage for later studies that link these discussions to information flows and return dynamics.

Among others, the importance of jump returns is underscored by Kapadia and Zekhnini (2019), who show that, on average, the annual mean return of a stock is cumulatively generated by price jumps occurring within only 4.5 days. Similarly, Jiang and Yao (2013) examine intermittent jumps triggered by information shocks over an extended horizon and document that return predictability associated with firm characteristics owes too much to price jumps such that size and liquidity variables lose their explanatory power, and the significance of the value factor declines sharply once extreme price movements are controlled for. Orłowski et al. (2024) show that the premium for market skewness risk can originate from jump and leverage effect components and document that skewness premia are higher in the overnight section. Using call-put implied volatility spread (CVol-PVol) as a proxy for jump risk and isolating it from volatility risk, Bali and Hovakimian (2009) find a positive association between CVol-PVol and future stock returns in the cross-section. As a priced factor, Pan (2002) documents a significant jump-risk premium in S&P500 index options over the period February 1996 - January 2005. Extreme shocks also intensify the interconnectedness in stock markets around the globe (Zhou and Liu (2023)). Building on this extending literature that underscores the economic significance of jumps, our study contributes by focusing on how overnight price jumps are instrumental for understanding follow-up return dynamics.

Firm-specific news is generally disclosed after the closing bell and priced in largely by individual investors as trading commences in the next morning (Lou et al. (2019)). Moreover, the main driving force of overnight returns is the information available to market participants (see, for instance, Jones et al. (1994) and Barclay and Hendershott (2003) among others). In a similar

fashion, Jiang et al. (2012) report that more than 95% of the earnings announcements in their sample of S&P500 stocks are released outside the trading hours after markets close. Michaely et al. (2014) likewise analyze these announcements and show that 92% take place outside regular market hours according to Newswire, and 81% according to I/B/E/S timestamps. As highlighted by Barardehi et al. (2022) and the references therein, order flow conveys the highest degree of information around the early hours of a trading day. Information flow can be firm-level (idiosyncratic) or market-level, and Boudoukh et al. (2019) document that idiosyncratic volatility triggered by firm-level public news accounts for 49.6% of the total overnight variance, whereas this explains only 12.4% of the trading hour (intraday) variance for S&P 500 firms. This body of work is unified in identifying information as the dominant force behind overnight returns.

Steffen (2023) documents a comprehensive literature review for extreme price movements and reversals in the intraday section with a special focus on flash crashes. However, literature on the overnight jump returns and investor overreaction is relatively intact. Similar to our reversal discussion, Lee (2023) documents that stocks with positive idiosyncratic jumps have lower follow-up returns, but the focus is not on the overnight jumps. The author also documents no significant association between negative price jumps and subsequent returns. Used as a proxy for information shocks, jumps in Jiang and Zhu (2017) are analyzed in the context of short-term underreaction in US equity markets in which the analysis rests on daily jump detection and decomposition of it into overnight and intraday sections.

Our study extends these findings by examining how overnight jumps, as discrete information shocks, shape the intensity and direction of short-term return patterns and broadly stands at the intersection of the jump, overreaction, and return predictability literature. There are specific studies (e.g. Baker et al. (2021), Jeon et al. (2022a), and Johnson et al. (2022) among others) where the direct focus is to analyze jumps and associated news flows and the types of information (e.g., idiosyncratic vs market information or scheduled vs unscheduled news flows) resulting in price jumps. However, overnight jumps in this study function as an ex-post metric for market judgment reflecting all idiosyncratic and market-wide discrete information shocks irrespective of their type and channel of dissemination. In other words, the focus of our research is the intensity of all forms of news flows and reflections in the price trajectory. Despite the indirect proxy for separating market-wide and firm-specific news flows, not distinguishing between different types of news events through direct event classification remains a limitation of our study.

3 Data and Methodology

3.1 Data

Since price jumps are low-probability episodes in nature, it is important to keep the database as large as possible.[5] We aim to tackle this rare-event challenge with a large sample of 9,283 stocks listed on the NYSE, AMEX, and NASDAQ during the whole July 1993 - December 2021 period or in between. Our daily raw data start as of June 15, 1992, on which the opening prices are first available in the Center for Research in Security Prices (CRSP) database, and our analyses start as of July 1993, when we first get our returns for the momentum variables along with the full-month jump incidences.

Our data sample consists of the entire CRSP database with some further filters. The study is conducted with common shares that are listed on the main US exchanges (NYSE, AMEX, NASDAQ). We make use of Permanent Company Number (PERMCO) and Permanent Security Number (PERMNO) as they are the primary CRSP identifiers to track companies and securities over the trading history, respectively. In our main analysis, we use PERMCO identifiers that are associated only with one PERMNO over the entire stock records. In the next step, we make sure that there are no trading breaks during the life of the company to avoid artificial jump identification. We keep stocks with at least three years of trading history and repeat our analyses with trading records longer than two years for robustness check. We do not shorten the data length further to ensure that momentum returns are calculated at least for a cycle of one complete year. Stocks that were delisted at any point during our whole analysis period remain included through their jump observations. Consequently, our results are not affected by survivorship bias or related selection issues. As the last data-sifting layer, we filter out observations with missing COMPUSTAT values. After these refinements, we cover 9283 stocks from the US markets. Sieved CRSP data are then merged with pertinent firm characteristics data from COMPUSTAT. We follow Fama and French (2008) and Jiang and Zhu (2017) to construct our variables and explain them below in turn.

CRSP's opening price reflects the opening auction price as long as there is a transaction in the auction. Opening auctions are designed to handle all collected orders at once, and a single

[5]For the significance of sample size in rare event studies, see Jiang and Yao (2013) on jumps and cross-sectional return predictability, Kelly and Jiang (2014) on extreme events and associated tail risk in stock returns and Boyer and Vorkink (2014) on skewness and investors' preferences towards lottery-like assets.

clearing price that maximizes the volume is announced as the auction price for numerous deals. It functions as a stabilizer and helps absorb the fluctuations by simultaneously processing all collected orders before the continuous auction. It has become an integral part of market design to alleviate the volatility in the opening hours and attenuate the noise. In that regard, announced CRSP opening prices reflect the initial market consensus around the accumulated overnight information. Missing opening prices are replaced with the previous day's close, resulting in zero overnight returns and no jump detection for those days.[6] If the closing price is missing, CRSP sets the bid-ask average as the closing price on that day. We retain these closing prices in the main analysis as they still reflect some sort of investor judgment about the market price. In our robustness check[7], however, we exclude jumps if the closing price is calculated through the bid-ask average.

Size (S): At the end of every June, we calculate market capitalization through the CRSP dataset. It is the natural logarithm of the last closing price times outstanding shares.

Book-to-Market Ratio (BM): Book value of the equity is received from the fiscal year ending figures in the previous calendar year, while the market value of the equity is calculated at the end of the last trading day in the preceding calendar year. The former is computed from COMPUSTAT by adding deferred taxes and investment tax credits to shareholders' equity and subtracting the preferred stock adjustments. Depending on availability, preferred stock rectification can be drained -with an order of precedence- through PSTKL or PSTKRV, or PSTK variable codes in COMPUSTAT. For shareholders' equity, SEQ or CEQ+PSTK or AT-LT variable codes can be used in order. TXDITC is the COMPUSTAT variable name for deferred taxes and investment tax credits.[8] Market value of the equity is computed with CRSP data.

Idiosyncratic Volatility (IVOL)[9]: We first run the Fama-French three-factor model with daily

[6]The number of days with missing opening price data constitutes only a small fraction of the total raw data set (3.62% of all daily observations all stocks combined). Setting the opening price equal to the previous day's closing price neutralizes the impact of these observations, whereas excluding such days could artificially induce jump detections by removing the transaction history associated with the closing price of the skipped day. This treatment is primarily to ensure that the jump detection algorithm runs without interruption. These days are not used in our predictive analysis, and the adopted approach will not impact the document results in the Tables. In technical terms, those zero-return days will become the additive identity in the Integrated Volatility summation.

[7]See Appendix B for all combinations of robustness check results.

[8]COMPUSTAT Acronyms: PSTK: Preferred/Preference Stock (Capital) - Total, PSTKL: Preferred Stock Liquidating Value, PSTKRV: Preferred Stock Redemption Value, TXDITC: Deferred Taxes and Investment Tax Credit, SEQ: Total Parent Stockholders' Equity, CEQ: Common/Ordinary Equity, AT: Total Assets, LT: Total Liabilities

[9]In their seminal paper, Ang et al. (2006) report that stocks with high idiosyncratic volatility oddly have lower

data frequency and save the regression residuals[10]. Monthly IVOL variables are created by calculating the standard deviations of these residuals over each separate period. Our predictability analyses require jump-isolated IVOL numbers as these discontinuous extreme returns already inflate IVOL and should be stripped from the classic figures before using them as regressors.[11] To do that, we create overnight-jump-days-isolated IVOL (IVOL-OJI) and use it in our main analysis [12]. As robustness checks, we also isolate all forms of discontinuities, either daily or overnight, and repeat our analyses with daily-jump-days-isolated idiosyncratic volatility (IVOL-DJI) and overnight-and-daily-jump-days-isolated idiosyncratic volatility (IVOL-ODJI) variables. Implications are still preserved with these alternative variables.

Illiquidity (AI): We use Amihud Illiquidity due to Amihud (2002), and it is the absolute daily return divided by daily trading volume in dollars. To calculate dollar trading volumes, we use the mid-point of the daily high-low range as the proxy multiplier. We control for illiquidity since it has been documented that expected excess stock returns embed some level of illiquidity premium. Following Jiang and Zhu (2017), we modify NASDAQ volume figures by multiplying them by 0.7.[13] This is to make trading volumes comparable across the stock exchanges since NYSE and AMEX are mostly centralized auction markets where customer orders directly interact with each other although NASDAQ is less-centralized with fragmented dealer market formation and volume counting procedure compelled by Securities and Exchange Commission (SEC) inflates the figures in this Exchange.

Momentum (MOM): It is the buy-and-hold return over an 11-month horizon backward with the preceding month skipped. Following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), we split it into two in the following manner during our analysis: $(t-1, t-5), (t-6, t-11)$.

Leverage (LEV): The Leverage variable is constructed by taking the natural logarithm of the

subsequent returns, and this empirical finding has been named as "idiosyncratic volatility puzzle". See Hou and Loh (2016) for a comprehensive recent discussion of present explanations in the literature and the extent to which this puzzle had been solved.

[10]The regressor factors are taken from Kenneth French's website.

[11]Without this isolation, the correlation between CJR_t^- and $IVOL$ moves from -0.50 to -0.67 and the correlation between CJR_t^+ and $IVOL$ soars from 0.45 to 0.71.

[12]We isolate the whole overnight jump day instead of the overnight component, as the follow-up return patterns are linked to the overnight movements.

[13]This adjustment factor is originally based on an SEC report on "order executions across equity market structures". See footnote 16 in the following link to that report https://www.sec.gov/pdf/ordrxmkt.pdf

ratio of total assets' book value on the fiscal year ending month in the preceding calendar year to market equity figures at the end of December in again the previous calendar year.

Previous Day Return (PREV): This variable is constructed to control for the information flow in the day preceding the overnight jumps. It is the preceding-day return in the isolated approach and the cumulated preceding-day return in the cumulative approach.

3.2 Methodology

3.2.1 Jump Identification

We model stock prices with a semi-martingale process embodying both diffusive continuous movements and jump components. Let X_t stand for the price process of a stock in a probability space with available information set $\mathcal{F}_t$ to all parties. For a unit period of [0, T] $(T \geq 0)$, it is a convention to specify the Ito semi-martingale process with price discontinuities as in the following jump-diffusion model:

$$X_t = X_0 + \int_0^t \mu_s d_s + \int_0^t \sigma_s dB_s + \sum_{k=1}^{N_j^t} J_i \quad ; \quad \forall t \in [0, T] \tag{1}$$

where the first three terms $(X_0 + \int_0^t \mu_s d_s + \int_0^t \sigma_s dB_s)$ constitute continuous stochastic price path with initial price (X_0), drift term (μ), diffusive volatility (σ) and standard Brownian motion (B). The last summation term injects the random price jumps into the model with counting process N_j and jump sizes $J = J_k$ for $k = 1, 2, ..., N_j^t$.

With equally spaced observations at times $t_0 < t_1 ... < t_{n-1} < t_n$ over the period $[0,\ T]$, one can calculate M distinct returns. Let $r_{m_i} = Y_{t_{i+\xi}} - Y_{t_i}$ be the return for an interval in which ξ determines the length of return intervals $\forall m \in [1,\ M]$ and $\forall \xi \in [0,\ T]$. Asymptotically, as ξ gets narrower, realized variance converges to quadratic variation. Furthermore, integrated volatility is detached from total quadratic variation via the realized bi-power variation due to Barndorff-Nielsen and Shephard (2004). It is also customary to link bi-power variation to realized variance to disentangle the jump variation. Specifically,

$$RV_T = \sum_{i=1}^{M} |r_{m_i}|^2 \quad \text{and} \quad \lim_{\xi \to 0} RV = QV = \int_0^t \sigma_s^2 ds \ + \sum_{i=0 \leq s \leq t} \Delta Y_s^2 \tag{2}$$

$$BV = \frac{\pi}{2}\frac{M}{M-1}\sum_{i=2}^{M}|r_{m_i}|\,|r_{m_{i-1}}| \quad \text{and} \quad \lim_{\xi\to 0} BV = IntV = \int_0^t \sigma_s^2 ds \tag{3}$$

where ΔY stands for instant log-price changes due to jumps and RV, QV, BV, and $IntV$ are respectively the realized variance, quadratic variation, bi-power variation, and integrated volatility. The terms $\frac{\pi}{2}\frac{M}{M-1}$ in bi-power variation act as a standardization factor (see Barndorff-Nielsen and Shephard (2003) for further discussion and Huang and Tauchen (2005) for extensions). Herewith, the jump variation component and its relative contribution to quadratic variation are straightforward in the following forms:

$$JV = RV - BV \quad \text{and} \quad \lim_{\xi\to 0} JV = \sum_{k=1}^{N_j^t} J_i \tag{4}$$

in which JV is the variation due to jumps.

To isolate integrated volatility from total quadratic variation in return series and determine the contribution of jumps to total variation, we use a non-parametric method developed by Lee and Mykland (2008). Dumitru and Urga (2012) compare alternative non-parametric jump tests and find that the techniques proposed by Andersen et al. (2007) and Lee and Mykland (2008) deliver the best performance. Let $\mathcal{L}_i$ be the test statistic for jump identification in Lee and Mykland (2008). In essence, it dissipates the concern for classifying a large return as a jump when it is essentially due to higher volatility during the period in question (and vice versa). Hence, $\mathcal{L}_i$ is formed as a standardized return metric in which the standardization is achieved by dividing each return by the square root of the accompanying integrated volatility.

$$\mathcal{L}_i = \frac{r_{m_i}}{\sqrt{IntV_{LM}}} \quad \text{with} \quad IntV_{LM} = \frac{\pi}{2}\frac{1}{M-2}\sum_{j=i-M+1}^{i-1}\left|r_{m_j}\right|\left|r_{m_{j-1}}\right| \tag{5}$$

where $IntV_{LM}$ stands for integrated volatility in Lee and Mykland (2008). Authors show that when there is no jump, the asymptotic distribution of $\mathcal{L}_i$ is a standard normal, whereas the presence of jumps leads to elevated test statistics. They offer the below metric to decide whether to reject the no-jump hypothesis or not. Variation in returns is due to jumps if,

$$\frac{max_{i\in\,\bar{A}_n}|\mathcal{L}_i| - C_n}{S_n} > \delta \tag{6}$$

where C_n and S_n are in the following mathematical notation with n being the number of obser-

vations and $c = \sqrt{2/\pi}$. The critical value is $\delta = -ln[-ln(1-\ \alpha)]$ in which α is the significance level. The window size K at the jump detection time is taken 16 as recommended in Lee and Mykland (2008) for daily datasets.

$$C_n = \frac{[2ln(n)]^{1/2}}{c} - \frac{ln4\pi + ln[ln(n)]}{2c[2ln(n)]^{1/2}} \quad \text{and} \quad S_n = \frac{1}{c[2ln(n)]^{1/2}} \tag{7}$$

Jump detection algorithm uses the past information within the detection window -the local neighborhood going backward- during the identification process and it does not suffer from any look-ahead bias. In other words, the "local neighborhood" corresponds to the window we are using. Hence, the very essence of jump detection is the past information within that window. The algorithm works through a rolling window setup by moving the window one step forward once a new observation is brought in. It is this new observation that is checked for the presence of a jump. In other words, a price movement is flagged as a jump by looking at its size and the preceding price movements within the window, the length of which is determined by the data frequency, and we set the window size K as 16 for daily data as recommended by Lee and Mykland (2008). The authors also effectively test the choice of window length and report that extending the window length does not yield any material improvement, but only adds to the computational burden.

Early literature and some recent work (e.g. Longin and Solnik (2001), Poon et al. (2004), Brogaard et al. (2018)) use percentile-based thresholds to classify extreme returns for negative and positive states. In general, the set of identified jumps will substantially overlap with the large price movements that exceed those percentile thresholds. Brogaard et al. (2018) examine HFT trading behavior during market jolts and identify extreme price movements as the returns corresponding to 99.9 percentile of return distribution. Their results remain consistent when they alternatively employ Lee and Mykland (2012) methodology. On the other hand, our focus is on distinguishing discontinuous returns from continuous returns rather than merely isolating extreme observations with certain cutoffs. The recent literature essentially utilizes modern techniques to dynamically separate discontinuous sharp movements from smooth continuous returns as the models that incorporate the jump classification have different financial implications and mathematical properties (Aït-Sahalia and Jacod (2009)). In technical terms, these jump detection techniques identify if a sharp price movement is a large realization of Brownian motion or a movement indicating a break in the price path that shall be labeled as jump. A purely

percentile-based approach may therefore miss discontinuous returns that fall below the chosen cutoff, while simultaneously misclassifying some large but continuous price movements as jumps, even though they arise from the diffusion component.

3.2.2 Data Construction

After we detect the jumps separately for overnight, intraday, and daily periods, we create different return time series for each interval. Intraday returns are simply calculated with closing and opening prices in the CRSP database. Since the CRSP daily return series are adjusted for distributions, we deduce overnight returns from daily and intraday returns instead of adjusting opening prices for distributions and generating a close-to-open return time series. Specifically;

$$r_i^{ovn} = r_i - r_i^{int} \tag{8}$$

$$COR = exp(\sum_{d=1} r_{i,d}^{ovn}) - 1 \quad \text{and} \quad CIR = exp(\sum_{d=1} r_{i,d}^{int}) - 1 \quad \text{and} \quad CDR = exp(\sum_{d=1} r_{i,d}) - 1 \tag{9}$$

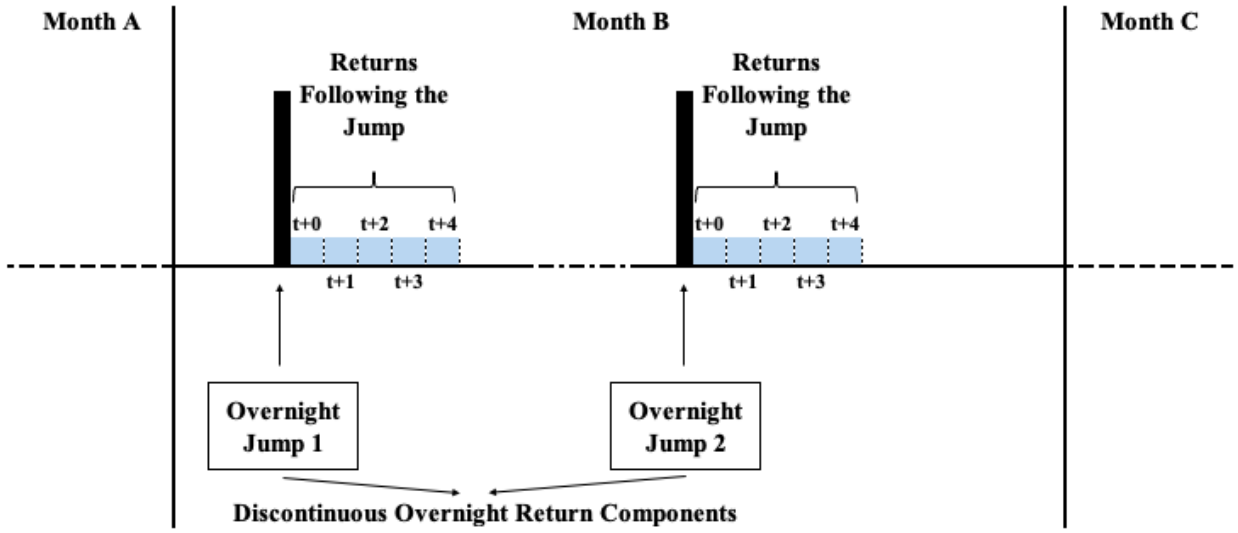
where $r_{i,d}^{ovn}$, $r_{i,d}^{int}$ and $r_{i,d}$ are respectively the overnight, intraday, and daily log returns of stock i on day d and COR, CIR and CDR are cumulative overnight returns, cumulative intraday returns and cumulative daily returns for a given period. To extract and cumulate the discontinuous overnight return components in a month just like Jiang and Yao (2013) do in yearly return setup for daily returns, we construct $CJR = exp(\sum_{k=1}^{N_j} r_{i,k}^{ovn}) - 1$ in which N_j stands for the counting process for overnight jumps in that month, $r_{i,k}^{ovn}$ corresponds to the overnight jump return at each jump incidence and CJR abbreviates monthly cumulative jump return.[14]

In addition to the isolated baseline approach where the cross-sectional regressions are constructed with individual jumps, we also construct cumulative jump, and follow-up returns and document the associated return predictability results. Within non-jump days, we focus on adjacent returns immediately after these overnight jumps and construct our dependent variables by cumulating these return numbers associated with each jump. Figure 1 illustrates a specific month with two randomly arriving overnight jumps and corresponding follow-up returns. In

[14] Jiang and Yao (2013) disentangle continuous and discontinuous return components for each year. They first calculate cumulative jump returns and subtract them from the total cumulative return to get the continuous return component within that year.

the diagram, black episodes represent overnight discontinuities and shaded days are the subsequent returns. For this example, we calculate CJR as formulated above and $CDR_{postjump} = exp(\sum_{d=1,k=1}^{d=5,k=2} r_{i,d,k}) - 1$ for stock i where d stands for the relevant day, k stands for the jump incidence, $r_{i,1,1} = r^{int}_{i,1,1}$ and $r_{i,1,2} = r^{int}_{i,1,2}$. For each stock, we construct separate monthly time series for both negative and positive cumulative overnight jumps.

Figure 1: Discontinuous Overnight Return Components and Data Construction



Notes: This diagram demonstrates a typical month where there are two sporadic overnight jumps. $t+0$ returns are respectively the $r^{int}_{i,1,1}$ and $r^{int}_{i,1,2}$, meaning that they are the first intraday returns following each overnight jump. The remaining $t+1$, $t+2$, $t+3$ and $t+4$ are the daily returns following the overnight jump day.

4 Empirical Findings

4.1 Descriptive Results

In this subsection, we analyze how stock returns evolve after overnight price jumps. Basic descriptive statistics for jumps are reported in Table 1. The exact number of nights is 24,305,513, implying that overnight jump days constitute only 1.57% of the total dataframe with 380,787 observations. In our robustness check, where the minimum trading history is two years and we exclude jumps if the CRSP sets the closing prices to the bid-ask average of the available best prices, the number of jumps becomes 320,944, constituting roughly the 1.31% of the analyzed number of days. These descriptive statistics are just a reaffirmation of the fact that jumps are rare events. For example, a stock with 20 years of trading history may have 30 overnight jumps scattered over the entire 240 months, meaning that there won't be any jump incidence on at least

210 months. A researcher cannot derive generalizable findings with this limited sample, and we discuss the importance of sample size in rare event studies. We aim to include as many stocks as possible to ensure a sufficiently large number of jumps each month, thereby generating robust findings. For these 9283 stocks at 342 months, the number of overnight jump observations is around 700 per month, even in the robustness dataset. With the collection of these 380,787 jump incidents in our main analyses, we are actually basing our research on a very large "jump dataset". Given the event-based nature of our study, the range of the sample and the large number of identified jumps allow us to draw systematic inferences about jump-related return dynamics rather than relying on isolated episodes.

Table 1: Descriptive Statistics for Isolated Jumps

Panel A: Overnight Return Jumps

	Numbers	Mean	Median	Int. Ret.<0	Int. Ret.>0	Int. Ret.=0	Opposite Sign Int. Return	Same Sign Int. Retun
Total	380,747	0.4%	-0.9%					
Negative	201,191	-6.9%	-4.8%	52,286	95,077	53,828	47%	26%
Positive	179,556	8.6%	5.2%	91,188	55,221	33,147	51%	31%

Panel B: Intraday Return Jumps

	Numbers	Mean	Median	Ovn. Ret.<0	Ovn. Ret.>0	Ovn. Ret.=0	Opposite Sign Ovn. Return	Same Sign Ovn. Return
Total	137,094	2.9%	2.6%					
Negative	62,998	-11.1%	-8.7%	26,878	34,905	1,215	55%	43%
Positive	74,096	14.9%	9.5%	41,338	31,109	1,649	56%	42%

Panel C: Daily Return Jumps

	Numbers	Mean	Median	Int. Ret.<0	Int. Ret.>0	Int. Ret.=0	Opposite Sign Int. Return	Same Sign Int. Return
Total	123,580	3.6%	3.7%					
Negative	58,108	-14.9%	-11.9%	50,948	4,168	2,992	7%	88%
Positive	65,472	20.0%	13.2%	3,967	59,262	2,243	6%	91%

Notes: Jump Statistics are tabulated for stocks with more than 3 years of trading history in the analysis period. Each panel shows the jump statistics for each corresponding component of the day. *Int. Ret.* in the table stands for intraday returns after overnight jumps. Our analyses cover 342 months over the period July 1993 - December 2021.

We report descriptive statistics and correlation numbers of our variables in Table 2 and Table 3.[15] Figure 3 demonstrates three empirical facts to us. First, it visually shows the overreaction during overnight negative and positive jumps by plotting the mean of follow-up cumulative daily returns in the left panel. This trend can also be visually inspected via Figure 2 as well. Second,

[15]Negative jumps are not the perfect equivalent of tail risk because of two reasons: First, even small price fluctuations outside tails may be marked as a jump during very calm periods. Second, tails also include high levels of negative returns that come in the form of volatility, whereas jumps correspond to specific returns triggered by information shocks, liquidity shocks, and other imbalances related to trading. That said, jump magnitudes are generally considerable, and negative jumps can be regarded as rarely and sporadically arriving proxies of tail risk. We support this argument by the high correlation of tail risk variables and $IVOL$ in Atilgan et al. (2020). Our CJR variable has a lower correlation with $IVOL$ for both positive and negative jumps as we isolate overnight jump days and use $IVOL_OJI$ in regression analyses.

Table 2: Negative Overnight Jump Stocks: Cumulative Approach Descriptive Statistics and Correlation Matrix

PANEL A: Descriptive Statistics - Negative Overnight Jumps

	CJR_t^-	$PREV_t$	$IVOL_OJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t
Mean	-0.09	0.02	0.03	19.39	-0.65	1.06	0.06	0.07	20.52
Median	-0.06	0.00	0.02	19.27	-0.56	0.78	0.02	0.03	0.49
St.Dev.	0.08	0.09	0.03	2.02	1.03	0.87	0.35	0.42	141.75
Min	-0.67	-0.26	0.00	14.74	-5.66	0.00	-0.77	-0.78	0.00
Max	-0.004	0.97	0.30	25.51	3.67	5.23	2.83	3.75	2381.98
Skewness	-3.02	4.21	4.30	0.30	-0.53	1.33	2.34	2.78	12.07
Kurtosis	14.81	57.51	39.09	-0.18	4.45	2.35	20.14	25.89	188.35
25th_Per.	-0.11	-0.01	0.02	17.91	-1.17	0.41	-0.12	-0.13	0.02
75th_Per.	-0.04	0.03	0.04	20.76	-0.04	1.54	0.18	0.20	4.14

PANEL B: Correlations

	CJR_t^-	$PREV_t$	$IVOL_OJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t
CJR_t^-	1.00								
$PREV_t$	-0.14	1.00							
$IVOL_OJI_t$	-0.50	0.27	1.00						
$SIZE$	0.25	-0.17	-0.42	1.00					
BM	-0.01	0.08	0.10	-0.37	1.00				
LEV	0.08	0.02	-0.04	-0.07	-0.09	1.00			
$RET_{t-1,t-5}$	0.10	-0.05	-0.13	0.03	-0.04	0.02	1.00		
$RET_{t-6,t-11}$	0.06	-0.05	-0.10	0.09	-0.16	0.00	0.00	1.00	
AI_t	-0.17	0.11	0.25	-0.24	0.12	0.02	-0.07	-0.05	1.00

Notes: This table tabulates descriptive statistics and correlations between the variables in our monthly cross-sectional regressions. We calculate the figures for each month, construct a time series, and average them. CJR_t^+ and CJR_t^- are respectively the monthly cumulated positive and negative jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for detailed explanations of variables.

the intraday portion of cumulative returns is more powerful after negative jumps as plotted in the right panel of Figure 3. A closer look into Figure 2 also reveals similar market behaviour: post-jump intraday returns wander mostly above zero after negative overnight jumps and below zero after positive overnight shocks, though the former is larger in scale. This aligns with the asymmetric intraday reaction in subplot 2b of Figure 3. Third, overnight jumps are preceded by a discernible average daily return in the opposite sign. We can see that the daily return is 1.1% on day $JD-1$ in the case of negative jumps and the trend is upward just like the post-jump period, and -0.6% in the case of positive jumps and the trend is downward just like the post-jump period. To control for this information from the previous day in isolated as well as cumulative

Table 3: Positive Overnight Jump Stocks: Cumulative Approach Descriptive Statistics and Correlation Matrix

PANEL A: Descriptive Statistics

	CJR_t^+	$PREV_t$	$IVOL_OJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t
Mean	0.11	-0.01	0.03	19.32	-0.65	1.07	0.05	0.07	16.74
Median	0.07	0.00	0.02	19.17	-0.55	0.78	0.01	0.02	0.38
St.Dev.	0.16	0.08	0.03	2.01	1.07	0.89	0.39	0.44	113.59
Min	0.004	-0.41	0.00	14.66	-5.97	0.00	-0.80	-0.80	0.00
Max	1.79	0.75	0.28	25.53	3.95	5.56	3.24	3.99	1881.77
Skew.	5.55	2.02	3.83	0.31	-0.53	1.42	2.56	2.85	12.26
Kurto.	53.01	44.02	29.77	-0.16	4.79	2.95	23.01	25.26	189.18
25th_Per.	0.04	-0.03	0.02	17.85	-1.18	0.41	-0.14	-0.15	0.02
75th_Per.	0.12	0.01	0.04	20.68	-0.04	1.55	0.18	0.21	3.19

PANEL B: Correlations

	CJR_t^+	$PREV_t$	$IVOL_OJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t
CJR_t^+	1.00								
$PREV_t$	-0.01	1.00							
$IVOL_OJI_t$	0.45	0.00	1.00						
$SIZE$	-0.28	0.06	-0.44	1.00					
BM	0.04	-0.03	0.08	-0.34	1.00				
LEV	-0.04	-0.01	-0.04	-0.06	-0.12	1.00			
$RET_{t-1,t-5}$	-0.11	0.03	-0.15	0.05	-0.05	0.00	1.00		
$RET_{t-6,t-11}$	-0.09	0.02	-0.12	0.10	-0.16	-0.01	0.02	1.00	
AI_t	0.14	-0.08	0.24	-0.23	0.11	0.02	-0.07	-0.05	1.00

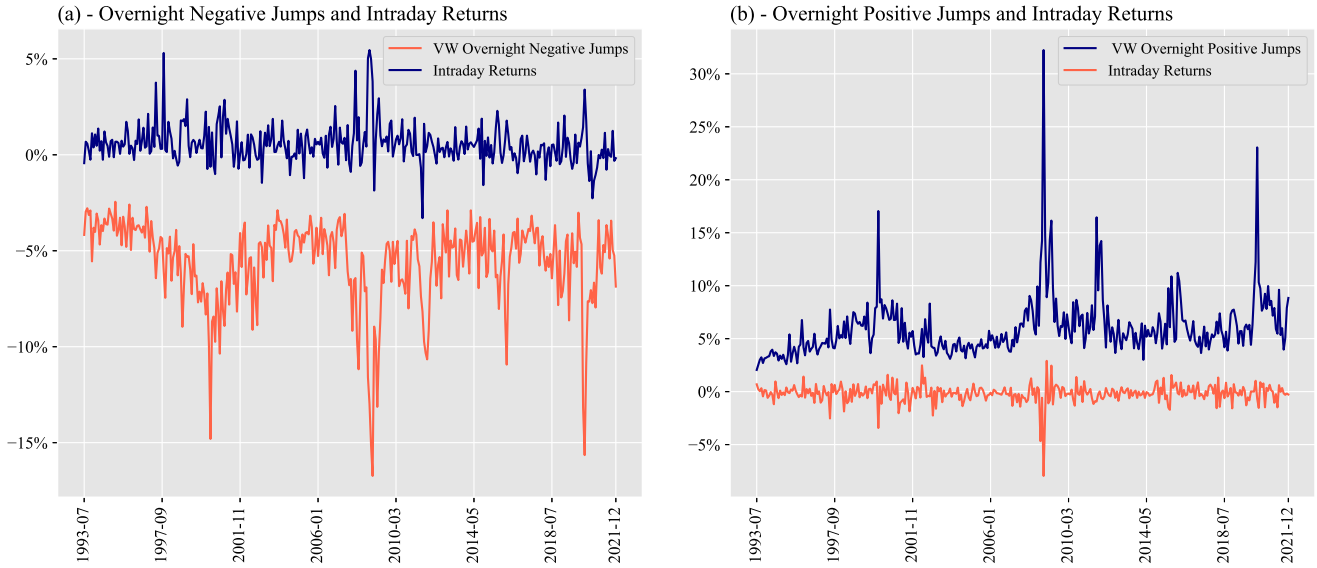
Notes: This table tabulates descriptive statistics and correlations between the variables in our monthly cross-sectional regressions. We calculate the figures for each month, construct a time series, and average them. CJR_t^+ and CJR_t^- are respectively the monthly cumulated positive and negative jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for detailed explanations of variables.

approach, we introduce $PREV_{t,d=-1}$ and populated results in Tables 4, 5, 6, 7, and 8 show that the previous day's information has varying levels of explanatory power for certain horizons in different subperiods and market states.

4.2 Return Predictability: Isolated Approach and Cumulative Approach

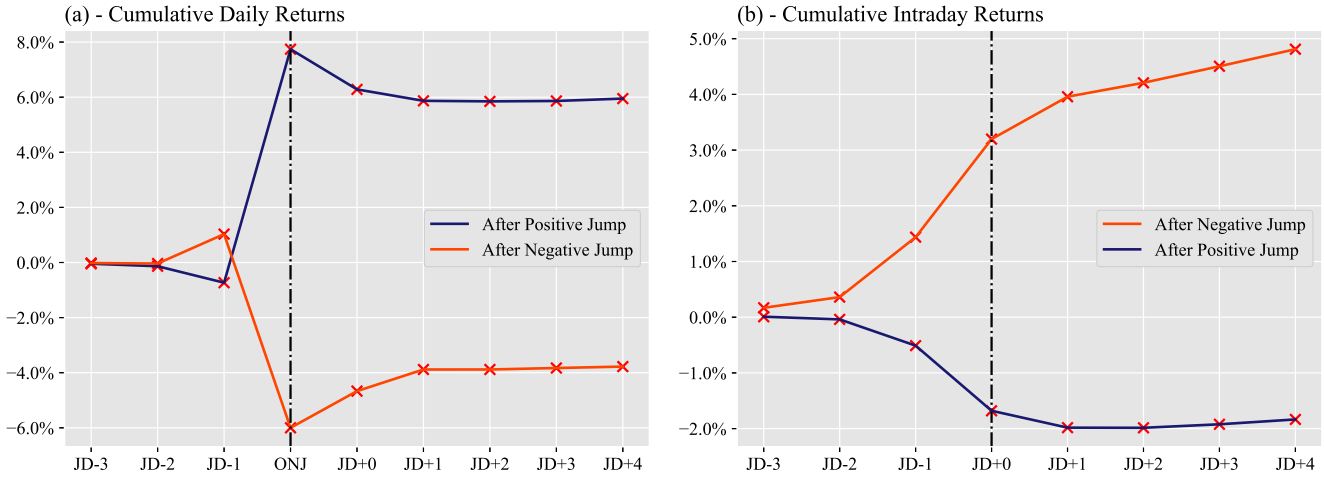
We start the baseline results by reporting the isolated return predictability approach without aggregating the jumps and their corresponding follow-up returns, and move to the cumulative approach as a complementary analysis. In the case of multiple jumps in a month in the isolated approach, each jump and its follow-up returns are treated as a different event set, and

Figure 2: Overnight Jumps and Intraday Returns



Notes: This graph shows value-weighted overnight jump returns and their value-weighted counterpart in the following intraday section.

Figure 3: Overreaction Path



Notes: This graph shows overreaction to overnight information shocks. The left panel plots the mean of the cumulative daily returns around negative and positive overnight price jumps, whereas the right panel unravels the mean of cumulative intraday returns.

these sets enter the cross-sectional regressions separately with month-specific characters. On the other hand, the cumulative approach aggregates all jumps within a calendar month into a single monthly observation (CJR). Although aggregating jump realizations within a month as schemat-

ically shown in Figure 1 may prompt questions about potential look-ahead bias, our empirical findings indicate that such bias does not materialize in the data due to the rarity of multiple jumps within a month and the tendency of repeated jump events for the same stock to exhibit similar jump and post-jump return dynamics. Specifically, when each jump is treated as a separate event, and follow-up returns are regressed on jump-day information and month-specific characteristics, the estimated coefficients are highly significant and comparable in magnitude, also in the cumulative approach. The alignment of results across these two approaches suggests that the cumulative specification does not introduce a timing distortion.

For the isolated approach, we estimate:

$$\begin{aligned} CDR_{i,t,d+1} = \alpha + \beta_1 JR_{i,t,d=0} + \beta_2 PREV_{i,t,d=-1} + \beta_3 IVOL_OJI_{i,t-1} + \beta_4 SIZE_i \\ + \beta_5 BM_i + \beta_6 LEV_i + \beta_7 RET_{i,t-1,t-5} + \beta_8 RET_{i,t-6,t-11} + \beta_9 AI_{i,t-1} + \varepsilon_{i,t} \end{aligned} \tag{10}$$

where $0 \leq d \leq 4$, $CDR_{i,t,d+1}$ is the post-jump daily returns for firm i realized until day d, and $JR_{i,t,d=0}$ is the overnight jump returns preceding the daily returns of our interest. For instance, $JR_{i,t,d=0}$ is the overnight jump returns of firm i for random jump days of a given month t, and $CDR_{i,t,3}$ corresponds to cumulative 3-day returns following these jumps within this month. The overnight period is regarded as $d = 0$ and the following intraday return is treated as $d = 1$. For the definition of other regressor variables, see Section 3.1.

We perform separate regressions for negative and positive jump incidences. For Table 4, we run the Equation 10 and the results show that negative and positive overnight jumps significantly predict the follow-up returns in both market states. Each month, we specifically run cross-sectional regressions of the stocks with overnight jump incidences. The dataset will overtly be an unbalanced panel version with irregular jump arrivals in the time dimension. The basket of stocks will also change each month depending on the availability of overnight information shocks. In the regressions, it is highly likely that the error terms will not be independent of each other since a positive/negative information shock may connote similar cases for the neighboring companies. So, we correct for the correlation of the errors as the standard errors will otherwise be misleading. To get the robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD), store coefficient estimates for 342 months, and calculate the mean and report Newey-West t-statistics with 12 lags.

The case that jumps following each other on adjacent days has low probability due to the

Table 4: Daily Return Predictability After Overnight Jumps - Isolated Approach

Negative Overnight Jumps

	Intercept	JR^-	$PREV$	$IVOL_OJI_{t-1}$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_{t-1}	$Adj.R^2$
$1D_t$	0.07	-0.42	-0.08	-0.14	-0.43	0.37	0.18	-0.17	0.75	0.06	0.11
	(2.96)	(-3.89)	(-4.09)	(-1.09)	(-2.86)	(3.25)	(2.76)	(-0.75)	(1.03)	(1.08)	
$2D_t$	0.09	-0.44	-0.11	-0.11	-0.54	0.47	0.29	-0.22	0.78	0.00	0.14
	(4.39)	(-4.42)	(-6.11)	(-0.9)	(-4.03)	(3.9)	(4.51)	(-0.92)	(1.11)	(0.41)	
$3D_t$	0.09	-0.45	-0.13	-0.13	-0.54	0.52	0.29	-0.43	0.77	0.00	0.13
	(4.39)	(-4.51)	(-6.82)	(-1.08)	(-4.09)	(4.06)	(4.08)	(-1.67)	(1.14)	(-0.17)	
$4D_t$	0.09	-0.43	-0.13	-0.12	-0.53	0.48	0.27	-0.35	0.68	-0.03	0.12
	(4.63)	(-4.6)	(-6.86)	(-1.04)	(-4.26)	(4.24)	(3.31)	(-1.45)	(1.08)	(-0.7)	
$5D_t$	0.09	-0.43	-0.13	-0.08	-0.51	0.51	0.31	-0.42	0.75	-0.07	0.12
	(4.27)	(-4.61)	(-6.47)	(-0.69)	(-4)	(4.28)	(3.29)	(-1.87)	(1.23)	(-0.82)	

Positive Overnight Jumps

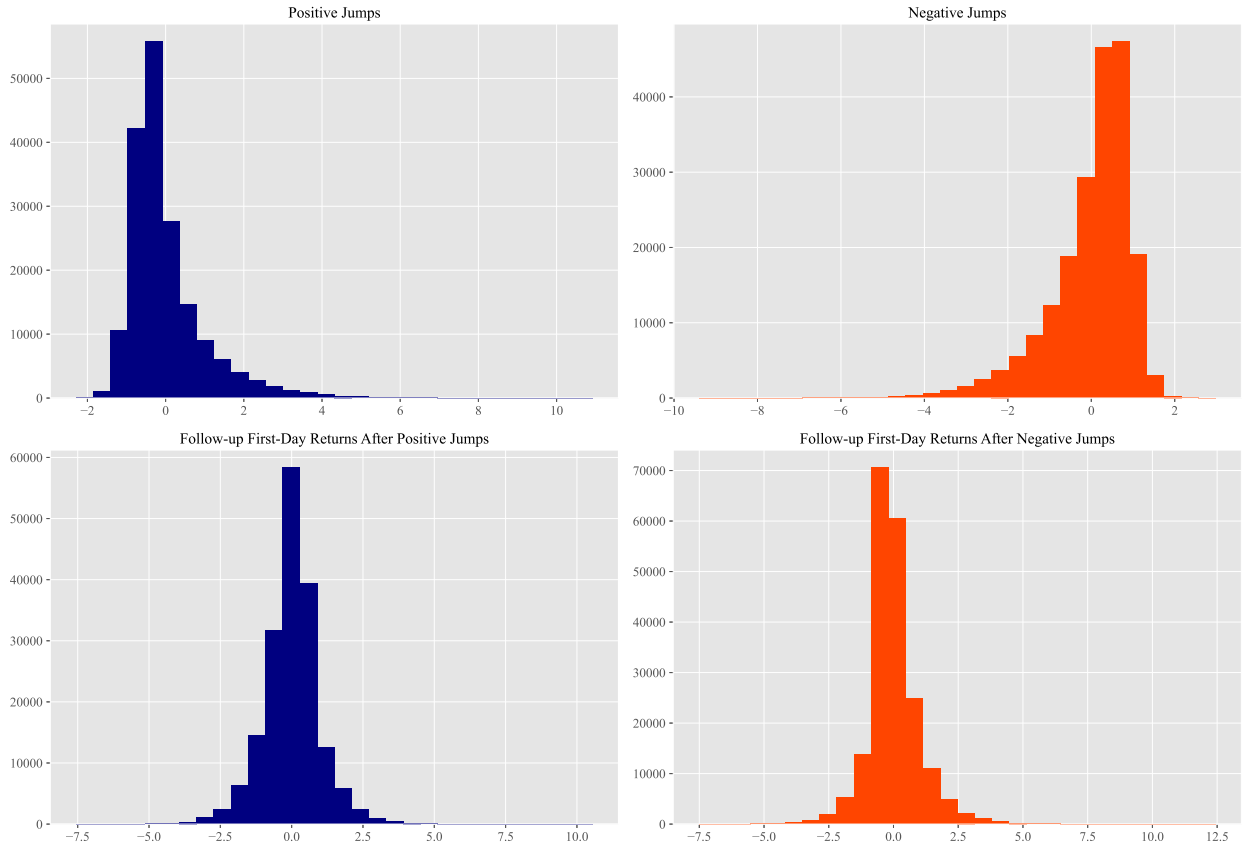
	Intercept	JR^+	$PREV$	$IVOL_OJI_{t-1}$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_{t-1}	$Adj.R^2$
$1D_t$	-0.06	-0.09	-0.03	-0.16	0.28	-0.01	-0.11	0.51	0.29	0.00	0.10
	(-7.34)	(-11.55)	(-3.34)	(-5.35)	(7.86)	(-0.36)	(-2.88)	(3.48)	(3.34)	(0.92)	
$2D_t$	-0.08	-0.11	-0.08	-0.19	0.39	-0.10	-0.21	0.57	0.23	0.00	0.10
	(-8.24)	(-11.6)	(-9.17)	(-5.31)	(9.13)	(-1.88)	(-5.07)	(3.46)	(1.92)	(-0.61)	
$3D_t$	-0.08	-0.12	-0.09	-0.22	0.41	-0.12	-0.18	0.58	0.23	0.00	0.10
	(-7.59)	(-12.87)	(-9.68)	(-6.29)	(8.57)	(-2.01)	(-3.88)	(3.7)	(1.72)	(-0.66)	
$4D_t$	-0.08	-0.12	-0.10	-0.25	0.43	-0.10	-0.19	0.63	0.16	0.00	0.09
	(-6.79)	(-11.93)	(-9.45)	(-6.25)	(7.73)	(-1.64)	(-3.3)	(4.16)	(0.95)	(-0.39)	
$5D_t$	-0.08	-0.13	-0.10	-0.26	0.43	-0.07	-0.21	0.64	0.14	0.00	0.09
	(-7.02)	(-11.59)	(-9.4)	(-6.09)	(8.19)	(-1.05)	(-3.67)	(4.26)	(0.76)	(0.55)	

Notes: This Table reports return predictability results when jumps are treated in isolation instead of calculating cumulative jump returns. Stocks having multiple jumps in a month appear with multiple observations in monthly regressions as opposed to once with their cumulative figures. Tabulated results are for the first five days after overnight jumps. The dependent variable $1D_t$ is the intraday return just after the overnight jump. We run monthly cross-sectional regressions and save the time series of coefficient estimates. To correct the correlation of the errors and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. $JR^{-,+}$ variables in the first and second part of the Table respectively stand for negative and positive jump returns, $PREV$ is the previous day return before jumps, $IVOL_OJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_{t-1} is the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_{t-1} are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables. Estimated model: $CDR_{i,t,d+1} = \alpha + \beta_1 JR_{i,t,d=0} + \beta_2 PREV_{i,t,d=-1} + \beta_3 IVOL_OJI_{i,t-1} + \beta_4 SIZE_i + \beta_5 BM_i + \beta_6 LEV_i + \beta_7 RET_{i,t-1,t-5} + \beta_8 RET_{i,t-6,t-11} + \beta_9 AI_{i,t-1} + \varepsilon_{i,t}$

obvious nature of jumps, but it does not have to be the case that there won't be any jump observation on successive days. In other words, jumps can be followed by other jumps, but this is rare in the sense that it would otherwise imply information shocks' arrival in chunks on different days. Above all, it becomes technically harder for returns to be marked as a jump if the recent preceding returns are also extreme. That is because of the fact that integrated volatility will become higher, and it will drag down the test statistic, and that will make the chances of the next return being marked as a jump low. In contrast, the chances of these returns being treated as *volatility* will be higher. As an extreme example, we can imagine a window of 15 days with an up and down return sequence of +4%, -4%, +4%,-4%+...+4%. The next -4% on the 16th day

will not be marked as a jump amid this fluctuating environment due to the inflated integrated volatility, and this return will be credited to volatility not to jump. We also note that, in case of multiple jumps within a month, the cumulative approach -treated as a complementary exercise to the baseline analysis- yields predictability results similar to those obtained under the isolated approach; the predictive evidence for jump returns in Table 4 for isolated approach is consistent with those reported in Table 6 for cumulative approach. The largest model specification for the cumulative approach in Table 5 is equivalent to the first-day specification in Table 6, and the other parsimonious model specifications in Table 5 provide similar predictive evidence. Certain fundamental reasons exist for the results to be sustained under this approach.

Figure 4: Standardized Jumps and Standardized Follow-up Returns

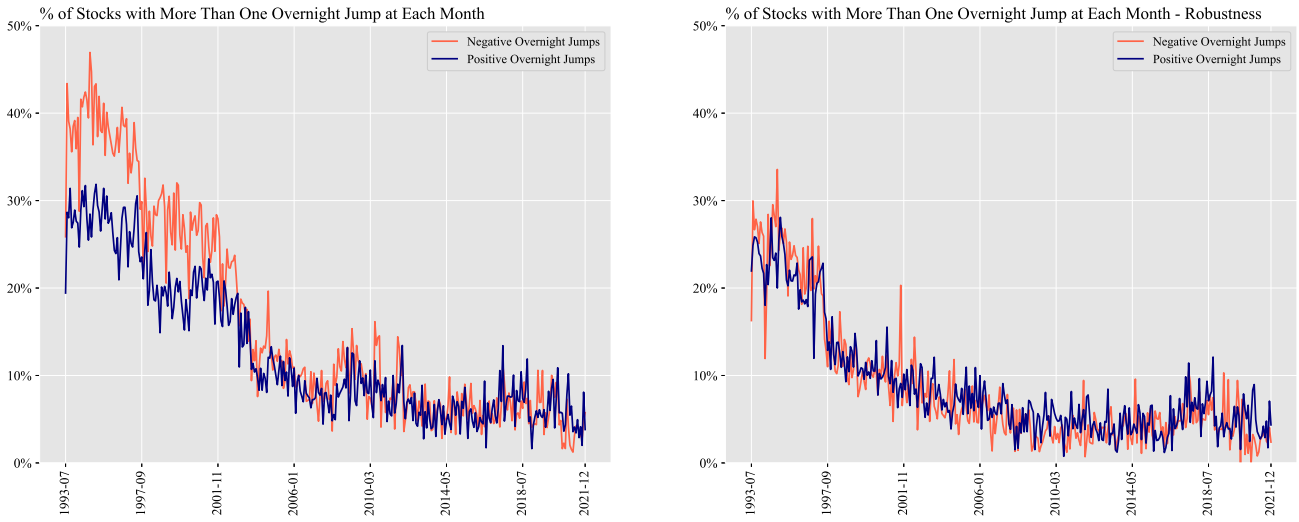


Notes: This Figure shows standardized positive and negative jumps as well as the follow-up returns. For each stock, we standardize positive and negative jumps separately, along with the first-day returns following the jumps. After repeating this for all stocks, we plot the whole standardized dataset.

Firstly, for stocks having multiple jumps in a month, it is highly likely that the jumps will resemble each other, though not perfectly be the same. That can also be thought of for the follow-up return patterns. To put it differently, stocks of certain characteristics will exhibit certain behav-

iors. For instance, Table 12 and Table 13 in Section 4.5 verify that stocks with low IVOL have low jump and reversal magnitudes on average, whereas high IVOL stocks are associated with higher jump and reversal figures. Hence, accumulating two jumps and accumulating follow-up returns will be akin to cumulating similar jumps and similar follow-up return components. To support this argument, we construct standardized positive and negative jump series for each stock, combine them for the whole dataset, draw histograms, and calculate excess kurtosis to see if the values are clustered around the mean with a peaked outlook. In Figure 4, standardized positive and negative jumps as well as the standardized follow-up returns on the first day have clear peakedness: standardized positive and negative jumps have respectively 6.07 and 3.50 excess kurtosis, and these figures are 3.10 and 5.46 for the follow-up returns after positive and negative jumps. In the same vein, 82% of positive jump observations are one σ away from the mean, and this is 78% for negative jumps. For the follow-up returns, figures are 77% and 79% for positive and negative jump incidences, respectively.

Figure 5: Percent of Stocks With More Than One Overnight Jump at Each month



Notes: This Figure plots the portion of stocks with more than one overnight jump over the entire data period. Each month, we determine the number of stocks that have at least one jump incidence and find the number of stocks with more than one jump observation. Then, the ratio of stocks with multiple overnight jumps in each month is graphed.

Second, a stock having multiple jumps in a month becomes a rare occasion, especially after 2000. Only 9.1% of the stocks have multiple positive jumps in a month in the main analyses after 2000, and that is around 6.0% in the robustness data. Figures are 10.4% and 5.3% in negative jumps, respectively for the main analyses and the robustness check. When we consider the

whole data period, these figures are 12.7% and 8.8% in positive jumps respectively for the main analyses and the robustness check, and it is 15.9% and 8.4% for negative jumps. In other words, the cumulative approach is functioning for a tiny portion of stocks, especially after 2000. The main trend is depicted in Figure 5.

For the cumulative approach, we estimate:

$$\begin{aligned} CDR_{i,t,d+1} = \alpha + \beta_1 CJR_{i,t,d=0} + \beta_2 PREV_{i,t,d=-1} + \beta_3 IVOL_OJI_{i,t} + \beta_4 SIZE_i \\ + \beta_5 BM_i + \beta_6 LEV_i + \beta_7 RET_{i,t-1,t-5} + \beta_8 RET_{i,t-6,t-11} + \beta_9 AI_{i,t} + \varepsilon_{i,t} \end{aligned} \tag{11}$$

where $0 \leq d \leq 4$, $CDR_{i,t,d+1}$ is the monthly cumulated post-jump daily returns for firm i, and $CJR_{i,t,d=0}$ is the cumulative overnight jump returns preceding the daily returns of our interest. For instance, $CJR_{i,t,d=0}$ is the monthly cumulated overnight jump returns of firm i for random jump days of a given month t, and $CDR_{i,t,3}$ corresponds to cumulative 3-day returns following these jumps within this month. The overnight period is regarded as $d = 0$ and the following intraday return is treated as $d = 1$. For the definition of other regressor variables, see Section 3.1.

For Table 5, we run Eq. 11 starting from the most parsimonious version and expand it by adding our control variables one at a time. In the first place, we look at cumulative-jump regression results for the first day. Table 5 shows the significance of $CJR_t^{-,+}$ in all forms of regression outputs with a negative sign for both negative and positive overnight jumps at 1% significance levels. Moreover, coefficients for negative and positive jump incidences are solid around -0.47 and -0.10 across different model specifications, and they are very close to the coefficients of isolated jump returns. In the largest model set-up (column 9), t-statistic values are 3.91 and 11.15 in absolute terms. Consistent with the isolated approach, our findings show that cumulative jump returns have a distinctive and highly significant predictive power for the follow-up equity returns. Among other control variables, momentum factors $RET_{i,t-1,t-5}$ and $RET_{i,t-6,t-11}$ do not have statistically significant loadings and predictive power in explaining cumulated return variations after negative overnight jumps. LEV is also not significant for the first day after sharp price declines, whereas it gains significance for the other days, implying that it is not a priced risk factor for the first day following overnight negative jumps.

Our regression results in Table 6 show that only CJR and $SIZE$ have statistically significant predictive power for cumulative returns up to 5 days following both overnight negative and positive jumps. Previous-day return ($PREV$), idiosyncratic volatility calculated over non-

Table 5: Overnight Jumps and Return Predictability - Cumulative Approach

PANEL A: Negative Jumps									
Dep. Variable:($1D_t$)	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**
CJR_t^-	-0.43	-0.44	-0.46	-0.46	-0.46	-0.46	-0.46	-0.47	-0.47
	(-4.23)	(-4.15)	(-3.86)	(-3.85)	(-3.86)	(-3.88)	(-3.89)	(-3.87)	(-3.91)
$PREV_t$		-0.04	-0.03	-0.04	-0.04	-0.04	-0.04	-0.04	-0.04
		(-1.27)	(-1.11)	(-1.55)	(-1.61)	(-1.69)	(-1.7)	(-1.73)	(-1.79)
$IVOL_OJI_t$			-0.14	-0.35	-0.35	-0.34	-0.33	-0.33	-0.31
			(-1.28)	(-2.6)	(-2.52)	(-2.43)	(-2.21)	(-2.21)	(-2.03)
$SIZE$				-0.53	-0.44	-0.42	-0.41	-0.43	-0.43
				(-4.94)	(-4.88)	(-4.56)	(-4.17)	(-3.81)	(-3.65)
BM					0.42	0.46	0.45	0.42	0.43
					(2.91)	(3.19)	(3.16)	(3.04)	(3.14)
LEV						0.18	0.16	0.15	0.15
						(1.47)	(1.25)	(1.13)	(1.12)
$RET_{t-1,t-5}$							-0.45	-0.47	-0.45
							(-0.9)	(-0.9)	(-0.83)
$RET_{t-6,t-11}$								0.42	0.37
								(0.63)	(0.6)
AI_t									0.01
									(1.79)
Intercept	-0.02	-0.03	-0.02	0.09	0.07	0.07	0.07	0.07	0.07
	(-1.89)	(-1.88)	(-1.95)	(5.52)	(5.01)	(4.43)	(4)	(3.81)	(3.6)
$Adj.R^2$	0.08	0.10	0.11	0.12	0.12	0.13	0.13	0.13	0.14
PANEL B: Positive Jumps									
Dep. Variable: ($1D_t$)	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**
CJR_t^+	-0.11	-0.12	-0.11	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10
	(-11.08)	(-12.14)	(-13.06)	(-11.97)	(-11.98)	(-11.97)	(-11.91)	(-11.85)	(-11.15)
$PREV_t$		0.04	0.03	0.03	0.03	0.03	0.03	0.03	0.03
		(4.02)	(3.09)	(2.44)	(2.42)	(2.41)	(2.39)	(2.38)	(2.51)
$IVOL_OJI_t$			-0.13	0.01	0.01	0.01	0.02	0.03	0.00
			(-3.67)	(0.3)	(0.29)	(0.25)	(0.54)	(0.61)	(-0.02)
$SIZE$				0.38	0.38	0.38	0.38	0.38	0.40
				(8.19)	(8.1)	(8.07)	(7.98)	(8)	(8.15)
BM					-0.02	-0.04	-0.02	0.01	-0.01
					(-0.45)	(-1)	(-0.47)	(0.15)	(-0.17)
LEV						-0.12	-0.11	-0.11	-0.11
						(-2.95)	(-2.85)	(-2.7)	(-2.86)
$RET_{t-1,t-5}$							0.55	0.56	0.56
							(3.64)	(3.6)	(3.69)
$RET_{t-6,t-11}$								0.36	0.36
								(3.87)	(3.89)
AI_t									0.01
									(2.38)
Intercept	0.00	0.00	0.00	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08
	(-3.41)	(-3.21)	(0.19)	(-7.51)	(-7.49)	(-7.43)	(-7.44)	(-7.49)	(-7.51)
$Adj.R^2$	0.07	0.10	0.12	0.12	0.13	0.13	0.13	0.13	0.14

Notes: To disentangle the discontinuous overnight components of monthly returns, we calculate cumulative overnight positive and negative jump returns for each firm. On a monthly scale, we also calculate cumulative post-jump intraday returns for the first day, as shown in Section 3.2.2. The dependent variable is the cumulated $1D_t$ returns (the intraday returns just after the overnight jumps). Then, we run monthly cross-sectional regressions and save the time series of coefficient estimates. To correct the correlation of the errors and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). From columns (2) to (9), we add each firm-specific control variable one at a time. This table reports results only for the first day after jumps, and results of the other days are available upon request. Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. CJR_t^- and CJR_t^+ are respectively the monthly cumulated negative and positive jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_t are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables. Model (9): $CDR_{i,t,d+1} = \alpha + \beta_1 CJR_{i,t,d=0} + \beta_2 PREV_{i,t,d=-1} + \beta_3 IVOL_OJI_{i,t} + \beta_4 SIZE_i + \beta_5 BM_i + \beta_6 LEV_i + \beta_7 RET_{i,t-1,t-5} + \beta_8 RET_{i,t-6,t-11} + \beta_9 AI_{i,t} + \varepsilon_{i,t}$

Table 6: Daily Return Predictability After Overnight Jumps - Cumulative Approach

PANEL A: Negative Jumps										
Dependent Variable	$1D_t$		$2D_t$		$3D_t$		$4D_t$		$5D_t$	
	coef.	**t**	**coef.**	**t**	**coef.**	**t**	**coef.**	**t**	**coef.**	**t**
$Intercept$	0.07	(3.6)	0.08	(4.55)	0.08	(4.43)	0.08	(4.66)	0.07	(4.28)
CJR_t^-	-0.47	(-3.91)	-0.52	(-4.45)	-0.50	(-4.38)	-0.45	(-4.27)	-0.44	(-4.1)
$PREV_t$	-0.04	(-1.79)	-0.01	(-0.5)	-0.06	(-2.66)	-0.05	(-2.58)	-0.05	(-2.38)
$IVOL_OJI_t$	-0.31	(-2.03)	-0.26	(-1.77)	-0.20	(-1.3)	-0.07	(-0.53)	0.02	(0.17)
$SIZE$	-0.43	(-3.65)	-0.50	(-4.67)	-0.49	(-4.67)	-0.48	(-4.84)	-0.45	(-4.65)
BM	0.43	(3.14)	0.58	(4.08)	0.63	(4.18)	0.60	(4.81)	0.66	(4.41)
LEV	0.15	(1.12)	0.31	(2.7)	0.33	(2.76)	0.28	(2.11)	0.32	(2.14)
$RET_{t-1,t-5}$	-0.45	(-0.83)	-0.19	(-0.37)	-0.45	(-0.87)	-0.36	(-0.8)	-0.49	(-1.16)
$RET_{t-6,t-11}$	0.37	(0.6)	0.60	(0.95)	0.56	(0.94)	0.58	(1.03)	0.65	(1.14)
AI_t	0.01	(1.79)	0.02	(3.87)	0.04	(4.72)	0.02	(1.95)	0.02	(1.59)
$Adj.R^2$	0.14		0.21		0.19		0.17		0.16	
PANEL B: Positive Jumps										
Dependent Variable	$1D_t$		$2D_t$		$3D_t$		$4D_t$		$5D_t$	
	coef.	**t**	**coef.**	**t**	**coef.**	**t**	**coef.**	**t**	**coef.**	**t**
Intercept	-0.08	(-7.51)	-0.13	(-8.64)	-0.15	(-9.12)	-0.17	(-8.64)	-0.17	(-8.45)
CJR_t^+	-0.10	(-11.15)	-0.10	(-10.53)	-0.11	(-10.78)	-0.10	(-10.76)	-0.10	(-11.24)
$PREV_t$	0.03	(2.51)	0.05	(2.32)	0.03	(1.48)	0.03	(1.41)	0.04	(1.95)
$IVOL_OJI_t$	0.00	(-0.02)	0.07	(0.95)	0.21	(2.72)	0.31	(3.39)	0.34	(3.31)
$SIZE$	0.40	(8.15)	0.62	(9.26)	0.71	(9.86)	0.79	(9.25)	0.81	(9.25)
BM	-0.01	(-0.17)	-0.01	(-0.13)	0.02	(0.32)	0.07	(1.08)	0.09	(1.17)
LEV	-0.11	(-2.86)	-0.18	(-4.59)	-0.13	(-2.6)	-0.11	(-1.79)	-0.14	(-1.92)
$RET_{t-1,t-5}$	0.56	(3.69)	0.77	(3.62)	0.88	(3.62)	0.99	(3.87)	1.03	(4.44)
$RET_{t-6,t-11}$	0.36	(3.89)	0.44	(3.78)	0.53	(4.1)	0.50	(3.58)	0.48	(3)
AI_t	0.01	(2.38)	0.00	(0.86)	0.01	(0.95)	0.00	(0.53)	0.01	(1.1)
$Adj.R^2$	0.14		0.15		0.14		0.13		0.12	

Notes: To disentangle the discontinuous overnight components of monthly returns, we calculate cumulative overnight positive and negative jump returns for each firm. On a monthly scale, we also calculate cumulative post-jump returns up to five days as shown in Section 3.2.2, and construct our dependent variables. Then, we run monthly cross-sectional regressions and save the time series of coefficient estimates. To correct the correlation of the errors and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). The dependent variable $1D_t$ is the cumulated intraday returns just after the overnight jumps. For the other days, the dependent variable represents cumulative returns up to that day after the jump incidence. Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. CJR_t^- and CJR_t^+ are respectively the monthly cumulated negative and positive jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_t are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables. Estimated model: $CDR_{i,t,d+1} = \alpha + \beta_1 CJR_{i,t,d=0} + \beta_2 PREV_{i,t,d=-1} + \beta_3 IVOL_OJI_{i,t} + \beta_4 SIZE_i + \beta_5 BM_i + \beta_6 LEV_i + \beta_7 RET_{i,t-1,t-5} + \beta_8 RET_{i,t-6,t-11} + \beta_9 AI_{i,t} + \varepsilon_{i,t}$

jump days ($IVOL_OJI$), book-to-market ratios (BM), leverage (LEV), momentum ($RET_{t-1,t-5}$ and $RET_{t-6,t-11}$) and firm-specific illiquidity (AI) also have highly significant coefficient estimates to explain the return behaviors around these short-term overreaction episodes but they are not priced for certain horizons and under particular market conditions. These findings conform to the gist of Kamara et al. (2016) as some factors may not be priced for certain horizons whereas some gain predictive power in the same time span. On the other hand, CJR_t is statistically significant at 1% significance level after both negative and positive overnight jumps on all days.

In our robustness checks, we conduct the analysis using isolated jump returns and shorten the minimum required trading history to two years, and exclude the jumps if the CRSP closing price

before the overnight jump day is set to the bid-ask averages. We maintain the return predictability of jump returns after negative and positive overnight jumps with high statistical significance. If we use $IVOL_DJI_t$ or $IVOL_ODJI_t$ as our idiosyncratic risk variable, the coefficients for JR_t^- and JR_t^+ are highly significant for all days for the whole analysis period. Thus, JR_t may function as a factor of overnight information shocks in predicting future stock returns.

Regarding the mechanics of overnight price jumps, one potential question for the documented overreaction and follow-up reversals is whether they represent a risk premium for liquidity-providing market makers. Grossman and Miller (1988) model that market makers require compensation for the immediacy they facilitate when market arrival times of buyers and sellers do not match. This risk is expected to increase as the order entry times between intermediated buyers and sellers are too much apart and inventory holding spans multiple days. However, several aspects of our empirical results and inherent features of markets do not provide strong support for the view that the documented reversals primarily reflect compensation for market makers' liquidity provision risk. First, our regression results evince that jump returns consistently predict future returns with negative loadings across all model specifications, subperiods, and market conditions, all stocks combined. If liquidity provision were the primary driver in the reversal, we would expect that mechanism to be uniform across all these states and market makers to be present regardless of the extent of shocks. A recent study due to Bellia et al. (2025) show that designated market makers provide liquidity only when the negative shock is confined to single stocks and they instead demand liquidity if the shock is affecting multiple equities. Concerns about trading against market participants with private information dominate market makers' liquidity provision decisions, causing them to refrain from fulfilling their role when that fear crowds out expected gains from liquidity provision. In that regard, their decisions are also susceptible to behavioral forces. Bellia et al. (2025) also highlight that designated market makers and HFTs often overlap in their study. A related study due to Brogaard et al. (2018) similarly document that HFT liquidity provision is contingent on the type of market movements; when the selling pressure extends over multiple stocks, inventory concerns and risk management dimensions turn them into liquidity demanders. Second, market-making capacity is not unlimited and cannot totally determine the price formation, especially during shock periods, as highlighted in Grossman and Miller (1988) in reference to crash of October 1987. Third, our findings of the highest economic and statistical significance of reversals being generally concentrated in stocks

with high IVOL_OJI figures and low-SIZE small-cap stocks support the behavioral claim in the sense that these stocks are more pronounced with heuristic biases and overreaction to information shocks. That said, while the behavioral overreaction interpretation may be consistent with the patterns observed in the data and with the literature on attention and discrete information arrivals, the empirical design does not explicitly test liquidity provision channels and inventory risk concerns using fund flows and inventory data. A dedicated study with such datasets would provide more conclusive evidence. This paper therefore focuses on documenting the empirical regularity around overnight jumps and subsequent return reversals rather than on establishing a definitive causal mechanism.

One may also think that market makers may price the stock accordingly to reconcile the risk taken over the night, and prices adjust in the next opening session and that creates an overnight jump. Trading volumes are dominated by High Frequency Traders (HFT) in the US, and it is well-known that HFTs typically don't hold overnight inventory and unload their positions before markets close. Zhang (2010) estimates that 78% of the trading volume in the US in 2009 are due to HFTs and that was almost 0% in 1995. HFTs are even more dominant in today's trading environment. Though there is devoted literature on inventory management and implications for asset pricing, market makers holding overnight positions and the reconciliation effect of this practice on opening prices might not have that much effect in the context of jumps. As can be verified from Table 12 and Table 13 in Section 4.5, the lowest jump magnitudes for Quintile 1 are -4.4%, -6.5% for negative jumps, and 5.6% and 6.9% for positive jumps, and it is unlikely that the aforementioned market maker practice will lead to such movements.

Regarding the observed asymmetry between negative and positive market states, it is empirically documented that negative shocks elicit more intense reactions than positive shocks, due to both theoretical and behavioral reasons. For stock returns, *leverage effect* is proposed as one of the economic reasons underlying this reaction: a sharp decline in the stock value leads to a higher debt-to-equity ratio, resulting in amplified risk perception and a more powerful reaction, whereas a positive shock lowers the debt-to-equity ratio and risk perception. Another theoretical explanation is the *volatility-feedback effect*, where the spike in volatility associated with a severe decline creates a riskier outlook, resulting in higher required returns. Since the dividends are unaffected during these short-window events, these higher discount rates in stock pricing result in further downward pressure on stock valuation. This asymmetric nature is also addressed in

volatility modelling by, among others, the Glosten et al. (1993) and Nelson (1991) (GJR-GARCH and the E-GARCH models respectively).

On the behavioral side, the literature suggests that this asymmetry in overreaction to news shocks is rooted in human psychology and the investors' cognitive biases. One of the most fundamental behavioral explanations is the loss aversion argued by Kahneman and Tversky (1979). The pain of losing a certain amount is stronger than the pleasure of gaining the same amount, which reflects that the loss function is steeper than the gain function. When there is a negative price shock, psychological distress evokes a more intense reaction, and investors liquidate their positions more rapidly to avoid further pain, and reaction is intensified especially if the wealth constraints are more compelling. This asymmetric reaction is also empirically displayed through the degree of reversal in Table 12 and Table 13 in Section 4.5 for negative and positive jumps. Benartzi and Thaler (1995) also argue that long-term investors are not devoid of this higher marginal sensitivity to losses than to gains, as they will frequently monitor their wealth. The attractiveness of the risky stock may wane abruptly amid wavering market conditions and this myopic loss aversion can prompt concerted position unwinding and intensify the overreaction and reversal phenomenon. Investors' attention is also categorical in the sense that it is more strongly drawn to market-wide or sector-specific information than to firm-specific news flows (Peng and Xiong (2006)). If the negative shocks coincide with systematic risk, they command the total market attention and induce synchronized action and sharp increases in correlations, impacting a wide cross-section of stocks and propagating contagion across the market.

Although the opening price is commonly handled within the opening auction mechanism, the potential role of bid–ask spreads is also worth mentioning. Bid-ask spreads are typically a few basis points for large-cap stocks, and it is relatively larger in small-cap stocks. Along with that, bid-ask spreads are structurally getting lower over the years. Bessembinder (2003) documented that decimalization at the beginning of 2001 substantially decreased the bid-ask spreads with the reduction in the minimum price increments. Hendershott et al. (2011) find that algorithmic trading narrowed the quoted and effective spreads, leading to improvement in the cost dimension. In the same vein, Menkveld (2013) and Malinova et al. (2018) document that high-frequency trading reduces the bid-ask spreads. Taken together, considering the behavioral dimensions on loss aversion, psychological forces and cognitive biases, formation of opening price through the opening auction that smooths out the noise, size of bid-ask spreads compared

to reversals across the stock universe and structural shrinkage in bid-ask spreads over the years with new market design and the proliferation of algorithmic trading, we believe that the role of bid–ask spread dynamics are unlikely to account for the magnitude and systematic nature of the overreaction and reversal patterns we document.

4.3 Different Time Periods

We test the predictive power of overnight jump returns across different time periods, taking into account specific factors. The early 2000s were crucial in terms of the structural transformation of the markets, and the proliferation of algorithmic trading coincides with this period. Widespread adoption of the Financial Information Exchange (FIX) Protocol and advances in hardware technology are coupled with new software developments. Latency-constrained infrastructures were replaced with fast processors and superior computational power brought agility to the market actors. Another crucial milestone was the transition to fiber networks which accelerated order transmission speed and paved the way for high-frequency trading and liquidity provision. Order books became more densely populated with increased quote updates, improving the absorption capacity and reducing the slippage costs. In the first years of the 2000s, the volume generated by algorithmic execution desks soared substantially, and Hendershott et al. (2011) indicate a positive association between liquidity and algorithmic trading activity for this period. A related advancement in liquidity provision was linked to the phased launch of the Autoquote facility by the NYSE between January 2003 and June 2003; this enabled algorithms to receive instant order book feedback and automatically adjust order placements based on the new information. Along with that, especially after the burst of the dot-com bubble, futures-based hedging practices improved, and institutions especially opted for using E-mini S&P500 futures contracts. Due to its liquidity and accessibility for a wide range of investor base, the open interest for E-mini contracts surpassed that of standard futures contracts around 2004. Enhanced usage of these derivative contracts also improved the cross-market activities and the trading in spot products. To test whether our predictability arguments remain valid under the changing dynamics of this new trading era, we repeat our analyses using two subsamples: July 1993–December 2003 and January 2004–December 2021. A recent study by Megaritis et al. (2021) also show that latent macroeconomic uncertainity impacts jumps in US stock markets. Hence, we further examine our model over the period August 2007–December 2013, which captures the Global Financial

Crisis, the subsequent European Debt Crisis, and the ensuing recovery and reform phases.[16] The unifying theme across these periods is the heightened market sensitivity.

Table 7: Predictability in Different Subperiods - Isolated Approach

	PANEL A:Negative Jumps						PANEL B:Positive Jumps					
	Jul.93-Dec.03		Jan.04-Dec.21		Aug.07-Dec.13		Jul.93-Dec.03		Jan.04-Dec.21		Aug.07-Dec.13	
	coef.	t	coef.	t	coef.	t	coef.	t	coef.	t	coef.	t
Intercept	0.01	(0.53)	0.11	(3.18)	0.18	(2.13)	-0.03	(-6.52)	-0.07	(-6.9)	-0.08	(-5.94)
$JR^{-,+}$	-0.39	(-11.93)	-0.43	(-2.57)	-0.89	(-2.33)	-0.11	(-10.4)	-0.07	(-8.65)	-0.08	(-7.64)
$PREV$	-0.03	(-2.9)	-0.11	(-3.85)	-0.19	(-3.34)	-0.04	(-2.42)	-0.02	(-2.38)	0.00	(-0.24)
$IVOL_OJI_{t-1}$	0.09	(3.45)	-0.27	(-1.39)	-0.58	(-1.26)	-0.16	(-4.85)	-0.16	(-3.65)	-0.23	(-3.71)
$SIZE$	-0.10	(-1.51)	-0.62	(-2.82)	-1.16	(-2.1)	0.17	(7.43)	0.35	(7.17)	0.39	(5.72)
BM	0.17	(4.47)	0.49	(2.81)	0.89	(2.22)	0.00	(0.1)	-0.02	(-0.4)	-0.01	(-0.1)
LEV	0.16	(4.33)	0.19	(1.92)	0.08	(0.35)	0.01	(0.3)	-0.17	(-3.5)	-0.07	(-0.98)
$RET_{t-1,t-5}$	0.16	(1.56)	-0.37	(-1.05)	-1.03	(-1.24)	0.37	(4.07)	0.59	(2.66)	1.40	(3.45)
$RET_{t-6,t-11}$	-0.07	(-0.4)	1.23	(1.08)	4.10	(1.51)	0.19	(2.69)	0.35	(2.68)	0.56	(3.66)
AI_{t-1}	0.00	(-2.85)	0.10	(1.12)	0.00	(0.33)	0.01	(5.29)	0.00	(0.37)	0.00	(1.5)
$Adj.R^2$	0.13		0.10		0.13		0.10		0.09		0.11	

Notes: This Table reports return predictability results when jumps are treated in isolation instead of calculating cumulative jump returns. Stocks having multiple jumps in a month appear with multiple observations in monthly regressions as opposed to once with their cumulative figures. Tabulated results are for the first day after overnight jumps. We run monthly cross-sectional regressions separately for the periods July 1993 - December 2003, January 2004 - December 2021, and August 2007 - December 2013, and save the time series of coefficient estimates. To account for cross-correlations and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). The dependent variable is $1D_t$, and it is the intraday return just after the overnight jump. Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. $JR^{-,+}$ variables respectively stand for negative and positive jump returns, $PREV$ is the previous day return before jumps, $IVOL_OJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_{t-1} is the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_{t-1} are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables. Estimated model: $1D_{i,t,d+1} = \alpha + \beta_1 JR_{i,t,d=0} + \beta_2 PREV_{i,t,d=-1} + \beta_3 IVOL_OJI_{i,t-1} + \beta_4 SIZE_i + \beta_5 BM_i + \beta_6 LEV_i + \beta_7 RET_{i,t-1,t-5} + \beta_8 RET_{i,t-6,t-11} + \beta_9 AI_{i,t-1} + \varepsilon_{i,t}$

Our test results for these subperiods are presented in Table 7 and Table 8 for the first day. The isolated approach predictive power interpretations for jump returns in Table 7 are aligned with the cumulative approach results documented in Table 8. The predictive power of isolated jump returns (JR) and cumulative jump returns (CJR) in all subperiods is maintained, and all factor loadings are statistically significant in these formations. However, there are three notable points to mention. Firstly, the coefficients of negative overnight jumps more than doubled for the period covering the GFC, European debt crisis, and reform phases, and that is in line with our expectations. As opposed to negative states, the factor loadings for positive overnight jumps does not deviate much from the global coefficient of the whole sample period. These periods are characterized by fluctuating market conditions, and the prevailing uncertainty affects the investor decision-making process in line with new risk management norms. It is a well-documented psychological fact that people are more sensitive to negative shocks than they are

[16] IMF Global Financial Stability Report 2008 points to mid-2007 as the first visible signs of financial turmoil. European Debt Crisis is considered the second phase of the Global Financial Crisis. After a series of IMF reviews and monitoring missions for certain European countries, EU Finance ministers jointly approved the Single Resolution Mechanism on December 18, 2013, as discussed in Office and Dept. (2016).

Table 8: Predictability in Different Subperiods - Cumulative Approach

	PANEL A: Negative Jumps						PANEL B: Positive Jumps					
	Jul.93-Dec.03		Jan.04-Dec.21		Aug.07-Dec.13		Jul.93-Dec.03		Jan.04-Dec.21		Aug.07-Dec.13	
	coef.	t	coef.	t	coef.	t	coef.	t	coef.	t	coef.	t
Intercept	0.00	(-0.06)	0.11	(4.94)	0.16	(3.68)	-0.03	(-8.11)	-0.11	(-8.68)	-0.11	(-6.67)
CJR_t	-0.43	(-14.86)	-0.49	(-2.61)	-1.00	(-2.31)	-0.14	(-9.36)	-0.08	(-10.42)	-0.08	(-7.79)
$PREV_t$	-0.01	(-0.62)	-0.06	(-1.73)	-0.14	(-1.96)	0.02	(1.08)	0.03	(2.32)	0.04	(1.95)
$IVOL_OJI_t$	0.07	(1.09)	-0.54	(-2.42)	-1.12	(-2.54)	-0.13	(-4.91)	0.07	(1.02)	-0.08	(-1.37)
SIZE	-0.08	(-0.8)	-0.63	(-4.05)	-1.05	(-2.95)	0.19	(9.16)	0.52	(8.98)	0.53	(6.28)
BM	0.24	(4.21)	0.54	(2.58)	1.00	(2.04)	0.00	(0.04)	-0.01	(-0.2)	0.00	(0.04)
LEV	0.23	(4.04)	0.10	(0.5)	-0.23	(-0.43)	0.00	(0.05)	-0.18	(-3.45)	-0.08	(-1.15)
$RET_{t-1,t-5}$	0.21	(1.62)	-0.84	(-1)	-2.93	(-1.63)	0.29	(2.55)	0.72	(3.23)	1.46	(3.75)
$RET_{t-6,t-11}$	-0.29	(-1.44)	0.75	(0.78)	2.90	(1.23)	0.23	(2.58)	0.44	(3.23)	0.67	(3.79)
AI_t	-0.01	(-3.39)	0.02	(2.32)	0.01	(1.49)	0.01	(4.45)	0.02	(2.01)	0.01	(2)
$Adj.R^2$	0.17		0.12		0.16		0.16		0.13		0.14	

Notes: To disentangle the discontinuous overnight components of monthly returns, we calculate cumulative overnight positive and negative jump returns for each firm. On a monthly scale, we also calculate cumulative post-jump returns as shown in Section 3.2.2, and construct our dependent variables. Then, we run monthly cross-sectional regressions separately for the periods July 1993 - December 2003, January 2004 - December 2021, and August 2007 - December 2013, and save the time series of coefficient estimates. To account for cross-correlations and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). The dependent variable is the cumulated $1D_t$ returns (the intraday returns just after the overnight jumps). Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. CJR_t^+ and CJR_t^- are respectively the monthly cumulated positive and negative jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, *SIZE* is the log of market cap at every June, *BM* is the log of book-to-market ratio, *LEV* is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for *SIZE*, *BM*, *LEV*, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_t are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables. Estimated model: $CDR_{i,t,d+1} = \alpha + \beta_1 CJR_{i,t,d=0} + \beta_2 PREV_{i,t,d=-1} + \beta_3 IVOL_OJI_{i,t} + \beta_4 SIZE_i + \beta_5 BM_i + \beta_6 LEV_i + \beta_7 RET_{i,t-1,t-5} + \beta_8 RET_{i,t-6,t-11} + \beta_9 AI_{i,t} + \varepsilon_{i,t}$

to positive catastrophic events. This is actually in line with the asymmetric volatility pattern of the news-impact curves reported by Pagan and Schwert (1990). Hence, having a higher degree of overreaction after negative jumps and increased factor loadings for JR_t^- and CJR_t^- is intuitive in our framework. Second, all coefficients are highly significant when the data sample is divided into two groups around December 2003. One visible change is the increase in t-statistics for negative overnight jumps for the first part of the data sample. We attribute this to higher noise and idiosyncratic volatility in the new era due to increased complexity and algorithmic execution that introduce additional sources of variation, reducing the precision compared to the first part of the data sample. Third, *PREV*, *IVOL_OJI*, *LEV*, *AI*, *SIZE*, $RET_{i,t-1,t-5}$ and $RET_{i,t-6,t-11}$ lose their predictive power for the follow-up returns or become priced risk factors in different sample splits and market states.

In addition, we repeat our analysis for acute market stress periods, including the burst of the dot-com bubble (March 2000–October 2002), the Global Financial Crisis (August 2007–June 2009), and the COVID-19 market crash (February 2020–May 2020). We collectively refer to these episodes as crisis periods. For both the isolated and cumulative approaches, our inferences are still retained in crisis and non-crisis periods. We report these findings in Table 9 and Table 10.

Table 9: Return Predictability in Crisis and Non-Crisis Periods - Isolated Approach

	PANEL A: Negative Jumps				PANEL B: Positive Jumps			
	Crisis Periods		Non-Crisis Periods		Crisis Periods		Non-Crisis Periods	
	coef.	t	coef.	t	coef.	t	coef.	t
$Intercept$	0.20	(1.72)	0.04	(4.41)	-0.08	(-3.33)	-0.05	(-6.74)
$JR^{-,+}$	-0.84	(-2.03)	-0.33	(-5.02)	-0.11	(-13.5)	-0.08	(-9.41)
$PREV_t$	-0.11	(-1.33)	-0.08	(-5.04)	-0.05	(-3.84)	-0.02	(-2.57)
$IVOL_OJI_{t-1}$	-0.60	(-1.09)	-0.04	(-0.55)	-0.11	(-1.45)	-0.17	(-5.27)
$SIZE$	-1.23	(-1.7)	-0.26	(-5.65)	0.39	(3.62)	0.26	(7.19)
BM	0.56	(1.65)	0.34	(3.55)	-0.03	(-0.38)	-0.01	(-0.28)
LEV	-0.13	(-0.79)	0.24	(3.57)	-0.15	(-1.37)	-0.10	(-2.83)
$RET_{t-1,t-5}$	-1.07	(-1.02)	0.02	(0.11)	1.28	(2.42)	0.34	(3.34)
$RET_{t-6,t-11}$	4.10	(1.19)	0.05	(0.19)	0.41	(2.06)	0.27	(2.73)
AI_{t-1}	0.00	(-2.12)	0.08	(1.09)	0.01	(2.93)	0.00	(0.51)
$Adj.R^2$	0.14		0.11		0.12		0.09	

Notes: This Table reports return predictability results when jumps are treated in isolation instead of calculating cumulative jump returns. Stocks having multiple jumps in a month appear with multiple observations in monthly regressions as opposed to once with their cumulative figures. Tabulated results are for the first day after overnight jumps. We run monthly cross-sectional regressions separately for crisis and non-crisis periods and save the time series of coefficient estimates. Crisis times collectively include the acute periods of the Dot-com crash (March 2000- October 2002), Global Financial Crisis (August 2007 - June 2009), and COVID-19 pandemic (February 2020 - May 2020). To account for cross-correlations and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). The dependent variable is $1D_t$, and it is the intraday return just after the overnight jump. Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. $JR^{-,+}$ variables respectively stand for negative and positive jump returns, $PREV$ is the previous day return before jumps, $IVOL_OJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_{t-1} is the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_{t-1} are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables. Estimated model: $1D_{i,t,d+1} = \alpha + \beta_1 JR_{i,t,d=0} + \beta_2 PREV_{i,t,d=-1} + \beta_3 IVOL_OJI_{i,t-1} + \beta_4 SIZE_i + \beta_5 BM_i + \beta_6 LEV_i + \beta_7 RET_{i,t-1,t-5} + \beta_8 RET_{i,t-6,t-11} + \beta_9 AI_{i,t-1} + \varepsilon_{i,t}$

Subperiod analysis is also instrumental in examining return predictability findings under different data qualifications. Although missing opening prices constitute only a small fraction of the raw dataset (3.62% of all daily observations across all stocks), such cases imply zero overnight returns, no jump detection, and therefore do not enter the jump dataset or subsequent regression analyses. Critically, the prevalence of missing opening prices declines sharply after the early 2000s (particularly after 2003).[17] This pattern is highly consistent with the subperiod analysis in Table 7 and with the broader discussion that the liquidity and trading volumes increased

[17]The average monthly number of missing opening prices across all stocks is 5,732 for July 1993 - December 2003, 724 for January 2004 - December 2021, 440 for January 2010 - December 2021, and 133 for January 2020 - December 2021.

Table 10: Return Predictability in Crisis and Non-Crisis Periods - Cumulative Approach

	PANEL A: Negative Jumps				PANEL B: Positive Jumps			
	Crisis Periods		Non-Crisis Periods		Crisis Periods		Non-Crisis Periods	
	coef.	t	coef.	t	coef.	t	coef.	t
$Intercept$	0.15	(2.43)	0.05	(3)	-0.10	(-2.9)	-0.08	(-7.16)
CJR_t	-0.95	(-2.1)	-0.36	(-5.13)	-0.10	(-11.17)	-0.10	(-9.37)
$PREV_t$	-0.10	(-0.94)	-0.03	(-1.93)	-0.01	(-0.93)	0.03	(3.08)
$IVOL_OJI_t$	-0.74	(-1.69)	-0.22	(-1.59)	-0.11	(-1.81)	0.02	(0.39)
$SIZE$	-1.04	(-2.16)	-0.30	(-4.05)	0.48	(3.15)	0.38	(7.83)
BM	0.47	(1.81)	0.42	(3.32)	-0.08	(-0.94)	0.01	(0.25)
LEV	-0.69	(-1.15)	0.32	(4.86)	-0.16	(-1.34)	-0.10	(-2.77)
$RET_{t-1,t-5}$	-2.95	(-1.24)	0.07	(0.23)	1.22	(2.16)	0.42	(3.9)
$RET_{t-6,t-11}$	3.38	(1.12)	-0.26	(-1.14)	0.40	(1.74)	0.35	(3.57)
AI_t	0.00	(-0.92)	0.02	(1.91)	0.01	(2.7)	0.01	(2.17)
$Adj.R^2$	0.17		0.13		0.14		0.14	

Notes: To disentangle the discontinuous overnight components of monthly returns, we calculate cumulative overnight positive and negative jump returns for each firm. On a monthly scale, we also calculate cumulative post-jump returns as shown in Section 3.2.2, and construct our dependent variables. Then, we run monthly cross-sectional regressions separately for crises and non-crises periods and save the time series of coefficient estimates. Crisis times collectively include the acute periods of the Dot-com crash (March 2000- October 2002), Global Financial Crisis (August 2007 - June 2009), and COVID-19 pandemic (February 2020 - May 2020). To account for cross-correlations and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). The dependent variable is $1D_t$, and it is the intraday return just after the overnight jump. Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. CJR_t^+ and CJR_t^- are respectively the monthly cumulated positive and negative jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_t are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables. Estimated model: $CDR_{i,t,d+1} = \alpha + \beta_1 CJR_{i,t,d=0} + \beta_2 PREV_{i,t,d=-1} + \beta_3 IVOL_OJI_{i,t} + \beta_4 SIZE_i + \beta_5 BM_i + \beta_6 LEV_i + \beta_7 RET_{i,t-1,t-5} + \beta_8 RET_{i,t-6,t-11} + \beta_9 AI_{i,t} + \varepsilon_{i,t}$

substantially following the proliferation of algorithmic trading and major improvements in market infrastructure. Consequently, the subperiod analysis for January 2004 - December 2021 also effectively reflects the results when the missing opening prices are generally absent.

4.4 Proxy Measure for News Type Analysis

The present study does not specifically identify the news types on a daily basis, and this is a limitation of our research. However, we build a proxy measure that may help us identify days with market-wide news flows and document the results with certain caveats. We calculate the number of positive and negative overnight jumps on each trading day over the entire sample period (July

1993–December 2021) and construct the ratio of positive-jump stocks to negative-jump stocks. When this ratio falls below 0.5 or exceeds 2, it suggests an imbalance in the cross-sectional distribution of jump directions, with either negative-jump stocks outnumbering positive-jump stocks by at least two-to-one or vice versa. Days meeting these criteria are classified as "market-wide negative jump days" and "market-wide positive jump days," respectively, and all remaining days are classified as "firm-specific jump days." Following the baseline approach, we estimate monthly cross-sectional regressions. For the market-wide news analysis, we retain only those jumps that occur on days classified as market-wide jump days within a given month. The remaining observations are assigned to the firm-specific news sample and used in a separate set of regressions. In months for which no market-wide jump days are identified, all observations are allocated to the firm-specific sample and are therefore used in the corresponding analysis. We further repeat the analysis using more conservative cutoff ratio values of 0.33 and 3.

On some of the days, the total number of jumps may be considerably lower (e.g., 6 positive jumps and 4 negative jumps) compared to other days. In other words, the large imbalance in the ratio of positive to negative jumps may arise even when the total number of jumps is relatively small and looking only at the ratio may be misleading as it overlooks the fact that market-wide news shocks impact a larger spectrum of the stocks. Accordingly, we impose an additional requirement that the total number of jumps on a given day be at least 30 for the day to be classified as a market-wide jump day. Our examination of the positive-to-negative jump ratios suggests that the cutoff values 0.5 and 2 (and more conservatively 0.33 and 3) are plausible thresholds for identifying days characterized by a large concentration of either negative or positive overnight jumps. For the first set of cutoff values (0.5 | 2.0), the estimated factor loading is larger on market-wide news days than on firm-specific news days for both positive and negative jump episodes. When the more conservative thresholds (0.33 | 3.0) are employed, the factor loadings for market-wide and firm-specific news flows become nearly identical during negative jump episodes. In contrast, for positive jump episodes, the factor loading associated with market-wide news days remains higher than that for firm-specific news days.

We also conduct subperiod analysis; factor loadings for market-wide news days are larger in both sets of cutoff values and in both positive and negative market states for the January 2004-December 2021 period. When the cutoff set is (0.5 | 2.0), the coefficient of JR becomes -0.61 for market-wide news days and -0.38 for firm-specific news days after negative overnight shocks

Table 11: Return Predictability and Underlying News Types - Isolated Approach

Panel A: Daily Return Predictability After Overnight Jumps with Jump-number Ratios 0.5 and 2

	Negative Jumps				Positive Jumps			
	Market-wide news		Firm-specific news		Market-wide news		Firm-specific news	
	coef.	t	coef.	t	coef.	t	coef.	t
$Intercept$	0.00	(-0.05)	0.11	(2.28)	-0.03	(-2.32)	-0.06	(-7.47)
$JR_t^{-,+}$	-0.49	(-4.42)	-0.38	(-3.25)	-0.10	(-3.88)	-0.08	(-11.41)
$PREV_t$	0.02	(0.64)	-0.07	(-3.27)	-0.01	(-0.3)	-0.03	(-3.2)
$IVOL_OJI_{t-1}$	-0.21	(-1.16)	-0.20	(-1.19)	-0.16	(-1.94)	-0.15	(-4.46)
$SIZE$	-0.07	(-0.9)	-0.60	(-2.24)	0.19	(2.63)	0.29	(8.06)
BM	0.06	(0.35)	0.44	(2.2)	0.03	(0.32)	0.00	(0.05)
LEV	0.08	(0.54)	0.10	(1.42)	-0.18	(-1.25)	-0.10	(-2.65)
$RET_{t-1,t-5}$	0.31	(0.5)	-0.81	(-1.31)	0.42	(1.71)	0.50	(3.27)
$RET_{t-6,t-11}$	-0.16	(-0.65)	1.50	(1.06)	0.00	(-0.01)	0.30	(3.16)
AI_{t-1}	0.69	(1.34)	0.05	(1.08)	0.74	(0.72)	0.00	(1.15)
$Adj.R^2$	0.22		0.11		0.22		0.10	

Panel B: Daily Return Predictability After Overnight Jumps with Jump-number Ratios 0.33 and 3

	Negative Jumps				Positive Jumps			
	Market-wide news		Firm-specific news		Market-wide news		Firm-specific news	
	coef.	t	coef.	t	coef.	t	coef.	t
$Intercept$	0.02	(1.7)	0.09	(2.33)	-0.03	(-1.14)	-0.06	(-7.32)
$JR_t^{-,+}$	-0.41	(-3.25)	-0.42	(-3.75)	-0.12	(-2.86)	-0.09	(-11.45)
$PREV_t$	-0.01	(-0.18)	-0.08	(-3.57)	-0.01	(-0.1)	-0.03	(-3.29)
$IVOL_OJI_{t-1}$	-0.05	(-0.68)	-0.18	(-1.16)	-0.11	(-0.86)	-0.16	(-4.83)
$SIZE$	-0.15	(-2.98)	-0.54	(-2.37)	0.16	(1.26)	0.28	(7.93)
BM	-0.30	(-1.34)	0.52	(2.44)	-0.05	(-0.37)	-0.01	(-0.3)
LEV	-0.11	(-0.86)	0.17	(2.37)	-0.13	(-0.42)	-0.12	(-3.04)
$RET_{t-1,t-5}$	1.30	(2.09)	-0.36	(-1.32)	0.34	(0.72)	0.51	(3.46)
$RET_{t-6,t-11}$	-0.39	(-0.9)	1.33	(1.03)	-0.81	(-1.4)	0.33	(3.68)
AI_{t-1}	1.84	(1.21)	0.05	(1.1)	-1.17	(-1.51)	0.00	(1.21)
$Adj.R^2$	0.24		0.11		0.27		0.10	

Notes: This table shows return predictability results for market-wide and firm-specific news flows identified based on a proxy measure. We first calculate the number of positive and negative overnight jumps on each trading day over the entire sample period (July 1993–December 2021) and construct the ratio of positive-jump stocks to negative-jump stocks. If the total number jumps is at least 30 and the ratio is between [0, 0.5] or [2, ∞), the relevant day is marked as "market-wide negative jump day" or "market-wide positive jump day" respectively. We repeat the analysis for ratio cutoffs [0, 0.33] or [3, ∞) and report the results in Panel B. For the market-wide news analysis, we retain only those jumps that occur on days classified as market-wide jump days within a given month. The remaining observations are assigned to the firm-specific news sample. In months for which no market-wide jump days are identified, all observations are allocated to the firm-specific sample. Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. $JR^{-,+}$ variables respectively stand for negative and positive jump returns, $PREV$ is the previous day return before jumps, $IVOL_OJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_{t-1} is the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_{t-1} are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables. Estimated model: $1D_{i,t,d+1} = \alpha + \beta_1 JR_{i,t,d=0} + \beta_2 PREV_{i,t,d=-1} + \beta_3 IVOL_OJI_{i,t-1} + \beta_4 SIZE_i + \beta_5 BM_i + \beta_6 LEV_i + \beta_7 RET_{i,t-1,t-5} + \beta_8 RET_{i,t-6,t-11} + \beta_9 AI_{i,t-1} + \varepsilon_{i,t}$

and it is respectively -0.10 and -0.07 for positive jumps. If we perform this subperiod analysis with conservative thresholds (0.33 | 3.0), the coefficients of JR this time converges with -0.46 for market-wide news days and -0.44 for firm-specific new days after negative overnight shocks. They remain as -0.10 and -0.07 for positive jumps. For the period July 1993 – December 2003, the factor loading associated with market-wide news is higher only under (0.33 | 3.0) cutoffs during positive market states. For the January 2004–December 2021 period, market-wide news days continue to exhibit higher coefficients when the ratio threshold is set to (0.5 | 2.0) and the minimum number of jumps is specified as either 15 or 45. When the ratio threshold is changed to (0.33 | 3.0), market-wide news days still exhibit substantially higher coefficients when the minimum requirement for total jump number is 45. However, when this minimum requirement is reduced to 15, the coefficients for market-wide news days become slightly lower than those observed for firm-specific news days.

The results, particularly for the January 2004 - December 2021 subsample that more closely represents modern trading conditions, are broadly consistent with our conjecture that market-wide news shocks induce stronger jump-reversal patterns. However, since the analyses rely on a proxy measure, several limitations are worthy of mention. Firstly, there are sometimes sector-specific shocks (such as regulatory changes or commodity price shocks that affect only a subset of firms, etc.), and such events may generate localized jumps without necessarily reflecting broad market-wide information shocks. As a result, the total number of negative and positive jump stocks may at times be similar, and the presence of a concerted reaction to the underlying news flow may be obscured. Secondly, market-wide shocks may negatively affect a large subset of stocks, but they may simultaneously have positive effects on other groups. For example, a catastrophic regional tension is clearly negative news for the overall market, particularly for firms in aviation, tourism, accommodation and hospitality, retail consumption, banking, construction, and exporters to the affected region. At the same time, such an event may also positively affect firms in defense industry, cybersecurity firms, domestic producers of commodities facing supply disruptions in the conflict region, and certain insurance companies, especially if trade routes are rerouted through longer and alternative corridors. As a result, relying solely on the cross-sectional count of positive and negative jumps may be insufficient to fully capture or correctly classify the underlying nature of news shocks. Third, we set a minimum threshold of 30 total jumps per day and test the results additionally for a minimum of 15 and 45 jumps each day. This

cutoff could be increased further at the cost of reducing the number of observations available for the news-type classification. In that respect, the choice of the total-jump threshold represents another challenging task.

4.5 Degree of Reversal and Predictability in Different Quintiles

Table 12: Degree of Reversal - Cumulative Negative Overnight Jumps

Panel A: Stocks are Sorted According to Jump-Day-Isolated Idiosyncratic Volatility Figures

	Jump Day			Jump Day+1			Jump Day+2		
	Reversal	**Jump**	**Reversal/Jump**	**Reversal**	**Jump**	**Reversal/Jump**	**Reversal**	**Jump**	**Reversal/Jump**
Quintile 1	1.0%	-4.4%	**0.23**	1.3%	-4.4%	**0.32**	1.4%	-4.4%	**0.33**
Quintile 2	1.2%	-6.2%	**0.20**	1.7%	-6.2%	**0.28**	1.7%	-6.2%	**0.28**
Quintile 3	1.7%	-8.0%	**0.20**	2.3%	-8.0%	**0.27**	2.2%	-8.0%	**0.27**
Quintile 4	3.0%	-10.3%	**0.24**	3.7%	-10.3%	**0.31**	3.7%	-10.3%	**0.31**
Quintile 5	4.5%	-16.0%	**0.26**	6.9%	-15.9%	**0.39**	6.9%	-15.9%	**0.40**
Q1-Q5	-3.5%	11.6%	**-0.03**	-5.5%	11.6%	**-0.07**	-5.5%	11.5%	**-0.07**
	(-10.11)	(46.9)	**(-1.59)**	(-12.67)	(46.81)	**(-3.03)**	(-12.18)	(46.88)	**(-22.04)**

Panel B: Stocks are Sorted According to Monthly Amihud Illiquidity Figures

	Jump Day			Jump Day+1			Jump Day+2		
	Reversal	**Jump**	**Reversal/Jump**	**Reversal**	**Jump**	**Reversal/Jump**	**Reversal**	**Jump**	**Reversal/Jump**
Quintile 1	0.4%	-6.5%	**0.07**	0.5%	-6.5%	**0.07**	0.5%	-6.5%	**0.08**
Quintile 2	0.8%	-7.8%	**0.12**	1.0%	-7.8%	**0.13**	1.0%	-7.8%	**0.14**
Quintile 3	1.5%	-8.4%	**0.19**	1.7%	-8.4%	**0.22**	1.7%	-8.4%	**0.22**
Quintile 4	3.7%	-9.6%	**0.33**	4.4%	-9.6%	**0.41**	4.2%	-9.6%	**0.40**
Quintile 5	5.1%	-12.6%	**0.41**	8.5%	-12.6%	**0.63**	8.6%	-12.6%	**0.64**
Q1-Q5	-4.6%	6.1%	**-0.34**	-8.0%	6.1%	**-0.56**	-8.1%	6.1%	**-0.57**
	(-15.86)	(22.14)	**(-15.06)**	(-20.48)	(22.06)	**(-21.69)**	(-20.05)	(21.97)	**(-20.84)**

Notes: Table shows the degree of reversals across different quintiles of stocks formed according to stock-specific risk figures. *Reversal* is the cumulative returns until each specified day after the jump incidence. *Reversal/Jump* is the fraction of jumps that is cumulatively reversed in respective days. At each month, we separately sort stocks in ascending order according to their Idiosyncratic Volatility levels calculated in isolation with overnight jump days ($IVOL_OJI$) during that month and their monthly Amihud Illiquidity (AI) figures and split the sorted stocks in quintiles. *Quintile 1* is for the stocks with lowest AI and $IVOL_OJI$ figures. For each month, we take the average of cumulative reversal returns within each quintile and construct different time series for them. Tabulated numbers are the time-series averages for each day after overnight negative jump incidence. $Q5 - Q1$ stands for the mean differences for each column variable with t-statistic values below in parentheses. Our analyses cover 342 months over the period July 1993 - December 2021.

The correction level in the mispricing is not homogeneous among stocks with different characteristics, which may be essential in impelling arbitrageurs to step in. Bunch of literature documents that there are *limits to arbitrage* (Shleifer and Vishny (1997), Hirshleifer (2001) and Kyle and Xiong (2001)) among many others) and arbitrage practices are not perfectly mechanical and not riskless. Willingness for price correction decays even further when the level of mispricing is intense. As pointed out by Atilgan et al. (2020) and relevant literature, idiosyncratic risk is regarded as one of the most crucial arbitrage costs, especially when combined with extreme noise trading. In Table 12 and Table 13, we delve into price reversals and their association with the stocks' idiosyncratic risks. In that regard, one may expect the fraction of jump returns being re-

Table 13: Degree of Reversal - Cumulative Positive Overnight Jumps

Panel A: Stocks are Sorted According to Jump-Day-Isolated Idiosyncratic Volatility Figures

	Jump Day			Jump Day+1			Jump Day+2		
	Reversal	**Jump**	**Reversal/Jump**	**Reversal**	**Jump**	**Reversal/Jump**	**Reversal**	**Jump**	**Reversal/Jump**
Quintile 1	-0.5%	5.6%	**0.08**	-0.5%	5.7%	**0.10**	-0.5%	5.7%	**0.08**
Quintile 2	-0.7%	7.1%	**0.08**	-0.8%	7.1%	**0.11**	-0.8%	7.1%	**0.10**
Quintile 3	-1.0%	8.9%	**0.10**	-1.3%	8.9%	**0.13**	-1.3%	8.9%	**0.13**
Quintile 4	-2.0%	11.8%	**0.16**	-2.4%	11.8%	**0.19**	-2.3%	11.8%	**0.19**
Quintile 5	-3.3%	22.3%	**0.16**	-3.9%	22.3%	**0.19**	-3.9%	22.3%	**0.19**
Q1-Q5	2.9%	-16.7%	**-0.08**	3.4%	-16.6%	**-0.09**	3.4%	-16.6%	**-0.10**
	(17.3)	(-27.2)	**(-9.42)**	(16.1)	(-27.16)	**(-8.52)**	(13.21)	(-27.12)	**(-8.06)**

Panel B: Stocks are Sorted According to Monthly Amihud Illiquidity Figures

	Jump Day			Jump Day+1			Jump Day+2		
	Reversal	**Jump**	**Reversal/Jump**	**Reversal**	**Jump**	**Reversal/Jump**	**Reversal**	**Jump**	**Reversal/Jump**
Quintile 1	-0.1%	6.9%	**0.01**	0.0%	6.9%	**-0.01**	0.2%	6.9%	**-0.03**
Quintile 2	-0.5%	9.0%	**0.05**	-0.4%	9.0%	**0.03**	-0.3%	9.0%	**0.02**
Quintile 3	-1.1%	10.4%	**0.10**	-1.1%	10.4%	**0.10**	-1.1%	10.4%	**0.09**
Quintile 4	-2.2%	12.4%	**0.18**	-2.5%	12.4%	**0.20**	-2.5%	12.4%	**0.20**
Quintile 5	-3.5%	16.9%	**0.22**	-4.9%	16.9%	**0.31**	-5.1%	16.8%	**0.31**
Q1-Q5	3.4%	-9.9%	**-0.21**	5.0%	-9.9%	**-0.32**	5.3%	-9.9%	**-0.35**
	(28.42)	(-23.35)	**(-25.23)**	(32.48)	(-23.27)	**(-30.12)**	(30.62)	(-23.23)	**(-27.42)**

Notes: Table shows the degree of reversals across different quintiles of stocks formed according to stock-specific risk figures. *Reversal* is the cumulative returns until each specified day after the jump incidence. *Reversal/Jump* is the fraction of jumps that is cumulatively reversed in respective days. At each month, we separately sort stocks in ascending order according to their Idiosyncratic Volatility levels calculated in isolation with overnight jump days ($IVOL_OJI$) during that month and their monthly Amihud Illiquidity (AI) figures and split the sorted stocks in quintiles. *Quintile 1* is for the stocks with lowest AI and $IVOL_OJI$ figures. For each month, we take the average of cumulative reversal returns within each quintile and construct different time series for them. Tabulated numbers are the time-series averages for each day after overnight negative jump incidence. $Q5 - Q1$ stands for the mean differences for each column variable with t-statistic values below in parentheses. Our analyses cover 342 months over the period July 1993 - December 2021.

versed to be lower for stocks with higher levels of idiosyncratic volatility. However, such a claim requires the knowledge of a fair price level to assess whether the reversed fraction eliminates the overreaction component and mispricing entirely.

We report results for the first three days after negative and positive jump incidences. Stocks are primarily sorted according to their $IVOL_OJI$ levels, as, all else being equal, it may be a powerful indicator for arbitrageurs whether to engage in price correction activity or not. At each month, stocks are sorted in ascending order according to their monthly $IVOL_OJI$ numbers. We split the sorted stock list into quintiles and analyze their jump and reversal patterns thoroughly. $Q5$ contains the riskiest and the most illiquid stocks, whereas $Q1$ encloses stocks with the lowest idiosyncratic volatility and illiquidity levels.

The reversed fraction of the jump is shown in column $Reversal/Jump$ with a positive sign. Panel A both in Table 12 and Table 13 tabulates results for quintile portfolios constructed according to ascending $IVOL_OJI$ levels, and Panel B shows the results for AI figures. Looking at the reversed jump fractions, it is visibly higher for negative jump episodes, which is compatible

Table 14: Regressions for Different Quintiles of IVOL_OJI - Cumulative Approach

Panel A: Negative Overnight Jumps

	Int.	CJR_t^-	$PREV_t$	$IVOL_OJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t	$Adj.R^2$
Q1	0.04	-0.31	-0.02	-0.81	-0.17	0.13	0.19	-0.29	0.32	0.16	0.13
	(5.62)	(-20.68)	(-0.88)	(-4.88)	(-5.28)	(2.69)	(3.45)	(-1.13)	(1.41)	(0.52)	
Q2	0.08	-0.22	-0.02	-0.31	-0.39	0.12	0.01	0.01	0.22	0.43	0.10
	(7.25)	(-14.3)	(-0.97)	(-1.19)	(-7.86)	(1.91)	(0.12)	(0.04)	(1.22)	(0.89)	
Q3	0.14	-0.24	-0.03	-0.48	-0.72	-0.04	0.13	-0.03	-0.12	0.44	0.10
	(8.02)	(-5.28)	(-1.94)	(-2.14)	(-6.12)	(-0.46)	(1.18)	(-0.13)	(-0.48)	(1.32)	
Q4	0.12	-0.44	0.01	-0.03	-0.81	1.07	-0.62	-0.41	2.89	0.81	0.11
	(3.22)	(-1.87)	(0.32)	(-0.1)	(-6.67)	(1.15)	(-1.21)	(-1.41)	(1)	(1.42)	
Q5	0.21	-0.51	-0.04	-0.18	-1.34	0.26	-0.09	0.08	-1.24	-0.48	0.18
	(4.51)	(-5.29)	(-2.05)	(-0.93)	(-5.26)	(0.71)	(-0.44)	(0.16)	(-1.39)	(-0.93)	

Panel B: Positive Overnight Jumps

	Int.	CJR_t^+	$PREV_t$	$IVOL_OJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t	$Adj.R^2$
Q1	-0.01	-0.05	0.04	0.05	0.07	-0.16	-0.25	-0.38	0.00	-0.24	0.05
	(-4.15)	(-7.04)	(2.9)	(0.54)	(4.42)	(-3.52)	(-4.89)	(-2.44)	(0)	(-0.94)	
Q2	-0.05	-0.07	-0.01	0.19	0.25	-0.13	-0.07	-0.08	0.22	-1.35	0.07
	(-5.96)	(-8.79)	(-0.41)	(1.11)	(7.01)	(-2.72)	(-1.28)	(-0.38)	(1.78)	(-1.66)	
Q3	-0.09	-0.12	0.02	-0.11	0.48	-0.02	-0.12	0.51	0.25	-0.47	0.08
	(-6.82)	(-12.14)	(1.83)	(-0.54)	(8.72)	(-0.29)	(-2.18)	(2.36)	(1.75)	(-1.6)	
Q4	-0.11	-0.13	0.01	-0.40	0.65	0.18	-0.03	0.83	0.35	0.03	0.11
	(-7.51)	(-11.12)	(0.93)	(-2.93)	(8.09)	(2.82)	(-0.42)	(4.26)	(2.4)	(0.48)	
Q5	-0.12	-0.13	0.01	0.18	0.53	0.02	-0.08	0.74	0.36	0.01	0.16
	(-3.56)	(-10.02)	(0.83)	(2.1)	(3.26)	(0.22)	(-0.6)	(1.9)	(1.62)	(0.02)	

Notes: This table shows the regression outputs for portfolios formed according to different $IVOL_OJI$ quintiles. At each month, we sort overnight jump stocks in ascending order based on the $IVOL_OJI$ numbers and form quintile portfolios. Then, we run monthly cross-sectional regressions and save the time series of coefficient estimates. The dependent variable is the cumulated $1D_t$ returns (the intraday returns just after the overnight jumps). To account for cross-correlations and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. CJR_t^+ and CJR_t^- are respectively the monthly cumulated positive and negative jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_t are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables. Estimated model: $CDR_{i,t,d+1} = \alpha + \beta_1 CJR_{i,t,d=0} + \beta_2 PREV_{i,t,d=-1} + \beta_3 IVOL_OJI_{i,t} + \beta_4 SIZE_i + \beta_5 BM_i + \beta_6 LEV_i + \beta_7 RET_{i,t-1,t-5} + \beta_8 RET_{i,t-6,t-11} + \beta_9 AI_{i,t} + \varepsilon_{i,t}$

with investors' inclination to overreact more intensely to negative shocks. That is in line with our expectations before the analysis. Degrees of reversal are similar and slightly above 20% across all quintiles for negative overnight jumps. 23% of the negative overnight jump is reversed for $Q1$ stocks in the intraday period right after the information shock, and this fraction is 26% for $Q5$ stocks. At the end of the second and third days after the jump, the reversal fraction is higher. For positive overnight jumps, we show that 8% of the jump is reversed in the first day after the overnight jump for $Q1$ stocks whereas this fraction is 16% for stocks with the highest idiosyncratic volatility numbers. The reversal fractions do not materially change after two and three days following the jump in all quintiles. Unreported results also indicate that return predictability re-

mains robust even when the jump-day open-to-close return component is excluded. Specifically, the regressions include only the returns realized at the end of second, third, fourth and fifth days by omitting the jump-day returns. Consistent with the observation that the largest reversal occurs on the jump day, the estimated factor loadings are smaller under this specification.

We replicate our analysis by sorting stocks according to their AI figures in that month and document our findings for negative and positive overnight jumps in Panel B of Table 12 and Table 13. The highest fraction of reversal for stock-specific illiquidity is observed in $Q5$ stocks, and magnitudes in all quintiles are higher for negative information shocks compared to positive overnight jumps, similar to the $IVOL_OJI$ sorting scheme.

Investors may consider the idiosyncratic volatility levels as a critical cost in their trading practices and refrain from exploiting the overreaction pattern for some specific stocks. Reversal degrees are quite similar in different $IVOL_OJI$ classifications after negative jumps and somewhat different after positive overnight shocks. Although we cannot be sure that the new prices after the reversals are at fair levels, we may expect that stocks in the $Q1$ group may be closer to the efficient price trajectory due to less noise and the clientele base being composed more of institutional and professional investors.

Another finding that conforms to the behavioral dimensions of sharp price movements is the overall levels of price reversals after negative and positive overnight information shocks. It is obvious from Table 12 and Table 13 that the degree of reversal is notably higher in all quintiles on all days after negative jumps, and that supports the premise of asymmetric reaction to negative and positive information shocks and aligns with investors' psychological inclination to be more sensitive to negative shocks. To put it differently, investors are overreacting more to unexpected negative news flows than they do to positive information shocks.

Drawing us a broad picture, return predictability results in Table 5 and Table 6 are for all stocks combined. To check the economic and statistical significance of CJR in different stock groups, we repeat our analyses by forming quintile portfolios based on different idiosyncratic risk levels. At each month, we sort jump stocks in ascending order according to $IVOL_OJI$ figures, split them into quintiles, and run cross-sectional regressions. Results are tabulated in Table 14. Panel A of Table 14 shows that cumulative overnight jump returns are highly significant in all quintiles for the first day following overnight jumps. The highest statistical significance is achieved in $Q1$ with a (-20.68) t-statistic and (-0.31) coefficient, whereas the highest economic

Table 15: Regressions for Different Quintiles of SIZE - Cumulative Approach

Panel A: Negative Overnight Jumps

	$Int.$	CJR_t^-	$PREV_t$	$IVOL_OJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t	$Adj.R^2$
Q1	-0.13	-0.78	-0.07	-0.35	0.01	0.29	0.69	0.15	0.45	0.01	0.26
	(-1.37)	(-5.48)	(-2.26)	(-2.68)	(1.21)	(1.45)	(0.89)	(0.19)	(0.71)	(0.99)	
Q2	0.35	-0.60	-0.12	-0.91	-0.02	0.48	0.21	-4.56	2.99	4.42	0.20
	(1.58)	(-2.61)	(-1.57)	(-2.12)	(-1.5)	(1.41)	(1.06)	(-1.2)	(1.15)	(2.89)	
Q3	0.12	-0.19	-0.01	-0.44	-0.01	0.11	-0.13	-0.34	0.01	30.71	0.14
	(3.46)	(-5.38)	(-0.44)	(-4.67)	(-3.33)	(1.12)	(-1.69)	(-1.11)	(0.02)	(2.51)	
Q4	0.08	-0.10	-0.01	-0.52	0.00	0.27	0.17	-0.44	-0.37	80.96	0.12
	(1.92)	(-5.01)	(-0.61)	(-5.89)	(-1.81)	(1.51)	(2.5)	(-1.57)	(-1.81)	(2.19)	
Q5	0.02	-0.05	-0.04	-0.55	0.00	-0.05	-0.14	-0.18	-0.08	585.51	0.10
	(1.5)	(-3.8)	(-2.1)	(-9.3)	(-0.76)	(-1.03)	(-1.34)	(-0.85)	(-0.43)	(3.52)	

Panel B: Positive Overnight Jumps

	$Int.$	CJR_t^+	$PREV_t$	$IVOL_OJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t	$Adj.R^2$
Q1	0.04	-0.14	0.00	0.17	0.00	0.15	0.22	0.82	0.79	0.01	0.21
	(1.02)	(-12.68)	(-0.02)	(2.1)	(-1.73)	(1.61)	(1.98)	(2.53)	(3.61)	(0.78)	
Q2	-0.08	-0.14	0.02	-0.02	0.00	0.18	-0.03	0.63	0.42	-3.07	0.18
	(-1.93)	(-8.74)	(1.15)	(-0.38)	(1.71)	(2.89)	(-0.37)	(2.24)	(1.77)	(-2.99)	
Q3	-0.04	-0.10	-0.01	0.02	0.00	0.10	-0.11	0.28	0.29	-33.02	0.14
	(-1.35)	(-6.12)	(-0.63)	(0.28)	(1.49)	(1.62)	(-1.37)	(1.67)	(1.58)	(-3.53)	
Q4	-0.02	-0.05	0.02	0.07	0.00	-0.04	-0.21	-0.09	-0.16	-75.52	0.10
	(-0.42)	(-5.19)	(1.43)	(0.82)	(0.56)	(-0.71)	(-1.99)	(-0.37)	(-1.23)	(-2.49)	
Q5	0.02	-0.03	-0.05	0.06	0.00	-0.25	-0.15	-0.31	0.22	-140.53	0.08
	(2.69)	(-4.41)	(-3.79)	(1.02)	(-3.09)	(-5.41)	(-2.82)	(-1.75)	(1.38)	(-1.59)	

Notes: This table shows the regression outputs for portfolios formed according to different $SIZE$ quintiles. At each month, we sort overnight jump stocks in ascending order based on the $SIZE$ figures and form quintile portfolios. Then, we run monthly cross-sectional regressions and save the time series of coefficient estimates. The dependent variable is the cumulated $1D_t$ returns (the intraday returns just after the overnight jumps). To account for cross-correlations and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. CJR_t^+ and CJR_t^- are respectively the monthly cumulated positive and negative jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_t are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables. Estimated model: $CDR_{i,t,d+1} = \alpha + \beta_1 CJR_{i,t,d=0} + \beta_2 PREV_{i,t,d=-1} + \beta_3 IVOL_OJI_{i,t} + \beta_4 SIZE_i + \beta_5 BM_i + \beta_6 LEV_i + \beta_7 RET_{i,t-1,t-5} + \beta_8 RET_{i,t-6,t-11} + \beta_9 AI_{i,t} + \varepsilon_{i,t}$

significance is observed in $Q5$ with a (-5.29) t-statistic and (-0.51) coefficient. For the positive jumps, CJR is highly significant for all quintiles, but the highest statistical significance is obtained in $Q3$. For economic significance, the last two stock groups with the highest $IVOL_OJI$ numbers ($Q4$ and $Q5$) have the highest estimate, whereas the $Q1$ has the lowest economic and statistical significance.

Another crucial firm-specific factor is the $SIZE$, and regression results for different quintiles are reported in Table 15. Figures show that CJR is highly significant in all quintiles for both negative and positive overnight jumps. Economic and statistical significance is higher in $Q1$ after both negative and positive overnight shocks. Figure 6 presents the statistical significance levels

of CJR for the quintile portfolios of all other factors on the first day. CJR is significant in all quintiles after both positive and negative overnight jumps.

We repeat the same procedure to evaluate how the predictive power changes across all quintile portfolios after one day, and report the corresponding CJR significance in Figure 7. Two points are worth noting. First, the results indicate that the predictive power generally remains significant across all days, but the t-statistics show a downward trend in magnitude as the time horizon expands. Second, for some stock quintiles formed according to $SIZE$ and AI factors, the predictive power disappears and eventually reverses in sign, suggesting that the initial overreaction evolves into a post-jump drift over time. Together with the discussions in Section 4.6, the declining t-statistics and the emergence of drift after a few days in some stocks support our argument that the overreaction to overnight information shocks is a short-term market phenomenon.

To check the robustness of these predictable patterns in isolated approach, we repeat our analysis with stocks with at least two years of trading history and eliminate the overnight jumps if the CRSP closing price is calculated as the average of bid and ask prices. We maintain the predictability for all quintile portfolios after negative jumps and positive jumps, but for $Q1$ after negative overnight jumps when stocks are sorted according to their AI magnitudes in ascending order as well as $Q3$ for $IVOL_OJI$ and $Q5$ for $SIZE$. All t-statistic values are depicted in Figure B1 and regression results when quintile portfolios are formed according to $IVOL_OJI$ and $SIZE$ factors.

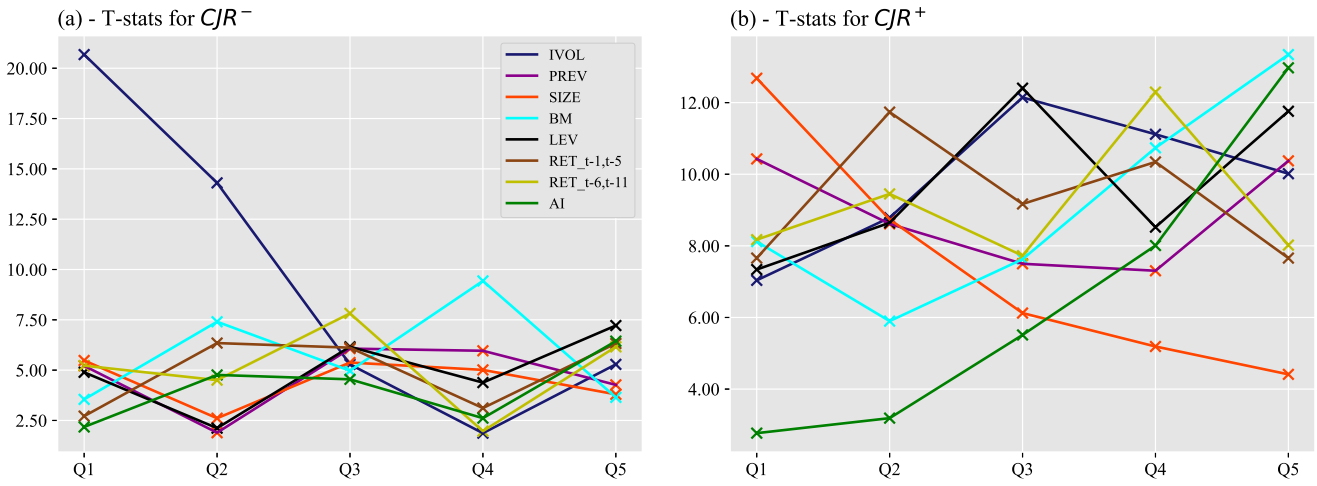
All in all, our findings show that cross-sectional return predictability around these short-window events is explained by our jump return factors that proxy information shocks and that predictability changes across different stock groupings.

4.6 Trading Strategies

Investors are implementing dynamic trading strategies with various expectations for the future. In our case, we check if a trading strategy based on jump classification can generate risk-adjusted returns or end up in losses.

We first construct value-weighted daily event portfolios and examine the contrarian trading strategy across different holding periods following jump incidents. On each trading day, we identify stocks with negative and positive overnight jumps, form value-weighted portfolios, and implement a long-short trading strategy taking long positions in negative jump stocks and short

Figure 6: CJR Significance in Quintile Portfolios



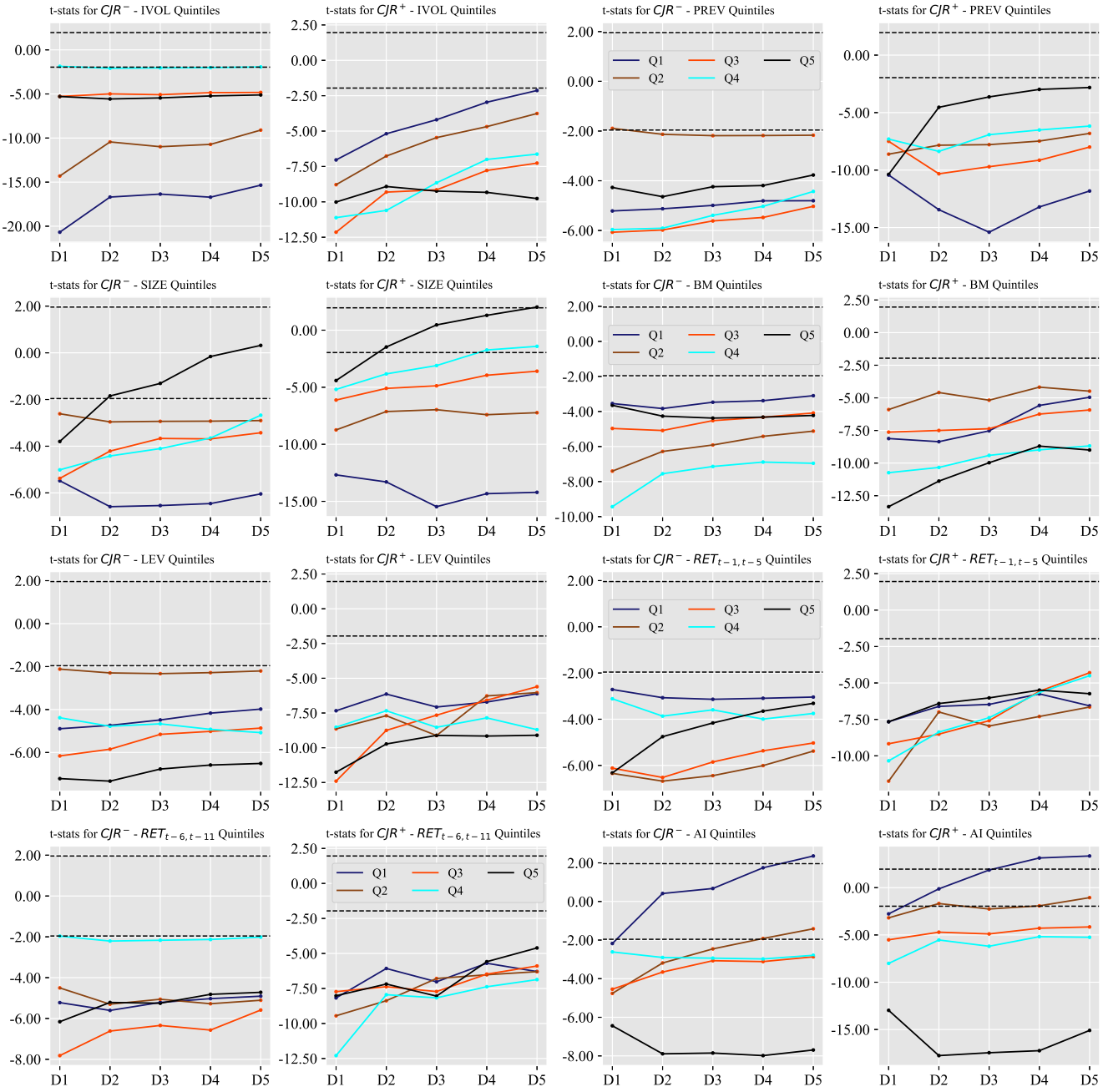
Notes: This graph shows the Newey-West adjusted t-statistics for CJR in 1-day predictability regressions. Each month, we sort overnight jump stocks in ascending order based on the levels of control variables and form quintile portfolios. Then, we run monthly cross-sectional regressions and save the time series of coefficient estimates. To account for cross-correlations and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). We calculate the Newey-West t-statistics with 12 lags for the time series of coefficient estimates. CJR_t^+ and CJR_t^- are respectively the monthly cumulated positive and negative jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), and AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month.

positions in positive jump stocks. We then calculate the value-weighted cumulative returns over the corresponding horizons. This contrarian strategy generates highly significant positive abnormal returns at short horizons. For the full sample period, the risk-adjusted return becomes 1.05%, with a Newey-West t-statistic of 11.38 (Table 16) for the first day. Consistent with the transient nature of the effect, both the magnitude of abnormal returns and the associated t-statistics decline as the holding period lengthens.

We further examine this trading strategy across different subperiods. As discussed in Section 4.3, the January 2004 to December 2021 period better reflects modern trading conditions, and yields higher risk-adjusted returns for the contrarian strategy. For the earlier period, July 1993–December 2003, the strategy still generates highly significant positive abnormal returns that generally diminish as the holding period increases; however, the risk-adjusted returns are consistently lower across all holdings horizons compared to the January 2004-December 2021 period. All results are reported in Table 16.

We repeat our analysis with monthly trading strategies for all overnight jump stocks in a

Figure 7: CJR Significance in Quintile Portfolios for Five Days



Notes: This graph shows the Newey-West adjusted t-statistics for CJR in predictability regressions for five days (D1, D2, D3, D4, D5). Each month, we sort overnight jump stocks in ascending order based on the levels of control variables and form quintile portfolios. Then, we run monthly cross-sectional regressions and save the time series of coefficient estimates. To account for cross-correlations and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). We calculate the Newey-West t-statistics with 12 lags for the time series of coefficient estimates. Dashed black horizontal lines correspond to 5% significance level. CJR_t^+ and CJR_t^- are respectively the monthly cumulated positive and negative jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), and AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month. Legends are placed in one subplot for each row. Subplots with missing legends have the same legends as the displayed ones.

Table 16: Short-term Contrarian Trading Strategy

	1-Day Holding	2-Day Holding	3-Day Holding	4-Day Holding	5-Day Holding
Jul.93 - Dec.21					
FF3 alpha	1.05%	0.93%	0.84%	0.79%	0.74%
t-stat	(11.38)	(9.91)	(8.62)	(8.06)	(6.79)
Jul.93 - Dec.03					
FF3 alpha	0.74%	0.65%	0.51%	0.37%	0.27%
t-stat	(11.56)	(7.69)	(5.85)	(3.87)	(2.70)
Jan.04 - Dec.21					
FF3 alpha	1.24%	1.10%	1.01%	1.02%	1.00%
t-stat	(8.73)	(7.80)	(7.13)	(7.17)	(6.20)

Notes: This table shows the results of short-term contrarian trading strategies. On each trading day, we identify stocks with negative and positive overnight jumps, form value-weighted daily event portfolios, and implement a contrarian trading strategy that takes long positions in negative jump stocks and short positions in positive jump stocks. We calculate the value-weighted cumulative returns over the corresponding horizons. Table reports FF3 alpha estimates with Newey–West t-statistics (12 lags) to account for autocorrelation. Results are presented for the full sample period and for two key subperiods that capture differing market conditions.

given month and derive the results with an iterative process. At the end of each month, we first calculate the cumulative overnight jump returns of stocks and sort them in ascending order according to these returns. Sorted stocks are split into deciles, with $D1$ having the lowest return and $D10$ with the highest return. Afterward, we calculate value-weighted portfolio returns for a one-month investment horizon distinctively for each decile.

The primary purpose of the trading strategy analysis is to demonstrate that the documented overreaction and subsequent return reversal are short-term phenomena that operate over a horizon of only a few days. Rather than only showing that a contrarian strategy following jump days generates positive abnormal returns at very short horizons, we also focus on demonstrating that a contrarian strategy implemented at the monthly frequency does not yield positive abnormal returns. Consistent with these interpretations, Figure 7 shows that the short-term negative return association weakens as time passes and, for some quintile portfolios, even turns into a positive association after a few days. The presence of loss at the one-month horizon should be viewed as complementary evidence, rather than a standalone test, and is informative only when interpreted together with the short-horizon positive abnormal returns and reversal patterns documented in the empirical findings.

As tabulated in Table 17 for the monthly horizon, a contrarian trading strategy for the stocks

Table 17: Monthly Trading Strategies Based on Cumulative Jump Classification

PANEL A - Long Strategy										
Decile	**D1**	**D2**	**D3**	**D4**	**D5**	**D6**	**D7**	**D8**	**D9**	**D10**
Raw Return	0.70%	0.90%	0.88%	1.13%	1.27%	0.99%	0.91%	0.90%	1.30%	1.34%
FF4 alpha	-0.003	-0.001	-0.001	0.002	0.005	0.001	0.000	0.000	0.003	0.003
t-stat	(-1.14)	(-0.18)	(-0.5)	(1.12)	(2.18)	(0.87)	(-0.07)	(-0.01)	(0.82)	(1.27)
PANEL B - Long & Short Strategy FF4 Abnormal Returns										
Decile	**-D1**	**-D2**	**-D3**	**-D4**	**-D5**	**-D6**	**-D7**	**-D8**	**-D9**	**-D10**
D1	-	-0.003	-0.002	-0.005	-0.008	-0.004	-0.003	-0.003	-0.006	-0.006
	-	(-0.72)	(-0.57)	(-1.62)	(-2.1)	(-1.37)	(-0.89)	(-0.97)	(-1.48)	(-2.03)

Notes: At the end of each month, we sort jump stocks according to their monthly cumulative overnight jump returns in ascending order, where $D1$ is the first decile with the lowest returns and $D10$ is the last decile with the highest returns. We form value-weighted portfolios for each decile with a one-month investment horizon (1M). This procedure is repeated every month, and the means of the portfolio returns are recorded continuously. We implement long and short trading strategies for each decile along with a long/short strategy among $D1$, $D5$, and $D10$ decile portfolios. Raw returns are the mean value of portfolio returns over the analysis period. The Table mainly reports FF4 alphas of trading strategies and Newey-West t-statistics with 12 lags. Our analyses cover 342 months over the period July 1993 - December 2021.

with the lowest negative overnight jump returns incurs -0.3% abnormal return with a Newey-West t-statistic of -1.14. However, a relative strength strategy for positive overnight jump stocks in $D10$ results in a 0.3% abnormal return with again an insignificant Newey-West t-statistic of 1.27. A frictionless long-short trading strategy that longs $D1$ and shorts $D10$ portfolios ends up -0.64% of abnormal return with a significant -2.03 Newey-West t-statistics. This combined trading strategy incurs a 0.8% abnormal loss with a significant t-statistic of -2.10 if we instead use $D1$ and $D5$ portfolios. These results cumulatively tell us that the overreaction-reversal pattern is a short-lived market episode. Betting on a one-month reversal pattern and constructing a contrarian trading strategy with extreme deciles will lead to risk-adjusted loss. The overreaction-reversal pattern in the wake of overnight information shocks morphs into drifting returns when the next-month investment portfolios are considered.

The one-to-five-day event-portfolio strategies, as well as other trading strategies used to estimate risk-adjusted returns rely on certain assumptions such as the absence of transaction costs and commissions, no short-selling constraints, and sufficient market liquidity to absorb the relevant order flow without generating slippage costs. The event-portfolio strategies also assume that the traders are well-equipped to accurately flag the sharp movements as overnight price discontinuities, execute trades promptly at the prices at which jumps are detected, and unwind the position at the closing price at the end of the respective holding horizon. In practice, these

assumptions may not hold entirely, and due to these market frictions, the documented FF3 alphas may not be fully attainable through live trading strategies. They represent an upper bound on the risk-adjusted returns achievable in practice. In that regard, this is analogous to the limits-to-arbitrage arguments of Shleifer and Vishny (1997) where the authors argue that the identification of an apparent riskless profit does not guarantee its full practical realization due to various implementation limits.

5 Theoretical Foundations and Implications

Let $Y_t = ln(X_t)$ in time dimension $t \geq 0$ and let the triple $(\Omega, \mathcal{A}_t, \mathcal{P})$ represent the probability space for which Ω is the possibly observable outcome space, $\mathcal{A}_t$ represents the σ-algebra associated with the subsets of Ω and $\mathcal{P}$ is the measure assigning probabilities on $\mathcal{A}_t$. Let also $\mathcal{F}_t$ be the information filtration as sub-σ-algebra on $\mathcal{A}_t$ generated by a random process. In Equation 1, we explained the jump-diffusion process as $\int_0^t dX_t = \int_0^t \mu_s d_s + \int_0^t \sigma_s dB_s + \sum_{k=1}^{N_j^t} J_i \quad ; \quad \forall t \in [0, T]$.

A general form of a diffusion process for the price of a stock with drift and volatility terms can be expressed as in Equation 12:

$$dX_t = \mu(X_t, t, ...)dt + \sigma(X_t, t, ...)dB_t \tag{12}$$

where the drift term $\mu(X_t, t, ...) = \Upsilon$ and the diffusive volatility term $\sigma(X_t, t, ...) = \eta$ can also be functions of price levels, time, and any other variables that govern the stochastic price process. Notation-wise, a geometric Brownian motion with discontinuity adjustments for information shocks can be expressed as in Equation 13.

$$dX_t / X_t = \mu dt + \sigma dB_t + J dN_t \tag{13}$$

where J is the jump size and N_t is the counting process for randomly arriving jump incidences. In the absence of jumps, Equation 13 collapses to $dX_t = \mu X_t dt + \sigma X_t dB_t$. We create an overreaction/underreaction term to decompose the jump returns and isolate the rational component of price movement following information shocks. With surprising information flow, we let Λ_t be the rational part of price movement and $\lambda_t = J_t - \Lambda_t$ be the movement due to psychological biases and both of these components are assumed to be orthogonal to dB_t with λ_t being a function of unobservable factors embedded all in Θ_t. If $\lambda_t = 0$, $\forall t \in [0, T]$, we conclude that all

available information (including shocks) is rationally processed, leading to accurate pricing. On the other hand, $\lambda_t > 0$, $\forall \Lambda_t > 0$ or $\lambda_t < 0$, $\forall \Lambda_t < 0$ means that market participants are prone to psychological biases and they overreact to information shocks whereas $\lambda_t < 0$, $\forall \Lambda_t > 0$ or $\lambda_t > 0$, $\forall \Lambda_t < 0$ implies that investors underreact to information shocks and create momentum in prices. Formally,

$$J_t \quad \text{level is flagged as} \quad \begin{cases} \textit{Overreaction}, & \text{if } \lambda_t > 0 \quad \text{and} \quad \Lambda_t > 0, \forall t \in [0,T] \\ \textit{Overreaction}, & \text{if } \lambda_t < 0 \quad \text{and} \quad \Lambda_t < 0, \forall t \in [0,T] \\ \textit{Rational} & \text{if } \lambda_t = 0, \forall t \in [0,T] \\ \textit{Underreaction}, & \text{if } \lambda_t > 0 \quad \text{and} \quad \Lambda_t < 0, \forall t \in [0,T] \\ \textit{Underreaction}, & \text{if } \lambda_t < 0 \quad \text{and} \quad \Lambda_t > 0, \forall t \in [0,T] \end{cases} \tag{14}$$

A geometric Brownian model with this sort of decomposition will be

$$\frac{dX_t}{X_t} = \mu dt + \sigma dB_t + \Lambda dN_t + \lambda dN_t \tag{15}$$

in which the $dX_t^{rational}/X_t^{rational} = \mu d_t + \sigma dB_t + \Lambda dN_t$ governs rational component whereas $dX_t^{biased}/X_t^{biased} = \lambda dN_t$ represents the deviation from the rational price path as separate diffusion processes. $\forall t \in [0,T]$, $X_t = X_t^{rational} + X_t^{biased}$ and $Y_t = ln(X_t^{rational} + X_t^{biased})$. Applying Itô's Lemma[18] to $Y_t = f(X_t^{rational}, X_t^{biased})$=$ln(X_t^{rational} + X_t^{biased})$ we get,

$$\begin{aligned} dY_t = (\mu - \frac{\sigma^2}{2} - \mu\Lambda_t dN_t - \mu\lambda_t dN_t)dt \\ +(\sigma - \sigma\Lambda_t dN_t - \sigma\lambda_t dN_t)dB_t \\ +(\Lambda_t + \lambda_t - \frac{{\Lambda_t}^2}{2} - \frac{{\lambda_t}^2}{2} - \Lambda_t\lambda_t)dN_t \end{aligned} \tag{16}$$

Equation 16 outlines the process that the log-prices follow, and it morphs into the widely-known form $dY_t = (\mu - \frac{\sigma^2}{2})dt + \sigma dB_t$ in the absence of jumps. It has useful interpretations. First, positive jumps ($\Lambda_t + \lambda_t > 0$) bring in additional downward drift in returns just like the diffusive volatility term. On the other hand, negative jumps ($\Lambda_t + \lambda_t < 0$) make upward adjustments in the drift term. Second, in the case of overreaction, λ_t amplifies the size of the drift adjustment whereas underreaction implies that λ_t drags down the size of the drift adjustment due to Λ_t and λ_t having opposite signs and leading to $\mid J_t \mid < \mid \Lambda_t \mid$. Similar inferences can be made for the effects in diffusive volatility term. Third, jump components wind up in the volatility term with

[18]See Appendix A for the derivation

the opposite sign meaning, for instance that negative jumps intensify the variance of stocks. That negative association is in line with Bandi and Reno (2016) which document that large negative price discontinuities generally couple with positive discontinuities in variance. Volatility is not necessarily bad in the context of asset pricing as long as its surge serves for rational pricing, as Fama (1989) stated after the market crash in October 1987[19]. In the context of the present study, we emphasize the role of the irrational jump component. The degree of reversal or the momentum magnitudes in follow-up returns - overreaction and underreaction - will be functions of that component λ_t. Broadly, mispricing-driven return predictability (see McLean and Pontiff (2016)) documented in the preceding chapter and its economic significance rest respectively on $\lambda_t \neq 0$ and γ, which determines the degree of reversal.

Equation 16 shows that the drift adjustment terms $\mu\Lambda_t dN_t$ and $\mu\lambda_t dN_t$ explicitly include the drift parameter μ (the same logic applies to the volatility adjustment term). This implies that the corrective drift is not a constant value but is proportional to the stock's underlying expected return or drift magnitude. If a stock has a higher μ, any disruption to its equilibrium price path (a jump rather than a large realization of the Brownian motion) results in a larger nominal adjustment to bring the process back to its mean. In other words, drift and volatility levels appear as scaling factors in corresponding adjustments. Thus, if a stock has inherently high levels of drift and volatility terms, their adjustment will be higher and vice versa. In a regular empirical asset pricing setting, small-cap stocks generally have higher expected returns to compensate for the risks ($\mu_{small-cap} > \mu_{large-cap}$). Thus, for small stocks, drift term adjustments will be higher for a given level of jump. For instance, in case of a positive jump, the bias term λ_t will reduce the drift term much more than it would do for large-cap stocks, lowering the expected log-return further. Empirical findings based on size factor in Table 15 are in line with this intuition; the negative loadings on jump returns are larger in magnitude for small-cap stocks across both positive and negative market episodes. In other words, a small-cap stock is expected to have a substantially lower return relative to large-cap stocks in the next period following a positive jump, while a negative jump is expected to be followed by a markedly higher subsequent return. The model essentially predicts that smaller stocks will overreact more intensely and, consequently, require a larger drift adjustment to correct the pricing.

[19]Eugene Fama highlights that "rational prices are not necessarily less volatile prices, and less volatile prices are not necessarily better than more volatile prices. The appropriate view of the October 1987 price shock depends critically on whether it was a rational response to changes in fundamental values"

Da et al. (2014) construct a measure for information discreteness (ID) and argue that discrete information tapers momentum dynamics whereas continuous information leads to investor inattention and drives return continuation. In their study, the proxy of information discreteness is quantified with ID=sgn(PRET)x[%neg - %pos], where PRET is the cumulative returns over the past twelve months, skipping the most recent month, and the %neg (%pos) is the percent of days with negative (positive) returns within this period, regardless of the return magnitudes. For example, if PRET is positive (meaning that sgn(PRET)=+1) but the number of days with negative returns is higher than the number of days with positive returns, the positiveness of PRET can be achieved by large discrete positive returns amid these negative-return days. Thus, information discreteness is assessed with a very large horizon of returns in regard to momentum dynamics. Authors argue that the positive coefficient of ID in their Fama-MacBeth regressions indicates a premium for jump risk or skewness. On the other hand, our study adopts an event-based approach in the short-term and analyzes returns around these information shocks in the light of behavioral dynamics. Along with that, Kamara et al. (2016) uncover that some of the risk factors are not priced in certain horizons but become a significant risk factor when the investment horizon changes. The quintile regression analyses in certain stock groupings displayed in Figure 7 indicate that the predictability weakens as the horizon of follow-up return expands. In other words, if the horizon of the investor is short-sided, the implications of a negative coefficient may be instrumental for investment planning, whereas longer horizons may require different strategies. Taken together, we argue that the reported return predictability is a short-lived mispricing that may not be reconciled with the persistent risk-premium interpretation.

An interesting question relates to the role of after-hours trading and how it may align with the implications of our findings. After-hours trading is typically associated with very low liquidity, very large bid-ask spreads and various order restrictions of Electronic Communication Networks. Barclay and Hendershott (2003) report that trading costs and information asymmetry are high in that period. Jiang et al. (2012) further argues that trading costs are four to five times larger compared to the regular trading hours and retail investors are discouraged from market participation during these hours due to the associated risks. In that regard, authors discuss that this thin trading period will be dominated by the professional and quasi-professional traders. In a jump event day, after-hours volume may be higher than the one during normal times, but these essential features of after-hours period restrict its representativeness for drawing reliable

and clear economic inferences. That makes the opening price the primary channel through which overnight information is impounded into prices. After all, although volumes are very thin, the direction of the price in after-hours period and opening auction is highly likely to be the same around these information shocks implying the possibility that our findings would have been even stronger if the after-hours trading would have been absent. That said, we believe that once after-hours period becomes more representative of the regular trading hours in terms of costs, market participation and volumes, its role in shaping overnight return dynamics is likely to become more pronounced.

Our study also has some implications for our understanding of market efficiency and for practitioners, especially the active portfolio managers, who look around for some insight into the future. First, there is still this ongoing debate on the concept to which the return predictability should be attributed. Is this concept the risk premium that is associated with some factors, or is it investors' behavioral biases flawing rationality? The present study contributes to cross-sectional return predictability literature by elaborating on overnight information shocks, which come about in the form of overnight price jumps and on subsequent reversals. Fama (1991) states that market efficiency is not testable because of the joint-hypothesis problem (it must be tested with a sound market equilibrium model) and the only testable thing is whether the information is reflected in prices *properly* or not. In order to claim market inefficiency, one should be sure that their model is not a bad model. In that regard, our findings and assertions may also be criticized. However, as widely documented, jumps are rare events and they come in as shocks in very short time periods. As clarified in Jiang and Yao (2013), large price movements around these tiny windows are due to information shocks and barely linked to risk premiums. Following this intuition, our study can also be classified as a short-window event study, just like the ones elaborating on return dynamics around earnings announcements, the literature on flash crashes that bounce back in a very short time, or other similar studies in the same spirit. To say the least, we cannot claim market inefficiency, but we can argue that the overnight information that surprises the market is not *properly* priced.

Second, we are curious if the reported return predictability will decay after the findings are published. If our reported return predictability is grounded on rational expectations and is a reflection of risk in the market, we can then expect this overreaction mechanism to persist as discussed in McLean and Pontiff (2016). If, on the other hand, this pattern is due to mispricing,

savvy investors can exploit this trend and then alleviate it in time. McLean and Pontiff (2016) documents a thorough analysis for 97 variables with cross-sectional predictive power, and the authors estimate 32% lower return after market participants become informed about the results of these publications.[20] Regarding this issue, we conjecture that this overreaction incidence will stay in the market mainly for two reasons. The first one is related to the heterogeneously clustered investor groups along the day. As documented in Lou et al. (2019), there is a persistent interplay between individual and institutional/professional investors. Opening-hour orders are dominated by individual investors, although the latter heavily trades in the second part of the day. This is actually in line with the settled market saying: *"The novice open the market and masters close it"*. Hence, unless the trading dynamics of these two groups converge with each other, we can expect this clientele effect to make this overreaction pattern perennial. Our second reasoning is linked to behavioral biases. It is a well-documented psychological fact that people overreact to information shocks and market anomalies are stronger when attention is high as argued in Nguyen and Pham (2021). They can either make their decisions based on the worst-case scenario amid uncertainty and risky conditions or become overoptimistic and credulous when confronted with positive news. Overall, we expect this return pattern to be persistent and open to exploitation by astute market participants who are free of psychological biases and vigilant for these opportunities. To be clear, our guess of the long-lasting nature of this trend around overnight shocks is not tied to the risk premium concept but rather to the competing and behavioral forces of different clientele that are dominant in different parts of a specific trading day.

The reversal trend after overnight information shocks may also matter for the discussions of *uncertainty resolution*. We document a bidirectional negative association between overnight jumps and future returns, and lower future realized returns may be regarded as compensation for reduced uncertainty. That said, associating the reversal trend with *uncertainty resolution* is challenging for a few reasons. First, if we were to measure stock-specific uncertainty from the dispersion in analyst coverage, the scope would be very limited, as those reports are mainly confined to the firms that are big and in the spotlight. If one were to calculate firm-specific ambiguity through the method used in Brenner and Izhakian (2018), it would not be encompassing due to thinly traded stocks in the intraday section and potentially produce misleading results. Measuring the market-specific uncertainty from a representative index and taking it as a global

[20] Academic research can also lead to new anomalies. See Horenstein (2021) for a detailed discussion on CAPM.

benchmark may also lead to misalignment for stocks without index membership. One possible way would be grouping stocks based on a factor, finding the highly correlated market index for each segment, and then applying Brenner and Izhakian (2018). We believe that this is a promising area for future research.

6 Conclusion

We first identify overnight information shocks and short-term return reversal and construct a factor (JR / CJR) that predicts subsequent returns for certain set-ups and stocks. Untouched to date, we contribute to short-term overreaction and return predictability discussions by specifically focusing on overnight-jump stocks. The initial price response to unexpected overnight information flows is excessive, and a part of that movement reverses in the subsequent trading periods. JR and CJR predict follow-up returns with statistical and economic significance for the whole data sample period and subperiods. It is also statistically significant for all quintile portfolios formed according to the different levels of all factors, except for a few portfolios in isolated and cumulative approaches. We construct contrarian trading strategies for short-term holding horizons (one to five days) as well as a one-month investment horizon to see if stocks with overnight positive jumps (winners) will experience relatively lower returns (losers) and vice versa. In the former trading strategy, daily-event portfolios generate highly significant positive abnormal returns (1.05%) on the first day with declining magnitudes as the holding horizon expands. For the monthly strategy, stocks are sorted according to their lagged cumulative monthly jump returns, and results of the long/short strategy with extreme decile portfolios show that this bet will induce a statistically significant loss (0.6%) rather than a profit.

Classifying the full universe of jump incidences into firm-specific and market-wide informational events and conducting separate analyses may help uncover additional dimensions of the overreaction mechanism. Our findings reflect the average market response to these information events, as the primary focus of this study is to document short-term reversal and associated return predictability. In spite of the constructed proxy measure, distinguishing between different types of informational events remains a limitation of the present study, and addressing this issue represents a promising avenue for future research and may yield further insights into the underlying dynamics. In addition, the level of idiosyncratic and market uncertainty surrounding overnight jumps and their association with short-term reversal mechanisms have not been ana-

lyzed. Dedicated research will hopefully enhance our knowledge of investors' decision-making around information shocks and uncertainty.

Appendix A — Diffusion Model

For the derivation of Equation 16, let first $f(p,q)$ be a multivariate function with variables p and q. Itô's Lemma for this function becomes

$$df = \frac{\partial f}{\partial p}dp + \frac{\partial f}{\partial q}dq + \frac{1}{2}\left[\frac{\partial^2 f}{\partial p^2}dp^2 + 2\frac{\partial^2 f}{\partial p \partial q}dpdq + \frac{\partial^2 f}{\partial q^2}dq^2\right] \tag{A1}$$

Similarly, let Y_t be the log of observed prices X_t which is a function of two different components $X_t^{rational}$ and X_t^{biased}: $Y_t = f(X_t^{rational}, X_t^{biased}) = ln(X_t^{rational} + X_t^{biased})$. Then,

$$\begin{aligned} dYt = \frac{\partial f}{\partial X_t^{rational}}dX_t^{rational} + \frac{\partial f}{\partial X_t^{biased}}dX_t^{biased} + \frac{1}{2}\Bigg[\frac{\partial^2 f}{\partial X_t^{rational2}}dX_t^{rational2} \\ +2\frac{\partial^2 f}{\partial X_t^{rational}\partial X_t^{biased}}dX_t^{rational}dX_t^{biased} + \frac{\partial^2 f}{\partial X_t^{biased2}}dX_t^{biased2}\Bigg] \end{aligned} \tag{A2}$$

Applying the first and second partial derivatives on $f(X_t^{rational}, X_t^{biased}) = ln(X_t^{rational} + X_t^{biased})$ yields

$$\begin{aligned} dYt = \frac{1}{X_t^{rational} + X_t^{biased}}dX_t^{rational} + \frac{1}{X_t^{rational} + X_t^{biased}}dX_t^{biased} \\ -\frac{1}{2}\Bigg[\frac{1}{(X_t^{rational} + X_t^{biased})^2}dX_t^{rational2} + 2\frac{1}{(X_t^{rational} + X_t^{biased})^2}dX_t^{rational}dX_t^{biased} \\ +\frac{1}{(X_t^{rational} + X_t^{biased})^2}dX_t^{biased2}\Bigg] \end{aligned} \tag{A3}$$

Remember that $\frac{dX_t}{X_t} = \mu dt + \sigma dB_t + \Lambda dN_t + \lambda dN_t$, $X_t = X_t^{rational} + X_t^{biased}$ and $dX_t = dX_t^{rational} + dX_t^{biased}$. If we let $\Psi = \frac{dX_t}{X_t}$ for brevity, Equation A3 becomes

$$\begin{aligned} dYt = \frac{\Psi}{dX_t}dX_t^{rational} + \frac{\Psi}{dX_t}dX_t^{biased} - \frac{1}{2}\Bigg[\frac{\Psi^2}{dX_t^2}dX_t^{rational2} + 2\frac{\Psi^2}{dX_t^2}dX_t^{rational}dX_t^{biased} \\ +\frac{\Psi^2}{dX_t^2}dX_t^{biased2}\Bigg] = \Psi\left[\frac{dX_t^{rational} + dX_t^{biased}}{dX_t}\right] - \frac{1}{2}\left[\frac{\Psi}{dX_t}dX_t^{rational} + \frac{\Psi}{dX_t}dX_t^{biased}\right]^2 \\ = \Psi - \frac{1}{2}\Psi^2 \end{aligned} \tag{A4}$$

Ψ^2 term can be formulated as

$$\begin{aligned}\Psi^2 = \mu^2 dt^2 + (2\mu dt + \sigma dB_t)\sigma dB_t + (2\mu dt + 2\sigma dB_t + \Lambda dN_t)\Lambda dN_t \\ +(2\mu dt + 2\sigma dB_t + 2\Lambda dN_t + \lambda dN_t)\lambda dN_t\end{aligned} \tag{A5}$$

where the components $\mu^2 dt^2$ and $2\mu dt\sigma dB_t$ will be zero as they contain the higher orders of dt. We also replace the term $\sigma^2 dB_t^2$ with $\sigma^2 dt$ as an adjustment for quadratic variation ($dB_t^2 = dt$). Hence, the diffusion process for the log of prices becomes

$$\begin{aligned}dY_t = \Psi - \frac{1}{2}\Psi^2 = (\mu - \frac{\sigma^2}{2} - \mu\Lambda dN_t - \mu\lambda dN_t)dt \\ +(\sigma - \sigma\Lambda dN_t - \sigma\lambda dN_t)dB_t \\ +\Lambda dN_t \\ +\lambda dN_t \\ -(\frac{\Lambda^2}{2} + \frac{\lambda^2}{2} + \Lambda\lambda)dN_t^2\end{aligned} \tag{A6}$$

where $dN_t^2 = dN_t \ \forall t \in [0, T]$, since the number of jumps at any time will either be 0 or 1. Then,

$$\begin{aligned}dY_t = (\mu - \frac{\sigma^2}{2} - \mu\Lambda dN_t - \mu\lambda dN_t)dt \\ +(\sigma - \sigma\Lambda dN_t - \sigma\lambda dN_t)dB_t \\ +(\Lambda + \lambda - \frac{\Lambda^2}{2} - \frac{\lambda^2}{2} - \Lambda\lambda)dN_t\end{aligned} \tag{A7}$$

Appendix B — Robustness Checks

We implement different layers of robustness checks by changing the minimum required length for trading history, by eliminating overnight jump days if they coincide with a CRSP closing price calculated as the average of bid and ask prices, and by changing the regressor for idiosyncratic risk ($IVOL_OJI$) to either $IVOL_DJI$ or to $IVOL_ODJI$. The type of robustness check is detailed for each table.

Table B1 documents the descriptive statistics for jumps derived by keeping the minimum length of trading history at 3 years. This time, we exclude the jumps if the overnight jumps come across with a CRSP closing price that is set to the bid-ask average. % of opposite and the same sign returns in the intraday period is quite similar for positive overnight jumps. % of opposite jumps in the intraday section is even magnified after negative jumps.

Table B2 documents the descriptive statistics and the correlations of the regression variables in our analyses of overnight negative jumps. CJR_t^- variable is constructed for the stocks with at least 2 years of trading history. We also exclude the jumps if the CRSP closing price before the overnight jump day is set to the bid-ask average. We also report the figures for $IVOL_OJI_t$ that is calculated in isolation with overnight jump days.

Table B3 documents the descriptive statistics and the correlations of the regression variables in our analyses of overnight positive jumps. CJR_t^+ variable is constructed for the stocks with at least 2 years of trading history. We also exclude the jumps if the CRSP closing price before the overnight jump day is set to the bid-ask average. We also report the figures for $IVOL_OJI_t$ that is calculated in isolation with overnight jump days.

Table B4 documents return predictability for different model specifications. We still use the $IVOL_OJI_t$ variable in our regression analysis but shorten the minimum trading history requirement to 2 years.

Table B5 documents return predictability results up to 5 days. We still use the $IVOL_OJI_t$ variable in our regression analysis but shorten the minimum trading history requirement to 2 years.

Table B6 documents the return predictability results in different periods. We still use the $IVOL_OJI_t$ variable in our regression analysis but shorten the minimum trading history requirement to 2 years. The subperiods are July 1993 - December 2003, January 2004 - December 2021,

and August 2007 - December 2013.

Table B7 documents return predictability results for different quintiles of $IVOL_OJI_t$ variable, but we shorten the minimum trading history requirement to 2 years and eliminate overnight jumps if the corresponding CRSP closing price is set to bid-ask average.

Table B8 documents return predictability results for different quintiles of the $SIZE$ variable, but we shorten the minimum trading history requirement to 2 years and eliminate overnight jumps if the corresponding CRSP closing price is set to bid-ask average.

Figure B1 depicts the significance of JR for different quintile portfolios of factors. We shorten the minimum trading history requirement to 2 years and eliminate overnight jumps if the corresponding CRSP closing price is set to the bid-ask average.

Table B9 documents the descriptive statistics and the correlations of the regression variables in our analyses of overnight negative jumps. CJR_t^- variable is constructed for the stocks with at least 2 years of trading history. We also exclude the jumps if the CRSP closing price before the overnight jump day is set to the bid-ask average. We also report the figures for $IVOL_DJI_t$ that is calculated in isolation with daily jump days.

Table B10 documents the descriptive statistics and the correlations of the regression variables in our analyses of overnight positive jumps. CJR_t^+ variable is constructed for the stocks with at least 2 years of trading history. We also exclude the jumps if the CRSP closing price before the overnight jump day is set to the bid-ask average. We also report the figures for $IVOL_DJI_t$ that is calculated in isolation with daily jump days.

Table B11 documents return predictability for different model specifications. In our regression analyses, we use the $IVOL_DJI_t$ variable calculated in isolation with daily jump days and shorten the minimum trading history requirement to 2 years.

Table B12 documents return predictability results up to 5 days. In our regression analyses, we use the $IVOL_DJI_t$ variable calculated in isolation with daily jump days and shorten the minimum trading history requirement to 2 years.

In Table B13, we document return predictability for different model specifications. In our regression analyses, we use the $IVOL_ODJI_t$ variable calculated in isolation with daily and overnight jump days and shorten the minimum trading history requirement to 2 years.

Table B14 documents return predictability results up to 5 days. In our regression analyses, we use the $IVOL_ODJI_t$ variable calculated in isolation with daily and overnight jump days and

shorten the minimum trading history requirement to 2 years.

Table B1: Descriptive Statistics for Isolated Jumps
Robustness Check: At least two years of trading history & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

Panel A: Overnight Jumps

	Numbers	Mean	Median	Int. Ret.<0	Int. Ret.>0	Int. Ret.=0	Opposite Sign Int. Return	Same Sign Int. Ret.
Total	320,944							
Negative	158,935	-7.3%	-5.0%	47,448	85,537	25,950	54%	30%
Positive	162,009	8.9%	5.5%	86,216	52,873	22,920	53%	33%

Panel B: Intraday Jumps

	Numbers	Mean	Median	Ovn. Ret.<0	Ovn. Ret.>0	Ovn. Ret.=0	Opposite Sign Ovn. Ret.	Same Sign Ovn. Ret.
Total	139,695							
Negative	64,405	-11.2%	-8.7%	35,735	27,381	1,289	43%	55%
Positive	75,290	14.7%	9.4%	42,254	31,296	1,740	56%	42%

Panel C: Daily Jumps

	Numbers	Mean	Median	Int. Ret.<0	Int. Ret.>0	Int. Ret.=0	Opposite Sign Int. Return	Same Sign Int. Ret.
Total	119,515							
Negative	55,629	-15.1%	-12.1%	49,574	4,018	2,037	7%	89%
Positive	63,886	19.9%	13.2%	3,932	58,015	1,939	6%	91%

Notes: Jump Statistics are tabulated for stocks with more than 2 years of trading history in the analysis period. Each panel shows the jump statistics for each corresponding component of the day. *Int. Ret.* in the table stands for intraday returns after overnight jumps. Our analyses cover 342 months over the period July 1993 - December 2021.

Table B2: Negative Overnight Jump Stocks: Cumulative Approach Descriptive Statistics and Correlation Matrix
Robustness Check: At least two years of trading history & $IVOL_OJI$ as the regressor & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

PANEL A: Descriptive Statistics - Negative Overnight Jumps

	CJR_t^-	$PREV_t$	$IVOL_OJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t
Mean	-0.08	0.01	0.03	19.58	-0.70	1.03	0.06	0.08	14.77
Median	-0.06	0.00	0.02	19.49	-0.60	0.76	0.02	0.03	0.22
St.Dev.	0.08	0.08	0.03	2.00	1.03	0.86	0.36	0.42	111.44
Min	-0.65	-0.26	0.00	14.86	-5.68	0.00	-0.77	-0.78	0.00
Max	-0.004	0.83	0.27	25.52	3.40	5.23	2.75	3.64	1799.14
Skewness	-3.10	3.63	3.99	0.24	-0.59	1.43	2.23	2.64	12.14
Kurtosis	15.81	49.27	32.99	-0.18	4.26	2.80	17.98	22.90	187.90
25th_Per.	-0.10	-0.01	0.02	18.14	-1.22	0.40	-0.12	-0.14	0.01
75th_Per.	-0.04	0.03	0.04	20.93	-0.09	1.45	0.18	0.21	2.39

PANEL B: Correlations

	CJR_t^-	$PREV_t$	$IVOL_OJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t
CJR_t^-	1.00								
$PREV_t$	-0.10	1.00							
$IVOL_OJI_t$	-0.48	0.24	1.00						
$SIZE$	0.22	-0.16	-0.44	1.00					
BM	0.01	0.07	0.10	-0.36	1.00				
LEV	0.08	0.02	-0.03	-0.05	-0.09	1.00			
$RET_{t-1,t-5}$	0.10	-0.05	-0.14	0.03	-0.04	0.01	1.00		
$RET_{t-6,t-11}$	0.06	-0.05	-0.10	0.09	-0.16	0.00	0.01	1.00	
AI_t	-0.12	0.10	0.27	-0.25	0.12	0.03	-0.07	-0.06	1.00

Notes: This table tabulates descriptive statistics and correlations between the variables in our monthly cross-sectional regressions. We calculate the figures for each month, construct a time series, and average them. CJR_t^+ and CJR_t^- are respectively the monthly cumulated positive and negative jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Our analyses cover 342 months over the period July 1993 - December 2021. See the Data section of the paper for detailed explanations of variables.

Table B3: Positive Overnight Jump Stocks: Cumulative Approach Descriptive Statistics and Correlation Matrix
Robustness Check: At least two years of trading history & $IVOL_OJI$ as the regressor & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

PANEL A: Descriptive Statistics - Positive Overnight Jumps

	CJR_t^+	$PREV_t$	$IVOL_OJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t
Mean	0.11	-0.01	0.03	19.44	-0.68	1.05	0.05	0.07	14.34
Median	0.07	0.00	0.02	19.32	-0.58	0.77	0.01	0.02	0.23
St.Dev.	0.15	0.08	0.03	1.99	1.06	0.88	0.38	0.44	104.27
Min	0.003	-0.41	0.00	14.77	-5.99	0.01	-0.80	-0.80	0.00
Max	1.76	0.69	0.27	25.53	3.78	5.57	3.12	3.90	1731.46
Skewness	5.62	1.75	3.72	0.27	-0.57	1.49	2.43	2.76	12.23
Kurtosis	53.73	42.04	27.70	-0.16	4.75	3.32	20.93	23.91	185.64
25th_Per.	0.04	-0.03	0.02	18.01	-1.21	0.40	-0.15	-0.15	0.01
75th_Per.	0.12	0.01	0.04	20.79	-0.06	1.49	0.18	0.21	2.28

PANEL B: Correlations

	CJR_t^+	$PREV_t$	$IVOL_OJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t
CJR_t^+	1.00								
$PREV_t$	-0.03	1.00							
$IVOL_OJI_t$	0.45	-0.03	1.00						
$SIZE$	-0.27	0.08	-0.46	1.00					
BM	0.04	-0.04	0.09	-0.34	1.00				
LEV	-0.04	-0.01	-0.04	-0.04	-0.12	1.00			
$RET_{t-1,t-5}$	-0.11	0.04	-0.16	0.05	-0.05	0.00	1.00		
$RET_{t-6,t-11}$	-0.09	0.02	-0.12	0.10	-0.16	-0.01	0.03	1.00	
AI_t	0.14	-0.11	0.25	-0.24	0.11	0.02	-0.07	-0.05	1.00

Notes: This table tabulates descriptive statistics and correlations between the variables in our monthly cross-sectional regressions. We calculate the figures for each month, construct a time series, and average them. CJR_t^+ and CJR_t^- are respectively the monthly cumulated positive and negative jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Our analyses cover 342 months over the period July 1993 - December 2021. See the Data section of the paper for detailed explanations of variables.

Table B4: Cross-Sectional Regressions for Return Predictability (Isolated Approach) Robustness Check: At least two years of trading history & $IVOL_OJI$ as the regressor & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

PANEL A: Negative Jumps

Dep. Variable:($1D_t$)	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**
JR_t^-	-0.43	-0.44	-0.43	-0.43	-0.43	-0.43	-0.43	-0.44	-0.44
	(-3.92)	(-3.92)	(-3.56)	(-3.57)	(-3.58)	(-3.61)	(-3.63)	(-3.59)	(-3.57)
$PREV_t$		-0.04	-0.04	-0.06	-0.06	-0.06	-0.06	-0.06	-0.06
		(-2.2)	(-2.85)	(-3.44)	(-3.59)	(-3.69)	(-3.72)	(-3.76)	(-3.88)
$IVOL_OJI_{t-1}$			0.01	-0.27	-0.26	-0.24	-0.23	-0.22	-0.22
			(0.08)	(-1.35)	(-1.29)	(-1.24)	(-1.16)	(-1.06)	(-1.06)
$SIZE$				-0.58	-0.49	-0.47	-0.47	-0.49	-0.46
				(-3.91)	(-3.73)	(-3.55)	(-3.41)	(-3.04)	(-2.89)
BM					0.41	0.45	0.44	0.44	0.43
					(3.48)	(3.77)	(3.8)	(3.64)	(3.58)
LEV						0.17	0.16	0.14	0.13
						(1.55)	(1.44)	(1.25)	(1.18)
$RET_{t-1,t-5}$							0.00	-0.10	-0.04
							(0)	(-0.39)	(-0.16)
$RET_{t-6,t-11}$								0.80	0.82
								(0.96)	(1.01)
AI_{t-1}									0.05
									(1.62)
$Intercept$	-0.02	-0.02	-0.02	0.10	0.08	0.08	0.08	0.08	0.08
	(-1.66)	(-1.66)	(-2.13)	(4.6)	(4.33)	(3.87)	(3.64)	(3.14)	(2.92)
$Adj.R^2$	0.07	0.08	0.10	0.11	0.11	0.11	0.12	0.12	0.12

PANEL B: Positive Jumps

Dep. Variable:($1D_t$)	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**
JR_t^+	-0.10	-0.10	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08
	(-12)	(-11.24)	(-11.55)	(-10.92)	(-11.01)	(-11.06)	(-11.01)	(-10.99)	(-10.56)
$PREV_t$		-0.01	-0.02	-0.02	-0.02	-0.02	-0.03	-0.03	-0.03
		(-1.4)	(-2.26)	(-2.76)	(-2.86)	(-2.9)	(-2.99)	(-3.11)	(-3.14)
$IVOL_OJI_{t-1}$			-0.25	-0.13	-0.13	-0.14	-0.13	-0.13	-0.16
			(-6.89)	(-4.16)	(-4.29)	(-4.25)	(-4.03)	(-4)	(-4.77)
$SIZE$				0.29	0.29	0.28	0.28	0.28	0.28
				(9.59)	(8.9)	(8.77)	(8.42)	(8.49)	(8.47)
BM					-0.03	-0.05	-0.04	-0.03	-0.03
					(-0.82)	(-1.38)	(-1.12)	(-0.71)	(-0.93)
LEV						-0.12	-0.12	-0.12	-0.13
						(-2.74)	(-2.78)	(-2.73)	(-2.94)
$RET_{t-1,t-5}$							0.52	0.53	0.54
							(3.52)	(3.47)	(3.61)
$RET_{t-6,t-11}$								0.32	0.29
								(3.65)	(3.44)
AI_{t-1}									0.00
									(0.77)
$Intercept$	0.00	0.00	0.00	-0.06	-0.06	-0.06	-0.06	-0.06	-0.06
	(-3.52)	(-3.76)	(2.07)	(-8.88)	(-8.39)	(-8.2)	(-8.06)	(-8.18)	(-8.05)
$Adj.R^2$	0.05	0.06	0.08	0.08	0.08	0.09	0.09	0.09	0.10

Notes: This Table reports return predictability results when jumps are treated in isolation instead of calculating cumulative jump returns. Stocks having multiple jumps in a month appear with multiple observations in monthly regressions as opposed to once with their cumulative figures. Tabulated results are for the first day after overnight jumps. The dependent variable $1D_t$ is the intraday return just after the overnight jump. We run monthly cross-sectional regressions and save the time series of coefficient estimates. To correct the correlation of the errors and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). From columns (2) to (9), we add each firm-specific control variable one at a time. Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. $JR^{-,+}$ variables in the first and second part of the Table respectively stand for negative and positive jump returns, $PREV$ is the previous day return before jumps, $IVOL_OJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_{t-1} is the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_{t-1} are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables.

Table B5: Daily Return Predictability After Overnight Jump (Isolated Approach)
Robustness Check: At least two years of trading history & $IVOL_OJI$ as the regressor & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

PANEL A: Negative Jumps

Dep. Variable	$1D_t$		$2D_t$		$3D_t$		$4D_t$		$5D_t$	
	coef.	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**
Intercept	0.08	(2.92)	0.09	(3.88)	0.09	(4.02)	0.09	(4.3)	0.09	(4.14)
JR_t^-	-0.44	(-3.57)	-0.43	(-3.82)	-0.44	(-3.92)	-0.42	(-4.04)	-0.42	(-4.03)
$PREV_t$	-0.06	(-3.88)	-0.10	(-6.02)	-0.12	(-7.04)	-0.12	(-7)	-0.13	(-6.83)
$IVOL_OJI_{t-1}$	-0.22	(-1.06)	-0.20	(-1.03)	-0.22	(-1.13)	-0.21	(-1.13)	-0.17	(-0.94)
$SIZE$	-0.46	(-2.89)	-0.52	(-3.68)	-0.52	(-3.85)	-0.53	(-4.04)	-0.51	(-3.92)
BM	0.43	(3.58)	0.46	(3.77)	0.49	(3.86)	0.48	(4.2)	0.52	(4.23)
LEV	0.13	(1.18)	0.19	(1.98)	0.20	(2.04)	0.19	(1.72)	0.22	(1.85)
$RET_{t-1,t-5}$	-0.04	(-0.16)	-0.10	(-0.37)	-0.31	(-1.03)	-0.28	(-1.03)	-0.31	(-1.22)
$RET_{t-6,t-11}$	0.82	(1.01)	0.76	(0.98)	0.71	(0.94)	0.68	(0.97)	0.76	(1.11)
AI_{t-1}	0.05	(1.62)	-0.07	(-0.72)	-0.04	(-0.56)	-0.07	(-0.7)	-0.09	(-0.77)
$Adj.R^2$	0.12		0.13		0.12		0.12		0.11	

PANEL B: Positive Jumps

Dep. Variable	$1D_t$		$2D_t$		$3D_t$		$4D_t$		$5D_t$	
	coef.	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**
Intercept	-0.06	(-8.05)	-0.07	(-7.69)	-0.07	(-7.28)	-0.07	(-6.18)	-0.07	(-6.49)
JR_t^+	-0.08	(-10.56)	-0.10	(-10.65)	-0.11	(-11.86)	-0.11	(-10.64)	-0.12	(-10.97)
$PREV_t$	-0.03	(-3.14)	-0.08	(-10.01)	-0.09	(-9.21)	-0.10	(-8.78)	-0.10	(-8.71)
$IVOL_OJI_{t-1}$	-0.16	(-4.77)	-0.20	(-6.17)	-0.22	(-6.57)	-0.24	(-6.61)	-0.25	(-6.26)
$SIZE$	0.28	(8.47)	0.35	(8.39)	0.38	(8.13)	0.38	(6.96)	0.39	(7.43)
BM	-0.03	(-0.93)	-0.10	(-1.89)	-0.10	(-1.9)	-0.10	(-1.57)	-0.06	(-0.88)
LEV	-0.13	(-2.94)	-0.19	(-4.26)	-0.16	(-3.31)	-0.17	(-2.85)	-0.18	(-3.43)
$RET_{t-1,t-5}$	0.54	(3.61)	0.64	(3.82)	0.63	(4.06)	0.66	(4.04)	0.70	(4.29)
$RET_{t-6,t-11}$	0.29	(3.44)	0.25	(2.07)	0.25	(1.95)	0.18	(1.18)	0.19	(1.14)
AI_{t-1}	0.00	(0.77)	0.00	(-0.2)	0.00	(0.42)	0.00	(0.45)	0.01	(0.78)
$Adj.R^2$	0.10		0.10		0.10		0.09		0.09	

Notes: This Table reports return predictability results when jumps are treated in isolation instead of calculating cumulative jump returns. Stocks having multiple jumps in a month appear with multiple observations in monthly regressions as opposed to once with their cumulative figures. Tabulated results are for the first five days after overnight jumps. The dependent variable $1D_t$ is the intraday return just after the overnight jump. We run monthly cross-sectional regressions and save the time series of coefficient estimates. To correct the correlation of the errors and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. $JR^{-,+}$ variables in the first and second part of the Table respectively stand for negative and positive jump returns, $PREV$ is the previous day return before jumps, $IVOL_OJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_{t-1} is the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_{t-1} are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables.

Table B6: Predictability in Different subperiods (Isolated Approach)
Robustness Check: At least two years of trading history & $IVOL_OJI$ as the regressor & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

	PANEL A: Negative Jumps						**PANEL B: Positive Jumps**					
	Jul.93 - Dec.03		**Jan.04-Dec.21**		**Aug.07 - Dec.13**		**Jul.93 - Dec.03**		**Jan.04 - Dec.21**		**Aug.07 - Dec.13**	
	coef.	**t**	**coef.**	**t**	**coef.**	**t**	**coef.**	**t**	**coef.**	**t**	**coef.**	**t**
Intercept	0.02	(1.21)	0.11	(2.86)	0.19	(2.03)	-0.04	(-6.22)	-0.07	(-6.93)	-0.08	(-5.07)
$JR_t^{-,+}$	-0.39	(-9.56)	-0.46	(-2.43)	-0.95	(-2.15)	-0.11	(-10.11)	-0.07	(-7.74)	-0.08	(-6.75)
$PREV_t$	-0.03	(-2.23)	-0.08	(-3.55)	-0.14	(-2.99)	-0.04	(-2.85)	-0.02	(-1.93)	-0.01	(-0.41)
$IVOL_OJI_{t-1}$	0.14	(2.99)	-0.43	(-1.36)	-0.99	(-1.23)	-0.15	(-3.66)	-0.17	(-3.48)	-0.26	(-3.55)
$SIZE$	-0.17	(-2)	-0.62	(-2.65)	-1.19	(-2.04)	0.20	(7.19)	0.33	(7.05)	0.40	(5.04)
BM	0.27	(4.76)	0.52	(2.86)	0.88	(2.11)	-0.01	(-0.41)	-0.05	(-0.85)	-0.01	(-0.11)
LEV	0.14	(2.45)	0.12	(0.72)	-0.22	(-0.54)	0.01	(0.21)	-0.21	(-3.49)	-0.11	(-1.53)
$RET_{t-1,t-5}$	0.28	(2.25)	-0.23	(-0.55)	-1.10	(-1.2)	0.40	(3.96)	0.62	(2.74)	1.36	(3.09)
$RET_{t-6,t-11}$	-0.07	(-0.31)	1.33	(1.06)	4.31	(1.38)	0.22	(3.18)	0.34	(2.61)	0.60	(3.78)
AI_{t-1}	0.00	(-0.84)	0.09	(1.69)	0.01	(1.85)	0.01	(5.13)	0.00	(0.15)	0.00	(0.64)
$Adj.R^2$	0.15		0.11		0.13		0.10		0.10		0.12	

Notes: This Table reports return predictability results when jumps are treated in isolation instead of calculating cumulative jump returns. Stocks having multiple jumps in a month appear with multiple observations in monthly regressions as opposed to once with their cumulative figures. Tabulated results are for the first day after overnight jumps. We run monthly cross-sectional regressions separately for the periods July 1993 - December 2003, January 2004 - December 2021, and August 2007 - December 2013, and save the time series of coefficient estimates. To account for cross-correlations and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). The dependent variable is $1D_t$, and it is the intraday return just after the overnight jump. Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. $JR^{-,+}$ variables respectively stand for negative and positive jump returns, $PREV$ is the previous day return before jumps, $IVOL_OJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_{t-1} is the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_{t-1} are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables.

Table B7: Regressions for Different Quintiles of IVOL_OJI (Isolated Approach)
Robustness Check: At least two years of trading history & $IVOL_OJI$ as the regressor & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

Negative Overnight Jumps

	$Int.$	JR_t^-	$PREV_t$	$IVOL_OJI_{t-1}$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_{t-1}	$Adj.R^2$
Q1	0.03	-0.17	-0.04	-0.15	-0.14	0.18	0.24	-0.23	0.10	0.91	0.10
	(3.71)	(-8.88)	(-2.18)	(-1.13)	(-3.93)	(3.42)	(3.68)	(-0.79)	(0.46)	(0.47)	
Q2	0.06	-0.13	-0.03	-0.20	-0.28	0.24	0.09	-0.53	0.01	0.79	0.08
	(5.38)	(-8.65)	(-1.49)	(-0.89)	(-5.57)	(2.53)	(1.17)	(-1.64)	(0.03)	(0.61)	
Q3	0.12	-0.43	-0.08	-0.39	-0.78	0.28	-0.56	-2.10	4.12	1.10	0.10
	(8.15)	(-1.42)	(-1.9)	(-0.54)	(-2.34)	(1.45)	(-0.85)	(-0.9)	(1.15)	(2.24)	
Q4	0.14	-0.18	-0.03	-0.05	-0.73	0.21	0.07	0.29	-0.34	0.55	0.11
	(7.35)	(-4.11)	(-2.02)	(-0.25)	(-7.84)	(3.02)	(0.71)	(1.01)	(-2.04)	(1.43)	
Q5	0.17	-0.58	-0.07	-0.07	-1.07	0.31	-0.25	1.12	0.29	0.23	0.17
	(4.42)	(-4.45)	(-3.5)	(-0.71)	(-4.92)	(1.14)	(-0.85)	(1.48)	(0.44)	(0.89)	

Positive Overnight Jumps

	$Int.$	JR_t^+	$PREV_t$	$IVOL_OJI_{t-1}$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_{t-1}	$Adj.R^2$
Q1	-0.02	-0.08	-0.01	0.06	0.09	-0.12	-0.19	-0.49	0.07	-0.25	0.07
	(-4.43)	(-10.78)	(-0.86)	(0.55)	(5.15)	(-2.11)	(-4.26)	(-2.81)	(0.44)	(-0.12)	
Q2	-0.04	-0.05	-0.03	0.08	0.18	-0.19	-0.20	-0.13	0.24	-0.70	0.07
	(-4.05)	(-6.79)	(-1.98)	(0.32)	(4.07)	(-2.96)	(-2.46)	(-0.58)	(1.03)	(-1.37)	
Q3	-0.07	-0.05	-0.01	-0.16	0.35	-0.05	-0.09	0.22	0.22	-0.27	0.08
	(-4.73)	(-4.95)	(-0.99)	(-0.63)	(6.07)	(-0.76)	(-1.35)	(0.81)	(1.32)	(-0.84)	
Q4	-0.11	-0.07	-0.01	-0.27	0.59	0.09	-0.11	0.67	0.22	-0.03	0.09
	(-7.81)	(-6.34)	(-1.1)	(-1.96)	(8.23)	(1.43)	(-0.86)	(2.98)	(1.2)	(-0.41)	
Q5	-0.06	-0.11	-0.04	-0.02	0.28	-0.07	-0.08	0.82	0.64	-0.02	0.12
	(-2.66)	(-8.99)	(-2.82)	(-0.25)	(2.6)	(-0.58)	(-0.95)	(2.54)	(1.66)	(-0.48)	

Notes: This table shows the regression outputs for portfolios formed according to different $IVOL_OJI$ quintiles when jumps are treated in isolation instead of calculating cumulative jump returns. At each month, we sort overnight jump stocks in ascending order based on the $IVOL_OJI$ numbers and form quintile portfolios. Then, we run monthly cross-sectional regressions and save the time series of coefficient estimates. The dependent variable is $1D_t$, and it is the intraday return just after the overnight jump. To account for cross-correlations and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. JR_t^- and JR_t^+ are respectively the negative and positive jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_{t-1} is the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_{t-1} are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables.

Table B8: Regressions for Different Quintiles of SIZE (Isolated Approach)
Robustness Check: At least two years of trading history & $IVOL_OJI$ as the regressor & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

Negative Overnight Jumps

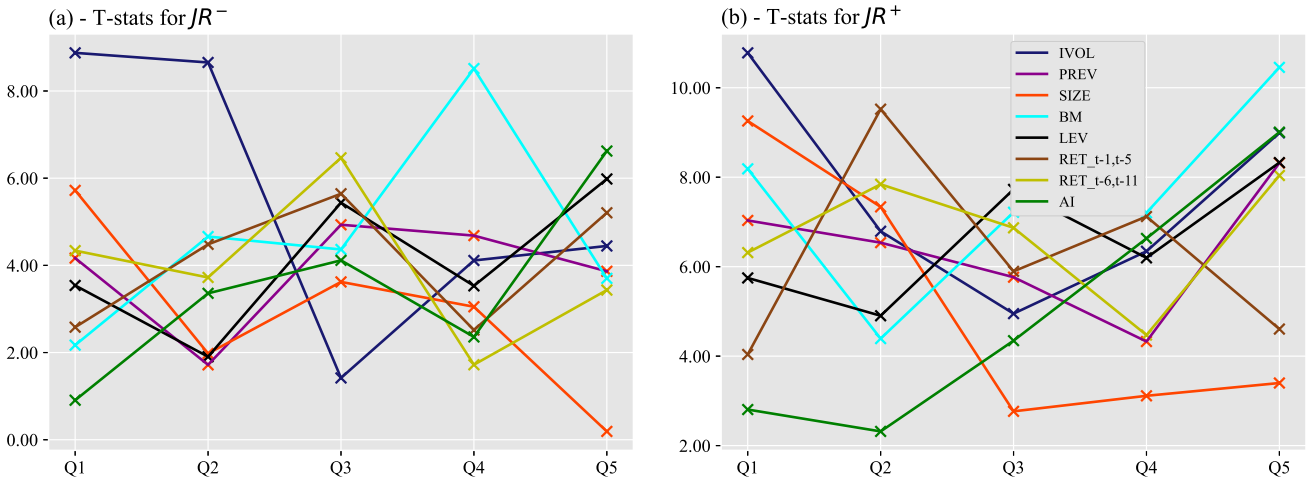
	$Int.$	JR_t^-	$PREV_t$	$IVOL_OJI_{t-1}$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_{t-1}	$Adj.R^2$
Q1	-0.07	-0.83	-0.13	-0.42	0.00	0.31	-0.27	0.17	1.13	0.01	0.26
	(-1.37)	(-5.72)	(-4.08)	(-2.06)	(1.18)	(1.96)	(-1.75)	(0.52)	(1.92)	(1.15)	
Q2	0.86	-0.56	-0.06	-0.92	-0.05	0.11	0.54	-0.14	3.76	10.21	0.17
	(1.13)	(-1.98)	(-2.69)	(-1.09)	(-1.13)	(0.62)	(1.25)	(-0.21)	(1.09)	(3.35)	
Q3	0.07	-0.12	-0.03	-0.14	0.00	0.03	-0.27	0.24	-0.17	46.55	0.11
	(1.92)	(-3.62)	(-1.94)	(-1.53)	(-1.91)	(0.46)	(-3.23)	(0.92)	(-0.8)	(3.54)	
Q4	0.00	-0.05	-0.03	-0.15	0.00	0.16	0.12	-0.17	-0.33	141.01	0.10
	(0.12)	(-3.05)	(-1.88)	(-2.45)	(-0.07)	(3.17)	(1.7)	(-0.67)	(-1.81)	(3.06)	
Q5	0.00	-0.003	-0.07	-0.12	0.00	-0.08	-0.15	0.02	0.13	508.07	0.09
	(-0.29)	(-0.19)	(-4.55)	(-2.28)	(0.7)	(-1.31)	(-1.65)	(0.12)	(0.65)	(3.31)	

Positive Overnight Jumps

	$Int.$	JR_t^+	$PREV_t$	$IVOL_OJI_{t-1}$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_{t-1}	$Adj.R^2$
Q1	0.10	-0.13	-0.06	-0.13	-0.01	0.14	0.01	0.87	0.80	0.00	0.16
	(2.15)	(-9.26)	(-3.05)	(-2.72)	(-2.37)	(1.04)	(0.06)	(2.89)	(2.62)	(0.38)	
Q2	-0.02	-0.11	-0.03	-0.19	0.00	0.01	-0.10	0.65	0.14	-2.97	0.14
	(-0.38)	(-7.34)	(-2.1)	(-3.02)	(0.25)	(0.17)	(-1.12)	(2.56)	(0.63)	(-2.65)	
Q3	-0.01	-0.04	-0.02	-0.11	0.00	-0.01	-0.19	-0.23	0.25	-10.31	0.10
	(-0.26)	(-2.77)	(-1.53)	(-1.74)	(0.28)	(-0.08)	(-2.26)	(-0.54)	(1.46)	(-1.4)	
Q4	-0.01	-0.03	-0.02	-0.01	0.00	-0.02	-0.21	-0.10	-0.25	-48.68	0.08
	(-0.34)	(-3.11)	(-1.25)	(-0.17)	(0.54)	(-0.28)	(-2.04)	(-0.46)	(-1.8)	(-1.51)	
Q5	0.02	-0.03	-0.07	0.10	0.00	-0.21	-0.10	-0.34	0.28	-260.16	0.08
	(2.29)	(-3.4)	(-5.72)	(2.04)	(-2.63)	(-4.46)	(-1.67)	(-2.05)	(1.94)	(-2.52)	

Notes: This table shows the regression outputs for portfolios formed according to different $SIZE$ quintiles when jumps are treated in isolation instead of calculating cumulative jump returns. At each month, we sort overnight jump stocks in ascending order based on the $SIZE$ figures and form quintile portfolios. Then, we run monthly cross-sectional regressions and save the time series of coefficient estimates. The dependent variable is $1D_t$, and it is the intraday return just after the overnight jump. To account for cross-correlations and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. JR_t^- and JR_t^+ are respectively the negative and positive jump returns, $PREV_t$ is the previous day returns before jumps, $IVOL_OJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_{t-1} is the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_{t-1} are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables.

Figure B1: JR Significance in Quintile Portfolios (Isolated Approach)
Robustness Check: At least two years of trading history & $IVOL_OJI$ as the regressor &
Overnight jumps eliminated if CRSP closing price is set to bid-ask average



Notes: This graph shows the Newey-West adjusted t-statistics for JR in 1-day predictability regressions. Each month, we sort overnight jump stocks in ascending order based on the levels of control variables and form quintile portfolios. Then, we run monthly cross-sectional regressions and save the time series of coefficient estimates. To account for cross-correlations and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). We calculate the Newey-West t-statistics with 12 lags for the time series of coefficient estimates. JR_t^- and JR_t^+ are respectively the negative and positive jump returns, $PREV_t$ is the previous day returns before jumps, $IVOL_OJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), and AI_{t-1} s the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month.

Table B9: Negative Overnight Jump Stocks: Cumulative Approach Descriptive Statistics and Correlation Matrix
Robustness Check: At least two years of trading history & $IVOL_DJI$ as the regressor & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

PANEL A: Descriptive Statistics in Negative Overnight Jumps

	CJR_t^-	$PREV_t$	$IVOL_DJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t
Mean	-0.08	0.01	0.03	19.58	-0.70	1.03	0.06	0.08	14.77
Median	-0.06	0.00	0.02	19.49	-0.60	0.76	0.02	0.03	0.22
St.Dev.	0.08	0.08	0.02	2.00	1.03	0.86	0.36	0.42	111.44
Min	-0.65	-0.26	0.00	14.86	-5.68	0.00	-0.77	-0.78	0.00
Max	-0.004	0.83	0.22	25.52	3.40	5.23	2.75	3.64	1799.14
Skew.	-3.10	3.63	3.07	0.24	-0.59	1.43	2.23	2.64	12.14
Kurto.	15.81	49.27	18.90	-0.18	4.26	2.80	17.98	22.90	187.90
25th_Per.	-0.10	-0.01	0.02	18.14	-1.22	0.40	-0.12	-0.14	0.01
75th_Per.	-0.04	0.03	0.04	20.93	-0.09	1.45	0.18	0.21	2.39

PANEL B: Correlations

	CJR_t^-	$PREV_t$	$IVOL_DJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t
CJR_t^-	1.00								
$PREV_t$	-0.10	1.00							
$IVOL_DJI_t$	-0.52	0.20	1.00						
$SIZE$	0.22	-0.16	-0.44	1.00					
BM	0.01	0.07	0.09	-0.36	1.00				
LEV	0.08	0.02	-0.04	-0.05	-0.09	1.00			
$RET_{t-1,t-5}$	0.10	-0.05	-0.14	0.03	-0.04	0.01	1.00		
$RET_{t-6,t-11}$	0.06	-0.05	-0.10	0.09	-0.16	0.00	0.01	1.00	
AI_t	-0.12	0.10	0.28	-0.25	0.12	0.03	-0.07	-0.06	1.00

Notes: This table tabulates descriptive statistics and correlations between the variables in our monthly cross-sectional regressions. We calculate the figures for each month, construct a time series, and average them. CJR_t^+ and CJR_t^- are respectively the monthly cumulated positive and negative jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Our analyses cover 342 months over the period July 1993 - December 2021. See the Data section of the paper for detailed explanations of variables.

Table B10: Positive Overnight Jump Stocks: Cumulative Approach Descriptive Statistics and Correlation Matrix
Robustness Check: At least two years of trading history & $IVOL_DJI$ as the regressor & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

PANEL A: Descriptive Statistics in Positive Overnight Jumps

	CJR_t^+	$PREV_t$	$IVOL_DJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t
Mean	0.11	-0.01	0.03	19.44	-0.68	1.05	0.05	0.07	14.34
Median	0.07	0.00	0.03	19.32	-0.58	0.77	0.01	0.02	0.23
St.Dev.	0.15	0.08	0.03	1.99	1.06	0.88	0.38	0.44	104.27
Min	0.003	-0.41	0.00	14.77	-5.99	0.01	-0.80	-0.80	0.00
Max	1.76	0.69	0.24	25.53	3.78	5.57	3.12	3.90	1731.46
Skew.	5.62	1.75	3.04	0.27	-0.57	1.49	2.43	2.76	12.23
Kurto.	53.73	42.04	18.25	-0.16	4.75	3.32	20.93	23.91	185.64
25th_Per.	0.04	-0.03	0.02	18.01	-1.21	0.40	-0.15	-0.15	0.01
75th_Per.	0.12	0.01	0.04	20.79	-0.06	1.49	0.18	0.21	2.28

PANEL B: Correlations

	CJR_t^+	$PREV_t$	$IVOL_DJI_t$	$SIZE$	BM	LEV	$RET_{t-1,t-5}$	$RET_{t-6,t-11}$	AI_t
CJR_t^+	1.00								
$PREV_t$	-0.03	1.00							
$IVOL_DJI_t$	0.49	-0.06	1.00						
$SIZE$	-0.27	0.08	-0.46	1.00					
BM	0.04	-0.04	0.08	-0.34	1.00				
LEV	-0.04	-0.01	-0.04	-0.04	-0.12	1.00			
$RET_{t-1,t-5}$	-0.11	0.04	-0.15	0.05	-0.05	0.00	1.00		
$RET_{t-6,t-11}$	-0.09	0.02	-0.12	0.10	-0.16	-0.01	0.03	1.00	
AI_t	0.14	-0.11	0.27	-0.24	0.11	0.02	-0.07	-0.05	1.00

Notes: This table tabulates descriptive statistics and correlations between the variables in our monthly cross-sectional regressions. We calculate the figures for each month, construct a time series, and average them. CJR_t^+ and CJR_t^- are respectively the monthly cumulated positive and negative jump returns, $PREV_t$ is the monthly cumulated previous-day returns before jumps, $IVOL_OJI_t$ is the idiosyncratic volatility calculated in isolation with overnight jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_t is the monthly Amihud Illiquidity measure constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Our analyses cover 342 months over the period July 1993 - December 2021. See the Data section of the paper for detailed explanations of variables.

Table B11: Cross-Sectional Regressions for Return Predictability (Isolated Approach) Robustness Check: At least two years of trading history & $IVOL_DJI$ as the regressor & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

PANEL A: Negative Jumps									
Dep.Variable:($1D_t$)	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**
JR_t^-	-0.43	-0.44	-0.43	-0.43	-0.43	-0.44	-0.44	-0.44	-0.44
	(-3.92)	(-3.92)	(-3.56)	(-3.57)	(-3.59)	(-3.61)	(-3.63)	(-3.59)	(-3.57)
$PREV_t$		-0.04	-0.04	-0.06	-0.06	-0.06	-0.06	-0.06	-0.07
		(-2.2)	(-2.81)	(-3.43)	(-3.58)	(-3.67)	(-3.7)	(-3.74)	(-3.85)
$IVOL_DJI_{t-1}$			-0.04	-0.35	-0.33	-0.32	-0.31	-0.29	-0.30
			(-0.23)	(-1.51)	(-1.45)	(-1.38)	(-1.31)	(-1.21)	(-1.23)
$SIZE$				-0.59	-0.50	-0.48	-0.47	-0.49	-0.46
				(-3.81)	(-3.62)	(-3.47)	(-3.32)	(-2.96)	(-2.81)
BM					0.44	0.47	0.47	0.47	0.45
					(3.29)	(3.58)	(3.64)	(3.46)	(3.4)
LEV						0.19	0.18	0.17	0.16
						(2.12)	(1.97)	(1.77)	(1.71)
$RET_{t-1,t-5}$							-0.10	-0.20	-0.14
							(-0.5)	(-0.65)	(-0.45)
$RET_{t-6,t-11}$								0.79	0.80
								(0.93)	(0.97)
AI_{t-1}									0.05
									(1.64)
$Intercept$	-0.02	-0.02	-0.02	0.10	0.09	0.08	0.08	0.09	0.08
	(-1.66)	(-1.66)	(-2.17)	(4.27)	(3.96)	(3.65)	(3.43)	(2.98)	(2.77)
$Adj.R^2$	0.07	0.08	0.10	0.11	0.11	0.11	0.11	0.12	0.12
PANEL B: Positive Jumps									
Dep. Variable:($1D_t$)	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**
JR_t^+	-0.10	-0.10	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08
	(-12)	(-11.24)	(-11.49)	(-10.89)	(-10.99)	(-11.02)	(-10.98)	(-10.96)	(-10.59)
$PREV_t$		-0.01	-0.02	-0.02	-0.02	-0.03	-0.03	-0.03	-0.03
		(-1.4)	(-2.31)	(-2.8)	(-2.91)	(-2.94)	(-3.06)	(-3.18)	(-3.19)
$IVOL_DJI_{t-1}$			-0.27	-0.14	-0.15	-0.15	-0.15	-0.15	-0.18
			(-7.2)	(-4.39)	(-4.57)	(-4.58)	(-4.43)	(-4.3)	(-5.05)
$SIZE$				0.29	0.28	0.27	0.28	0.28	0.28
				(9.86)	(9.07)	(8.96)	(8.59)	(8.73)	(8.63)
BM					-0.04	-0.06	-0.05	-0.03	-0.04
					(-0.99)	(-1.59)	(-1.34)	(-0.92)	(-1.19)
LEV						-0.12	-0.13	-0.12	-0.13
						(-2.79)	(-2.85)	(-2.82)	(-3.05)
$RET_{t-1,t-5}$							0.51	0.52	0.52
							(3.4)	(3.36)	(3.46)
$RET_{t-6,t-11}$								0.32	0.30
								(3.56)	(3.34)
AI_{t-1}									0.00
									(0.3)
$Intercept$	0.00	0.00	0.00	-0.06	-0.06	-0.05	-0.05	-0.05	-0.05
	(-3.52)	(-3.76)	(2.61)	(-9.14)	(-8.53)	(-8.36)	(-8.22)	(-8.41)	(-8.19)
$Adj.R^2$	0.05	0.06	0.07	0.08	0.08	0.09	0.09	0.09	0.10

Notes: This Table reports return predictability results when jumps are treated in isolation instead of calculating cumulative jump returns. Stocks having multiple jumps in a month appear with multiple observations in monthly regressions as opposed to once with their cumulative figures. Tabulated results are for the first day after overnight jumps. The dependent variable $1D_t$ is the intraday return just after the overnight jump. We run monthly cross-sectional regressions and save the time series of coefficient estimates. To correct the correlation of the errors and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). From columns (2) to (9), we add each firm-specific control variable one at a time. Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. $JR^{-,+}$ variables in the first and second part of the Table respectively stand for negative and positive jump returns, $PREV$ is the previous day return before jumps, $IVOL_DJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with daily jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_{t-1} is the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_{t-1} are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables.

Table B12: Daily Return Predictability After Overnight Jump (Isolated Approach)
Robustness Check: At least two years of trading history & $IVOL_DJI$ as the regressor & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

PANEL A: Negative Jumps

Dep. Variable	$1D_t$		$2D_t$		$3D_t$		$4D_t$		$5D_t$	
	coef.	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**
Intercept	0.08	(2.77)	0.09	(3.65)	0.09	(3.78)	0.10	(4.11)	0.09	(3.98)
JR_t^-	-0.44	(-3.57)	-0.43	(-3.81)	-0.44	(-3.91)	-0.42	(-4.03)	-0.42	(-4.01)
$PREV_t$	-0.07	(-3.85)	-0.10	(-6.02)	-0.12	(-7.07)	-0.12	(-7)	-0.13	(-6.82)
$IVOL_DJI_{t-1}$	-0.30	(-1.23)	-0.26	(-1.16)	-0.28	(-1.25)	-0.28	(-1.32)	-0.23	(-1.1)
$SIZE$	-0.46	(-2.81)	-0.52	(-3.56)	-0.51	(-3.72)	-0.53	(-3.95)	-0.51	(-3.83)
BM	0.45	(3.4)	0.49	(3.57)	0.52	(3.65)	0.50	(3.98)	0.54	(4.13)
LEV	0.16	(1.71)	0.21	(2.62)	0.22	(2.56)	0.21	(2.17)	0.24	(2.2)
$RET_{t-1,t-5}$	-0.14	(-0.45)	-0.19	(-0.62)	-0.39	(-1.17)	-0.37	(-1.21)	-0.38	(-1.28)
$RET_{t-6,t-11}$	0.80	(0.97)	0.74	(0.94)	0.69	(0.9)	0.66	(0.93)	0.75	(1.08)
AI_{t-1}	0.05	(1.64)	-0.07	(-0.72)	-0.04	(-0.56)	-0.07	(-0.7)	-0.10	(-0.77)
$Adj.R^2$	0.12		0.13		0.12		0.12		0.11	

PANEL B: Positive Jumps

Dep. Variable	$1D_t$		$2D_t$		$3D_t$		$4D_t$		$5D_t$	
	coef.	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**
Intercept	-0.05	(-8.19)	-0.07	(-7.84)	-0.07	(-7.21)	-0.07	(-5.93)	-0.07	(-6.23)
JR_t^+	-0.08	(-10.59)	-0.10	(-10.72)	-0.11	(-11.97)	-0.11	(-10.68)	-0.12	(-11.11)
$PREV_t$	-0.03	(-3.19)	-0.08	(-10.04)	-0.09	(-9.26)	-0.10	(-8.85)	-0.10	(-8.9)
$IVOL_DJI_{t-1}$	-0.18	(-5.05)	-0.23	(-6.73)	-0.26	(-7.17)	-0.29	(-6.27)	-0.30	(-6.45)
$SIZE$	0.28	(8.63)	0.34	(8.55)	0.37	(8.09)	0.37	(6.77)	0.37	(7.24)
BM	-0.04	(-1.19)	-0.11	(-2.11)	-0.12	(-2.11)	-0.11	(-1.77)	-0.07	(-1.12)
LEV	-0.13	(-3.05)	-0.20	(-4.31)	-0.17	(-3.39)	-0.18	(-2.97)	-0.19	(-3.57)
$RET_{t-1,t-5}$	0.52	(3.46)	0.62	(3.71)	0.59	(3.82)	0.59	(3.51)	0.63	(3.73)
$RET_{t-6,t-11}$	0.30	(3.34)	0.24	(1.92)	0.25	(1.8)	0.17	(1.02)	0.18	(1.01)
AI_{t-1}	0.00	(0.3)	0.00	(-0.7)	0.00	(-0.03)	0.00	(-0.29)	0.00	(0.37)
$Adj.R^2$	0.10		0.10		0.10		0.09		0.09	

Notes: This Table reports return predictability results when jumps are treated in isolation instead of calculating cumulative jump returns. Stocks having multiple jumps in a month appear with multiple observations in monthly regressions as opposed to once with their cumulative figures. Tabulated results are for the first five days after overnight jumps. The dependent variable $1D_t$ is the intraday return just after the overnight jump. We run monthly cross-sectional regressions and save the time series of coefficient estimates. To correct the correlation of the errors and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. $JR^{-,+}$ variables in the first and second part of the Table respectively stand for negative and positive jump returns, $PREV$ is the previous day return before jumps, $IVOL_DJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with daily jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_{t-1} is the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_{t-1} are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables.

Table B13: Cross-Sectional Regressions for Return Predictability (Isolated Approach) Robustness Check: At least two years of trading history & $IVOL_ODJI$ as the regressor & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

PANEL A: Negative Jumps									
Dep.Variable:($1D_t$)	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**
JR_t^-	-0.43	-0.44	-0.43	-0.43	-0.43	-0.44	-0.44	-0.44	-0.44
	(-3.92)	(-3.92)	(-3.54)	(-3.56)	(-3.58)	(-3.6)	(-3.62)	(-3.57)	(-3.56)
$PREV_t$		-0.04	-0.04	-0.06	-0.06	-0.06	-0.06	-0.06	-0.07
		(-2.2)	(-2.84)	(-3.42)	(-3.56)	(-3.66)	(-3.68)	(-3.71)	(-3.82)
$IVOL_ODJI_{t-1}$			-0.04	-0.37	-0.36	-0.34	-0.33	-0.32	-0.33
			(-0.23)	(-1.52)	(-1.46)	(-1.4)	(-1.33)	(-1.24)	(-1.26)
$SIZE$				-0.61	-0.52	-0.50	-0.50	-0.52	-0.49
				(-3.63)	(-3.45)	(-3.31)	(-3.16)	(-2.83)	(-2.7)
BM					0.42	0.45	0.45	0.44	0.43
					(3.45)	(3.76)	(3.83)	(3.66)	(3.59)
LEV						0.18	0.18	0.16	0.15
						(1.95)	(1.78)	(1.58)	(1.54)
$RET_{t-1,t-5}$							-0.11	-0.22	-0.16
							(-0.51)	(-0.67)	(-0.48)
$RET_{t-6,t-11}$								0.79	0.81
								(0.92)	(0.96)
AI_{t-1}									0.05
									(1.64)
$Intercept$	-0.02	-0.02	-0.02	0.11	0.09	0.09	0.09	0.09	0.08
	(-1.66)	(-1.66)	(-2.2)	(4.05)	(3.81)	(3.48)	(3.28)	(2.86)	(2.68)
$Adj.R^2$	0.07	0.08	0.10	0.11	0.11	0.11	0.12	0.12	0.12
PANEL B: Positive Jumps									
Dep.Variable:($1D_t$)	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**
JR_t^+	-0.10	-0.10	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08
	(-12)	(-11.24)	(-11.43)	(-10.9)	(-11)	(-11.03)	(-11)	(-10.97)	(-10.58)
$PREV_t$		-0.01	-0.02	-0.02	-0.03	-0.03	-0.03	-0.03	-0.03
		(-1.4)	(-2.33)	(-2.81)	(-2.91)	(-2.95)	(-3.08)	(-3.2)	(-3.21)
$IVOL_ODJI_{t-1}$			-0.29	-0.15	-0.16	-0.17	-0.16	-0.16	-0.20
			(-7.31)	(-4.66)	(-4.82)	(-4.82)	(-4.66)	(-4.56)	(-5.31)
$SIZE$				0.29	0.28	0.27	0.27	0.27	0.27
				(9.82)	(9.03)	(8.89)	(8.52)	(8.63)	(8.55)
BM					-0.04	-0.06	-0.05	-0.04	-0.05
					(-1)	(-1.61)	(-1.36)	(-0.97)	(-1.23)
LEV						-0.13	-0.13	-0.13	-0.14
						(-2.83)	(-2.88)	(-2.85)	(-3.09)
$RET_{t-1,t-5}$							0.51	0.52	0.52
							(3.36)	(3.32)	(3.42)
$RET_{t-6,t-11}$								0.32	0.29
								(3.54)	(3.32)
AI_{t-1}									0.00
									(0.28)
$Intercept$	0.00	0.00	0.00	-0.06	-0.06	-0.05	-0.05	-0.05	-0.05
	(-3.52)	(-3.76)	(2.66)	(-9.06)	(-8.46)	(-8.26)	(-8.11)	(-8.27)	(-8.07)
$Adj.R^2$	0.05	0.06	0.07	0.08	0.08	0.09	0.09	0.09	0.10

Notes: This Table reports return predictability results when jumps are treated in isolation instead of calculating cumulative jump returns. Stocks having multiple jumps in a month appear with multiple observations in monthly regressions as opposed to once with their cumulative figures. Tabulated results are for the first day after overnight jumps. The dependent variable $1D_t$ is the intraday return just after the overnight jump. We run monthly cross-sectional regressions and save the time series of coefficient estimates. To correct the correlation of the errors and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). From columns (2) to (9), we add each firm-specific control variable one at a time. Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. $JR^{-,+}$ variables in the first and second part of the Table respectively stand for negative and positive jump returns, $PREV$ is the previous day return before jumps, $IVOL_ODJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with overnight and daily jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_{t-1} is the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_{t-1} are multiplied by 100. Our analyses cover 342 months over the period July 1993 -

Table B14: Daily Return Predictability After Overnight Jump (Isolated Approach)
Robustness Check: At least two years of trading history & $IVOL_ODJI$ as the regressor & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

PANEL A: Negative Jumps

Dep. Variable	$1D_t$		$2D_t$		$3D_t$		$4D_t$		$5D_t$	
	coef.	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**
$Intercept$	0.08	(2.68)	0.10	(3.48)	0.10	(3.62)	0.10	(3.87)	0.10	(3.76)
JR_t^-	-0.44	(-3.56)	-0.43	(-3.8)	-0.44	(-3.9)	-0.42	(-4.01)	-0.42	(-4)
$PREV_t$	-0.07	(-3.82)	-0.10	(-5.96)	-0.12	(-7.01)	-0.12	(-6.97)	-0.13	(-6.78)
$IVOL_ODJI_{t-1}$	-0.33	(-1.26)	-0.29	(-1.2)	-0.30	(-1.29)	-0.30	(-1.33)	-0.24	(-1.12)
$SIZE$	-0.49	(-2.7)	-0.55	(-3.39)	-0.54	(-3.55)	-0.55	(-3.73)	-0.53	(-3.63)
BM	0.43	(3.59)	0.46	(3.8)	0.50	(3.89)	0.48	(4.2)	0.52	(4.31)
LEV	0.15	(1.54)	0.20	(2.39)	0.22	(2.39)	0.20	(2.01)	0.23	(2.08)
$RET_{t-1,t-5}$	-0.16	(-0.48)	-0.22	(-0.66)	-0.42	(-1.19)	-0.39	(-1.21)	-0.40	(-1.29)
$RET_{t-6,t-11}$	0.81	(0.96)	0.75	(0.93)	0.70	(0.89)	0.67	(0.92)	0.76	(1.07)
AI_{t-1}	0.05	(1.64)	-0.07	(-0.72)	-0.04	(-0.56)	-0.07	(-0.7)	-0.10	(-0.77)
$Adj.R^2$	0.12		0.13		0.12		0.12		0.11	

PANEL B: Positive Jumps

Dep.Variable	$1D_t$		$2D_t$		$3D_t$		$4D_t$		$5D_t$	
	coef.	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**	**coef.**	**t-stat**
$Intercept$	-0.05	(-8.07)	-0.06	(-7.85)	-0.07	(-7.19)	-0.07	(-5.89)	-0.07	(-6.24)
JR_t^+	-0.08	(-10.58)	-0.10	(-10.58)	-0.11	(-11.79)	-0.11	(-10.57)	-0.12	(-10.96)
$PREV_t$	-0.03	(-3.21)	-0.08	(-10.04)	-0.09	(-9.28)	-0.10	(-8.85)	-0.10	(-8.9)
$IVOL_ODJI_{t-1}$	-0.20	(-5.31)	-0.25	(-6.9)	-0.27	(-7.31)	-0.31	(-6.56)	-0.32	(-6.5)
$SIZE$	0.27	(8.55)	0.34	(8.57)	0.36	(8.09)	0.36	(6.75)	0.37	(7.25)
BM	-0.05	(-1.23)	-0.11	(-2.12)	-0.12	(-2.11)	-0.11	(-1.78)	-0.07	(-1.11)
LEV	-0.14	(-3.09)	-0.20	(-4.32)	-0.17	(-3.4)	-0.18	(-2.99)	-0.19	(-3.57)
$RET_{t-1,t-5}$	0.52	(3.42)	0.62	(3.7)	0.59	(3.88)	0.60	(3.58)	0.64	(3.79)
$RET_{t-6,t-11}$	0.29	(3.32)	0.23	(1.93)	0.24	(1.79)	0.17	(1.02)	0.18	(1.03)
AI_{t-1}	0.00	(0.28)	0.00	(-0.7)	0.00	(-0.02)	0.00	(-0.26)	0.00	(0.35)
$Adj.R^2$	0.10		0.10		0.10		0.09		0.09	

Notes: This Table reports return predictability results when jumps are treated in isolation instead of calculating cumulative jump returns. Stocks having multiple jumps in a month appear with multiple observations in monthly regressions as opposed to once with their cumulative figures. Tabulated results are for the first five days after overnight jumps. The dependent variable $1D_t$ is the intraday return just after the overnight jump. We run monthly cross-sectional regressions and save the time series of coefficient estimates. To correct the correlation of the errors and get robust standard errors, we use the cluster command with the CRSP Standard Industrial Classification Code (SICCD). Table populates averaged monthly coefficient estimates and their Newey-West t-statistics with 12 lags. $JR^{-,+}$ variables in the first and second part of the Table respectively stand for negative and positive jump returns, $PREV$ is the previous day return before jumps, $IVOL_ODJI_{t-1}$ is the previous-month idiosyncratic volatility calculated in isolation with overnight and daily jump days, $SIZE$ is the log of market cap at every June, BM is the log of book-to-market ratio, LEV is the log of total assets' book value divided by the log of market equity, $RET_{t-1,t-5}$ and $RET_{t-6,t-11}$ are set as the lagged momentum returns split for different horizons following Grinblatt and Moskowitz (2004) and Jiang and Zhu (2017), AI_{t-1} is the monthly Amihud Illiquidity measure for the previous month constructed as the mean of daily figures in a month which is later multiplied by 1,000,000. Estimated regression coefficients for $SIZE$, BM, LEV, $RET_{t-1,t-5}$, $RET_{t-6,t-11}$ and AI_{t-1} are multiplied by 100. Our analyses cover 342 months over the period July 1993 - December 2021. See Section 3.1 for the detailed explanations of variables.

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